

GRADENEXT · SMART LAB

Simulation Experiment Book

Grade 6 · Unit A · Systems and Subsystems

California Integrated Science, Grade 6 · Domain: Science Practice / Systems Thinking · 5 topics · 27 experiments

NGSS performance expectations anchored: Supporting unit (feeds later PEs)

Specification standard: v1.0 · Assembled 04 September 2026

Build specifications for interactive simulation experiments · one per syllabus subtopic · for developers, artists and curriculum designers

How to read this book. Every subtopic in the GradeNext California Science syllabus has exactly one simulation experiment specification. Each specification is a *build order*: it tells a developer what to code, an artist what to model, and a curriculum designer what the student is meant to discover. Nothing here is a worksheet; every experiment is a running, manipulable scene with a control panel, more than one scenario, and something measurable coming out.

Anatomy of a specification

Section	What it is for	Who reads it first
Header fields	Experiment name, render mode, simulation engine, interaction level, session length, NGSS anchor	Product, curriculum
Theme & scene	What the student sees on load: setting, camera, palette, lighting, foreground/background, where the panel sits	Art, UX
Objects & components	Every entity the engine must instantiate, including invisible ones (fields, colliders, probes). Build notes give geometry, appearance, counts, behaviour. The <i>Editable</i> column says which objects the student can drag, place, swap, resize or connect directly	3D/2D art, engine
How it works — the model	The rules the engine runs: equations, rate laws, state machines, datasets; what updates per tick; what is deliberately simplified and why	Engine, science review
Control panel	Every control with widget type, range or options, default, unit and visible effect. At least one control is <i>structural</i> (changes which objects exist or how they are arranged)	Engine, UI
Scenarios	Named presets that reconfigure the same simulation to ask a different question. Presets reference control names from the panel	Curriculum, QA
Student activities	Numbered actions the student performs and what they record	Curriculum
Outputs & measurement	Readouts, graphs, tables, badges the sim produces and when they update	Engine, UI, assessment
What the student should realise	The single idea, the misconception it displaces, and the sentence the student should be able to say	Curriculum, assessment

Glossary of tags

Render mode · *2D Canvas* — vector/sprite scene on a flat canvas, the mode of the existing collision and projectile sims · *3D Scene* — camera, lights, meshes and materials · *2.5D Layered* — parallax layers or cutaways with depth ordering but no free camera · *Hybrid 2D+3D* — a 3D viewport alongside a 2D graph, diagram or map panel · *Data Dashboard* — datasets, filters, plot canvases and claim-building tools as the primary scene.

Simulation engine · *Rigid-body* (masses, forces, collisions) · *Particle system* (many small agents with simple rules) · *Field/vector* (scalar or vector fields sampled in space) · *Molecular* (atoms, bonds, valence, CPK colouring) · *Agent-based* (organisms or entities with behaviours) · *Fluid/thermal* (heat flow, buoyancy, convection) · *Ray/wave* (light and sound propagation) · *Data-driven model* (real or realistic datasets driving the scene) · *State machine* (staged processes with transitions) · *Procedural geology* (layers, uplift, erosion, plate motion).

Interaction level · *Explore* (look, zoom, toggle overlays) · *Manipulate* (drag, place, swap objects) · *Investigate* (control variables, measure, record) · *Design* (build, test, score, iterate) · *Argue-from-data* (select evidence, build and score claims).

Object class · *Actor* (moves or acts) · *Structure* (static or slowly changing body) · *Particle* (instanced small entity) · *Field* (invisible spatial quantity) · *Instrument* (measures something) · *Overlay* (visual layer on top of the scene) · *Environment* (backdrop, terrain, container) · *UI-Probe* (cursor-following tool).

Control widgets · Slider · Stepper · Dropdown · Toggle · Radio · Colour · Dial · Drag-handle · Multi-select · Numeric field · Timeline scrubber.

Engineering conventions

- **Experiment ID** is G<grade>-<subtopic> (for example G6-B2.1). It is stable and unique across the whole programme.
- **Scenario presets** use control=value pairs separated by semicolons. Control names in presets are shortened versions of the panel labels; the JSON companion resolves them.
- **Every control carries a unit.** Where a control is unitless the unit column reads "—".
- **The JSON companion** for this unit (json/G<grade>_Unit<letter>.json) contains every field of every specification in structured form, with numeric ranges parsed into min/max and option lists into options[], ready for ingestion by the Smart Lab content pipeline.
- **Ages 11–14.** All simulated hazards are clearly virtual. No procedure here is a real-lab instruction.

Unit map

ID	Subtopic	Experiment	Render	Engine	Level	Session
A1.1	What makes a system a system	Unplug the Aquarium	Hybrid 2D+3D	Data-driven model (stock-and-flow) + Agent-based	Investigate	12–18 min
A1.2	Subsystems nested within systems	The Bike Inside the Bike	Hybrid 2D+3D	Rigid-body + State machine	Investigate	18–25 min
A1.3	Interactions among a system's parts	Cut the Link, Cook the Tomatoes	Hybrid 2D+3D	Data-driven model (system dynamics) + Fluid/thermal	Investigate	18–25 min
A1.4	Emergent properties	No Bee Is In Charge	3D Scene	Agent-based	Investigate	18–25 min
A1.5	Systems across scales, from a cell to a planet	From Chloroplast to Coastline	2.5D Layered	Data-driven model (coupled multi-scale) + Agent-based	Investigate	18–25 min
A2.1	Drawing a system's boundary	Where You Draw the Line	Hybrid 2D+3D	Data-driven model (flow network) + State machine	Investigate	12–18 min
A2.2	Open vs closed systems	Three Jars, One Lamp	3D Scene	Data-driven model (mass and energy balance) + Fluid/thermal	Investigate	12–18 min
A2.3	Inputs and outputs	On the Dyno: In One End, Out the Other	Hybrid 2D+3D	Data-driven model (process balance) + State machine	Investigate	12–18 min
A2.4	Tracing matter and energy through a system	Follow One Drop, Follow One Joule	2.5D Layered	Agent-based (tracer particles on a flow network) + Data-driven model	Investigate	18–25 min
A2.5	Choosing a boundary for a purpose	The Zero-Emission Bus Argument	Data Dashboard	Data-driven model + Agent-based	Argue-from-data	18–25 min
A3.1	Why scientists build models	The Experiment You Cannot Run	Hybrid 2D+3D	Data-driven model + State machine	Investigate	12–18 min
A3.2	Diagrams and flowcharts as models	The Diagram That Runs	Hybrid 2D+3D	State machine + Agent-based	Design	18–25 min
A3.3	Physical and digital models	Two Bays: The Warehouse and the Solver	Hybrid 2D+3D	Fluid/thermal + Data-driven model	Investigate	18–25 min
A3.4	What a model leaves out on purpose	The Model That Flattened the Sierra	Hybrid 2D+3D	Data-driven model + Fluid/thermal	Argue-from-data	18–25 min
A3.5	Building and revising a model of a system	Three Versions of a Kelp Forest	Hybrid 2D+3D	Data-driven model (stock-and-flow) + Agent-based	Design	18–25 min
A4.1	The geosphere	Peel the Planet: Reading the Rock Shells	3D Scene	Procedural geology + Ray/wave	Investigate	18–25 min
A4.2	The hydrosphere	Ninety-Seven, Two, and a Splash	Hybrid 2D+3D	Data-driven model + Particle system	Investigate	18–25 min
A4.3	The atmosphere	Ride the Balloon to the Edge of Air	3D Scene	Fluid/thermal + Data-driven model	Investigate	12–18 min
A4.4	The biosphere	The Living Skin: Painting Life onto Bare Rock	Hybrid 2D+3D	Agent-based + Data-driven model	Design	18–25 min
A4.5	Interactions among Earth's four spheres	Pull One Thread: The Four-Sphere Web	2.5D Layered	Data-driven model + Field/vector	Investigate	12–18 min
A4.6	Modeling an Earth-system event	One Spark, Sixty Years: A Sierra Watershed	Hybrid 2D+3D	State machine + Data-driven model + Particle system	Argue-from-data	18–25 min
A5.1	Lab safety and working like a scientist	The Bench That Bites	3D Scene	State machine + Fluid/thermal	Investigate	18–25 min

ID	Subtopic	Experiment	Render	Engine	Level	Session
A5.2	Variables and fair tests	Four Chambers, One Question	Hybrid 2D+3D	Data-driven model + State machine	Investigate	18–25 min
A5.3	SI units and measurement	Read It Right: The Metrology Bench	Hybrid 2D+3D	Rigid-body + State machine	Investigate	18–25 min
A5.4	Organizing and graphing data	The Plotting Bench: Charts That Lie	Data Dashboard	Data-driven model	Investigate	18–25 min
A5.5	Claim, evidence and reasoning	Argument Bridge: The Delta Fish Kill	Hybrid 2D+3D	Data-driven model + Rigid-body	Argue-from-data	18–25 min
A5.6	Designing an investigation of a system	Run My Plan: The Ecocolumn Trial	Hybrid 2D+3D	Data-driven model + State machine	Design	18–25 min

A1 · Systems and subsystems

A1.1 · What makes a system a system

Experiment name: Unplug the Aquarium

Render mode: Hybrid 2D+3D

Simulation engine: Data-driven model (stock-and-flow) + Agent-based

Interaction level: Investigate

Session length: 12–18 min

NGSS anchor: Supporting — MS-LS2-3

THEME & SCENE

A 60-litre classroom aquarium at eye level against a plain slate wall, lit from above by one LED bar so the water throws slow caustics across the bench top. Six zebra danio hold station in the filter outflow; five ribbons of Vallisneria sway; the gravel is dark with biofilm. In the foreground, close enough to click, a four-way power strip with three plugged cables running up to the pump, the heater and the light. To the right of the glass, a translucent stock-and-flow overlay redraws the same aquarium as labelled tanks joined by pipes and valves: ammonia, nitrite, nitrate, dissolved oxygen, heat. On a second bench behind, deliberately unlit, the identical components lie loose in a tray. Control panel docks right; a dipping test probe follows the cursor.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Glass tank	Structure	60×30×35 cm rim-polished box, 6 mm glass, refractive water volume filled to 32 cm, silicone seams visible at corners	No
2	Gravel bed and biofilm	Structure	4 cm of 3 mm pea gravel, 40 % of the bacteria population lives here; surface darkens from buff to olive as biofilm stock rises	Yes: resize
3	Canister filter and pump	Actor	20 cm cylinder under the bench, impeller spinning at 1200 rpm, intake strainer and spray-bar return; outflow jet drives a visible surface ripple	Yes: connect
4	Filter media cartridge	Structure	Removable basket holding a blue sponge block and 300 ceramic rings; carries 60 % of the nitrifying bacteria stock	Yes: swap
5	Heater rod	Actor	25 cm glass tube, 100 W, suction-cupped to the rear glass, amber LED pulses when the element is on; thermostat with 0.5 °C hysteresis	Yes: connect
6	LED light bar	Actor	55 cm alloy strip on the rim, 6500 K, dimmable, on a daily timer; drives plant photosynthesis rate	Yes: connect
7	Zebra danio	Actor	3 cm agents, 6 by default, silver-blue with four dark stripes; shoaling steering behaviour, gill beat rate tied to dissolved oxygen, colour desaturates under stress	Yes: place
8	Vallisneria plants	Actor	5 clumps of 25 cm ribbon leaves, jade, anchored in gravel; leaf count grows with nitrate and light; pearls oxygen bubbles when photosynthesising	Yes: place
9	Nitrifying bacteria colony	Field	Invisible population stock, split across media and gravel; doubling time 20 h, dies back without flow or oxygen; visualised only as biofilm darkening	No
10	Ammonia / nitrite / nitrate stocks	Field	Three invisible scalar stocks in mg/L, spatially uniform, drawn as filling tanks in the overlay	No
11	Dissolved oxygen field	Field	Invisible scalar, 0–10 mg/L, higher near the surface ripple and near pearling leaves	No
12	Water temperature field	Field	Invisible scalar with 4 h thermal inertia; relaxes toward 19 °C room temperature when the heater is unplugged	No
13	Power strip and cables	Instrument	Four-socket white strip, 3 plugs seated; pulling a plug animates the cable falling slack and cuts that device's flows to zero	Yes: connect

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
14	Dipping test probe	UI-Probe	Handheld wand; hovering over any point reads that point's ammonia, nitrite, nitrate, oxygen and temperature	Yes: drag
15	Stock-and-flow overlay	Overlay	Tanks, pipes and valve icons mirroring the live model; pipe width scales with flow rate, valves close to grey when a flow is zero	No
16	Loose parts tray	Structure	Second bench holding the same pump, heater, light, media, gravel, plants and fish jar, all unconnected; items can be dragged into the tank to assemble it	Yes: drag

HOW IT WORKS — THE MODEL

A true stock-and-flow engine runs at one tick per simulated minute. Stocks: ammonia, nitrite, nitrate, oxygen, heat, bacteria, plant biomass, fish health. Fish excrete ammonia at a rate proportional to biomass and feeding; uneaten food decays into ammonia over 8 h. Bacteria convert ammonia to nitrite and nitrite to nitrate at rate $k \cdot B \cdot S / (S + K_m)$, and B grows only where oxygenated water actually flows through media, so unplugging the pump both starves the colony and stops delivering substrate to it. Plants draw nitrate and release oxygen in light. Fish and bacteria consume oxygen continuously. Fish health falls when ammonia exceeds 0.25 mg/L or oxygen drops below 4 mg/L. The failure to avoid: the filter must never act as a magic life-support box that simply keeps fish alive. Every death has to be traceable along the chain, and in the loose-parts scenario every coupling coefficient is genuinely set to zero so the identical parts sit there doing nothing.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Power strip	Multi-select	Pump / Heater / Light	Pump, Heater, Light	—	Which devices are connected; unplugging zeroes that device's flows
Filter media	Dropdown	Mature seeded media / New sterile sponge / None	Mature seeded media	—	Starting bacteria population, from full to zero
Heater setpoint	Slider	18–30	25	°C	Target water temperature and heater duty cycle
Light hours	Slider	0–16	8	h/day	Photosynthesis window, oxygen production, plant growth
Fish stocked	Stepper	0–12	6	count	Ammonia production rate and oxygen demand
Feeding rate	Slider	0–4.0	1.5	g/day	Ammonia input from waste and from uneaten food decay
Plants	Stepper	0–10	5	count	Nitrate uptake and daytime oxygen input
Weekly water change	Stepper	0–50	25	% volume	Dilutes every dissolved stock once per simulated week
Time compression	Dial	1×–5000×	1000×	—	Simulated days per real second
Overlay mode	Radio	Tank only / Stocks and flows / Both	Both	—	Whether the pipe-and-valve diagram is drawn over the scene

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	A running tank	power=Pump, Heater, Light; media=Mature seeded media; fish=6; feeding=1.5	Over 14 simulated days, which stocks stay steady and which cycle? Name three parts that affect each other.
S2	Pull the plug	power=Heater, Light; media=Mature seeded media; fish=6	The pump is unplugged and nothing else changes. Trace the chain from the dead impeller to the first stressed fish.
S3	A heap of parts	power=none; media=None; fish=0; plants=0; loose parts tray in use	Every component is present on the bench and none is connected. Is this an aquarium? What exactly is missing?

#	Scenario	Preset	The question the student answers
S4	New tank syndrome	media=New sterile sponge; fish=12; feeding=3.0; time=5000×	The filter is running from minute one. Why does ammonia still spike, and what has to grow before it falls?

STUDENT ACTIVITIES

1. Run S1 for 14 simulated days. Record ammonia, nitrite, nitrate and oxygen on days 1, 7 and 14 with the dipping probe.
2. Unplug the pump and run again. Record the hour at which ammonia first crosses 0.25 mg/L, and the hour at which the first fish shows stress colour.
3. Rebuild from the loose parts tray: drag in the tank, gravel, media, pump, heater, light, plants and fish, plugging each cable as you go. Record the number of active flows in the overlay after each part you connect.
4. Set feeding to 4.0 g/day with plants at 0, then repeat with plants at 10. Compare nitrate on day 14 and say which part removed it.
5. Write down the three parts you would keep if you could keep only three, and run that configuration to test whether the tank survives 14 days.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Ammonia, nitrite, nitrate	Line graph	mg/L	Three traces on shared axes; the classic cycling curve emerges over 30 days
Dissolved oxygen	Live numeric	mg/L	Updated every simulated hour, sampled at the probe position
Water temperature	Live numeric	°C	Current value plus heater on/off state
Fish health index	Bar chart	0–100	One bar per fish, red below 40
Active flows	Counter	count	Number of non-zero couplings between parts right now; reads 0 in the heap scenario
System check	Pass-fail badge	—	Green only when parts, connections and a maintained function are all present
Run log	Data table	mixed	Exportable CSV: one row per simulated hour, every stock and flow

WHAT THE STUDENT SHOULD REALISE

Students believe a system is a group of related things. The parts tray holds every component of a working aquarium and does nothing, while pulling one plug on the assembled tank kills fish that the pump never touched. Parts alone are a heap; parts plus interactions plus a maintained function are a system. The student should be able to say: *"It is not the parts that make it a system, it is that the parts do something to each other."*

A1.2 · Subsystems nested within systems

Experiment name: The Bike Inside the Bike

Render mode: Hybrid 2D+3D

Simulation engine: Rigid-body + State machine

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: Supporting — MS-ETS1-4

THEME & SCENE

A rider on a road bike climbing a bright, dusty canyon grade, framed dead side-on so the whole machine reads as a diagram. The bike is drawn X-ray: frame tubes ghosted to 30 % opacity so the chain, cassette teeth, cable housings, brake pads and hub bearings show through in hard-edged colour. Behind, the canyon wall scrolls; a gradient strip along the bottom shows where on the climb the rider is. Whichever subsystem is currently selected lifts into full opacity inside a soft cyan containment frame while everything else dims. A green power ribbon runs from the rider's legs through crank, chain, cassette, hub and tyre to the road, narrowing at each loss. Left of the scene, a collapsible system tree; control panel docked bottom-right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Frame and fork	Structure	56 cm alloy diamond frame, 8 tubes modelled separately, ghosted; flexes 2 mm under peak pedal torque	No
2	Rider	Actor	Rigged cyclist, cadence-driven leg cycle 40–110 rpm, torso rocks above 250 W, head turns to check the road	Yes: swap
3	Crank, pedals and chainrings	Actor	172.5 mm cranks, 50/34 T rings, teeth individually modelled; rotates with the chain, not independently	Yes: swap
4	Chain	Actor	114 articulated links, steel grey, each pin a hinge constraint; sags 6 mm on the lower run, jumps sprockets on a shift	Yes: connect
5	Cassette and freehub	Actor	11 sprockets 11–34 T, bronze wear shading on the most-used cogs; freehub ratchets audibly when coasting	Yes: swap
6	Rear derailleur and jockey wheels	Actor	Parallelogram cage, two 11 T jockey wheels, spring return; travels across the cassette on cable pull	Yes: place
7	Shift cable and housing	Structure	1.1 mm stainless inner in a black outer, routed along the down tube; can be severed at any point along its run	Yes: connect
8	Brake lever and brake cable	Structure	Alloy lever with a 12 mm throw, 1.6 mm inner cable to the caliper; severing drops lever resistance to zero	Yes: connect
9	Caliper, pads and rim track	Actor	Dual-pivot caliper, two 55 mm pads; pad-to-rim gap 1.5 mm, glows orange with accumulated braking heat	Yes: swap
10	Wheel, tyre, spokes, hub	Structure	700×28 C tyre on a 32-spoke wheel; hub broken out into cones, bearings, axle; one bearing can be set to seized	Yes: swap
11	Bearing friction field	Field	Invisible per-bearing coefficient set by grease state; feeds the drivetrain loss ladder	No
12	Road and gradient profile	Environment	1 km climb, gradient editable per 100 m segment, chip-seal texture, heat shimmer at the top	Yes: drag
13	Power flow ribbon	Overlay	Ribbon from rider to contact patch, width $\propto$ watts, notched and narrowed at each loss point with a labelled figure	No
14	System tree panel	UI-Probe	Collapsible tree: Bicycle > Drivetrain / Braking / Steering / Wheels / Frame / Rider, each expanding to components and parts; components can be dragged between branches and are scored	Yes: drag
15	Containment frame	Overlay	Cyan rounded box drawn around the current selection, with a level label reading System, Subsystem, Component or Part	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Failure propagation highlight	Overlay	When a part fails, every enclosing level that loses function pulses red in the tree, from the part upward	No

HOW IT WORKS — THE MODEL

Rigid-body drivetrain plus a nested state machine. Rider power P is split by a loss ladder: chain friction (1–4 % depending on lubrication), bearing drag (per-bearing coefficient), tyre rolling resistance ($\propto 1/\text{pressure}$), and air drag at $\frac{1}{2}\rho C_d A v^2$. What remains accelerates a mass of rider plus bike against gravity component $mg \cdot \sin \theta$, so speed on the grade follows from a simple power balance updated 60 times a second. Every component carries a state: healthy, worn, or failed. Failure propagates strictly upward through the containment tree, never sideways by magic: a seized rear hub bearing raises one friction coefficient, that raises drivetrain loss, that drops wheel power to near zero, and only then does the tree mark Bicycle as non-functional. Braking runs its own chain of lever travel, cable tension, pad force, friction coefficient and heat. Cutting the brake cable leaves 95 % of the machine perfectly healthy, which is exactly the point.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Zoom level	Radio	Whole bike / Subsystem / Component / Part	Whole bike	—	Camera framing and which level the containment frame encloses
Selected subsystem	Dropdown	Drivetrain / Braking / Steering / Wheels / Frame / Rider	Drivetrain	—	Which branch lifts to full opacity and gets its own readouts
Sever a link	Multi-select	Brake cable / Shift cable / Chain / Rear hub bearing / Three spokes	none	—	Sets that part to failed and starts upward propagation
Rider power	Slider	40–350	120	W	Power entering the drivetrain at the pedals
Gradient	Slider	–8 to +12	6	%	Road slope over the current 100 m segment
Gear	Stepper	1–11	6	sprocket	Cassette sprocket selected, sets pedal-to-wheel ratio
Tyre pressure	Slider	1.5–6.5	4.0	bar	Rolling resistance and comfort; low pressure adds visible tyre squash
Brake pad condition	Dropdown	New / Worn / Glazed / Missing	New	—	Pad friction coefficient and therefore stopping distance
Chain lubrication	Slider	0–100	70	%	Chain friction loss, from 1 % to 4 % of rider power
Timed descent	Toggle	On / Off	Off	—	Runs a scored 1 km descent that requires the braking subsystem to work

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Healthy climb	power=180; gradient=6; gear=9; lubrication=70; sever=none	Expand the tree to four levels. How many parts sit below "Drivetrain", and what does the drivetrain do that no single part of it does?
S2	Cut the brake cable	sever=Brake cable; descent=On; gradient=–8	Every other part is healthy. Which levels of the tree turn red, and why does the whole bicycle count as failed?
S3	One seized bearing	sever=Rear hub bearing; power=250; gradient=6	A part four levels down has stopped. Follow the power ribbon and record where the 250 W is going instead.
S4	Single-speed rebuild	sever=Shift cable; gear=6 fixed; drag derailleur out of the tree	Removing a whole subsystem still leaves a rideable bike. What can it no longer do, and on which gradient does that first matter?

STUDENT ACTIVITIES

1. Expand the system tree fully and record the four levels for one named part, from the part up to the whole bicycle.

2. Set rider power to 180 W on a 6 % gradient and record the wattage at each notch of the power ribbon. Calculate how many watts never reach the road.
3. Sever the brake cable, run the timed descent, and record which tree levels turn red and which stay green.
4. Seize the rear hub bearing and record speed before and after. Then argue in one sentence whether a bearing is a system.
5. Drag the derailleur and shift cable out of the Drivetrain branch, re-run the climb at 12 % gradient, and record what the rider can no longer do.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Speed	Live numeric	km/h	Updated per frame, with the current gradient shown alongside
Power loss ladder	Bar chart	W	Stacked losses: chain, bearings, tyre, air, plus power delivered to the road
Stopping distance	Live numeric	m	From 30 km/h with the current pad and cable state; reads "no stopping" when severed
Tree health	Pass-fail badge	—	One badge per tree level, green or red, updating as failure propagates
Levels affected	Counter	count	How many tree levels a single part failure knocked out
Climb time	Timer	s	Time for the 1 km segment, logged per run
Run log	Data table	mixed	Exportable CSV: configuration, severed parts, losses, times

WHAT THE STUDENT SHOULD REALISE

Students treat "the system" as a fixed thing and its parts as merely smaller things. Here every level is a system in its own right: the drivetrain has its own inputs, outputs and failures, and the level you call "the system" is a choice you make for your question. A cut cable stops a 90 kilogram machine. The student should be able to say: *"Every part of this bike is itself made of parts, and a failure at any level climbs upward."*

A1.3 · Interactions among a system's parts

Experiment name: Cut the Link, Cook the Tomatoes

Render mode: Hybrid 2D+3D

Simulation engine: Data-driven model (system dynamics) + Fluid/thermal

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: Supporting — MS-PS3-4

THEME & SCENE

Eleven in the morning in July inside a Central Valley polytunnel near Fresno. The camera looks down a row of 120 staked tomato plants, leaves backlit through 200-micron poly film, the far end of the tunnel dissolving into heat shimmer. Ridge vents crack open overhead on visible motor arms; a wet wall breathes at the far end with a slow fan; a shade cloth waits on a motorised roll; drip lines glisten at the root zone. Warm ochre and washed green, with hard white light. On the right, a dark 2D link board draws every component as a node and every interaction as a labelled arrow carrying a sign, a gain and a delay. Arrows can be clicked, cut, slowed or amplified. Control panel sits below the board.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Polytunnel shell	Structure	30 m × 9 m hoop house, 14 galvanised hoops, 200 µm poly film transmitting 78 % of incident light and trapping long-wave heat	Yes: resize
2	Tomato plants	Actor	120 staked agents at 45 cm spacing, leaf area 0.2–2.4 m ² each, growing; leaves droop through four visible wilt stages, fruit trusses set above 18 °C	Yes: place
3	Ridge vent motors	Actor	Two 12 m vent panels on rack-and-pinion arms, 0–100 % open, 40 s full travel, faint motor whine	Yes: connect
4	Wet wall and fan	Actor	9 m corrugated cellulose pad kept wet by a header pipe, 1.1 m diameter extract fan opposite; cools by evaporation, raises humidity	Yes: connect
5	Shade cloth roll	Actor	40 % knitted shade fabric on a motorised roller under the ridge; unrolls in 25 s, cuts solar gain and light for photosynthesis together	Yes: connect
6	Propane unit heater	Actor	12 kW hanging heater with a squirrel-cage fan, orange burner glow when firing, night duty only	Yes: connect
7	Drip irrigation manifold	Actor	6 lateral lines, 2 L/h emitters at each plant, pressure gauge at the header; wets a visible soil disc	Yes: connect
8	CO ₂ regulator and tank	Actor	50 kg cylinder and a perforated distribution tube along the row; releases in pulses when enrichment is on	Yes: place
9	Controller box	Instrument	Wall-mounted panel with air temperature, humidity and CO ₂ readouts; issues open, close, fire and mist commands on thresholds	Yes: swap
10	Air temperature stock	Field	Scalar stock in °C with a 12-minute thermal time constant; vertically stratified, 2 °C hotter at the ridge	No
11	Humidity stock	Field	Scalar stock in % RH, fed by transpiration and the wet wall, drained by venting	No
12	Soil water stock	Field	Scalar stock in mm available water, filled by drip, drained by transpiration; sets whether plants can transpire at all	No
13	Causal link board	Overlay	9 nodes and 12 labelled arrows; each arrow shows sign (+ or –), gain and delay, and can be cut, restored or edited	Yes: connect
14	Link scissors and delay dial	UI-Probe	Cursor tool: click an arrow to sever it (it goes dashed grey), or drag its delay dial to add lag	Yes: drag
15	Loop detector	Instrument	Traces closed paths through the intact arrows and labels each B (balancing) or R (reinforcing), with the loop highlighted on the board	No
16	Airflow and shimmer overlay	Overlay	Streamlines through vents and along the row, plus a false-colour heat wash from 20 °C blue to 48 °C white	No

HOW IT WORKS — THE MODEL

A genuine system-dynamics engine: four stocks (air temperature, humidity, soil water, CO₂) updated every simulated minute by explicit flows, and every flow is owned by an arrow on the link board. Solar gain adds heat at $850 \text{ W/m}^2 \times \text{transmission} \times (1 - \text{shade})$. Venting removes heat proportional to opening area and the inside-outside temperature difference. Transpiration removes heat as latent energy at a rate proportional to leaf area, temperature and available soil water, and adds the same water to humidity. The controller closes two balancing loops: temperature raises vent opening, which lowers temperature; humidity raises vent opening as well. Every arrow carries sign, gain and a delay implemented as a first-order lag, so adding lag to a balancing loop produces genuine oscillation, not a scripted wobble. Cutting an arrow leaves every component present and running, which is the whole argument. Plant stress integrates minutes spent above 35 °C leaf temperature.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Link editor	Multi-select	temp → vent / temp → transpiration / transpiration → humidity / humidity → vent / shade → solar gain / soil water → transpiration / CO ₂ → growth / temp → heater	all intact	—	Which interactions exist; cut arrows go dashed and their flow drops to zero
Delay on selected link	Slider	0–30	2	min	Lag between cause and effect on the arrow currently selected
Gain on selected link	Slider	0.2–3.0	1.0	×	How strongly that arrow's cause moves its effect
Outside air temperature	Slider	5–45	38	°C	Driving heat difference across the film and through open vents
Solar input	Slider	0–1000	850	W/m ²	Heat and light entering through the roof
Vent control mode	Radio	Automatic / Manual / Disconnected	Automatic	—	Whether the controller drives the vents, the student does, or nothing does
Manual vent opening	Slider	0–100	0	% open	Vent area when in manual mode
Irrigation	Slider	0–8	4	L/plant/day	Soil water refill rate, which sets the ceiling on transpiration cooling
Shade cloth	Toggle	Deployed / Retracted	Retracted	—	Cuts solar gain by 40 % and light for photosynthesis by 40 %
Time compression	Dial	1×–600×	120×	—	Simulated minutes per real second

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	An ordinary July day	all links intact; outside=38; solar=850; vents=Automatic; irrigation=4	Name every closed loop the detector finds and say which of them holds the temperature down.
S2	One arrow cut	link temp → vent cut; everything else unchanged	Every part is still present and powered. Why does the tunnel reach 49 °C by two o'clock?
S3	The same link, delayed	link temp → vent intact; delay=25 min	The link works, just late. Describe the shape the temperature graph now makes and measure its period.

#	Scenario	Preset	The question the student answers
S4	Dry roots	irrigation=0; all links intact; vents=Automatic	The cooling link runs through the soil. With vents working perfectly, why do leaf temperatures still climb?

STUDENT ACTIVITIES

1. Run S1 for one simulated day and record peak air temperature, peak humidity and the loop labels the detector reports.
2. Cut the temp → vent arrow and re-run. Record peak temperature, then list every part that is still working normally.
3. Restore that arrow and set its delay to 25 minutes. Measure the period and amplitude of the temperature oscillation from the graph.
4. Set irrigation to 0 and predict, before running, whether vents alone can hold the temperature. Run it and record leaf temperature at 14:00.
5. Set the gain on humidity → vent to 3.0 and describe what the tunnel now does at dawn, when humidity is high but temperature is low.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Air and leaf temperature	Line graph	°C	Two traces over 24 simulated hours, with a 35 °C stress threshold drawn
Humidity	Line graph	% RH	Plotted on the same time axis as temperature
Soil water available	Live numeric	mm	Current value plus hours of transpiration remaining
Active loops	Data table	—	Every closed loop found, its members, and its B or R label
Oscillation period and amplitude	Live numeric	min and °C	Computed from the last three peaks; blank until three peaks exist
Heat stress accumulated	Counter	°C·min above 35	Integrated leaf-temperature excess, the yield-damage proxy
Day-end yield score	Pass-fail badge	—	Green if stress stays under 400 °C·min and fruit set continues
Run log	Data table	mixed	Exportable CSV: every stock, every flow, every link state, per simulated minute

WHAT THE STUDENT SHOULD REALISE

Students explain system behaviour by listing parts: a greenhouse is hot because it has a heater. Here every part stays present and powered while one arrow is cut, and the tunnel cooks anyway. Then the same arrow, merely delayed, makes the temperature swing instead of settle. Behaviour lives in the interactions, not the inventory. The student should be able to say: *"The parts did not change. What changed was what they were allowed to do to each other."*

A1.4 · Emergent properties

Experiment name: No Bee Is In Charge

Render mode: 3D Scene

Simulation engine: Agent-based

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: Supporting — MS-LS2-2

THEME & SCENE

Nine in the morning at a Sierra-foothills apiary. The camera opens inside the hive, close on a comb face where six hundred bees crawl over honey-gold hexagons in a slow boiling mass, with a dance floor near the entrance where returning foragers cut figure-eights. Pull the camera back through the entrance slot and the scene becomes a warm top-down field 400 metres across: four flower patches in an almond orchard and a wildflower verge, each a different density of bloom, with bee traffic drawn as faint amber threads between them and the hive. Dust hangs in the light. A colony readout board floats top-left; a single-bee inspector card appears wherever the student clicks. Control panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Worker bee agent	Actor	12 mm body, amber and charcoal bands, blurred wing quads at 200 Hz; up to 2000 instances; five-state machine: rest, scout, forage, dance, follow	Yes: place
2	Queen	Actor	18 mm elongated abdomen, unmarked except for a white dot; walks the brood comb laying one egg per cell; issues no instructions to anyone	No
3	Comb cells	Structure	Hex grid at 5.4 mm pitch across two frames; per-cell state: empty, nectar, capped honey, pollen, brood; caps render as pale wax discs	Yes: place
4	Wax builder bees and wax scales	Actor	Bees hanging in festoons, each depositing 0.8 mm wax scales; new cells appear at the growing comb edge without any template	No
5	Flower patches	Environment	Four circular patches, radius 8–20 m, nectar concentration stock per patch, bloom density drawn as sprite count; deplete on visits and regrow	Yes: drag
6	Hive nectar store	Field	Colony-level stock in grams, filled by returning foragers, drained by consumption; shown as a filling bar and as capped cells	No
7	Waggle dance floor	Structure	12 cm patch of comb just inside the entrance; dances render as figure-eights whose run angle encodes bearing to the sun and whose run duration encodes distance	No
8	Odour field	Field	Invisible diffusing scalar grid, 2 m cells, deposited by foragers at flowers and decaying with a 6-minute half-life	No
9	Hive air temperature field	Field	Invisible scalar across the comb volume; heated by bee metabolism, cooled by fanning and water evaporation	No
10	Fanning and clustering behaviour	Actor	Sub-state of the worker agent: a bee fans at the entrance above its own 36 °C threshold, joins a huddle below 33 °C; no bee reads the colony average	No
11	Guard bees and entrance	Actor	6 bees at a 90 mm entrance slot, challenging arrivals; traffic counter sits under the sill	No
12	Rule card panel	UI-Probe	Five cards, one per individual behaviour rule; each can be switched off, which removes that behaviour from every bee at once	Yes: swap
13	Colony readout board	Overlay	Floating panel: share of foragers at each patch, total nectar, hive core temperature, decision time	No
14	Single-bee inspector	UI-Probe	Click any bee to lock the camera to it and open a card listing everything that bee knows: one patch, one bearing, one nectar memory	Yes: drag
15	Visit heat map	Overlay	Toggleable false-colour wash over the field, counting foraging visits per square metre over the last 10 minutes	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Weather event driver	Environment	Rain or cold-snap events that suppress flight and drop outside temperature over 3 simulated minutes	Yes: swap

HOW IT WORKS — THE MODEL

Pure agent-based. Every bee runs the same five rules and holds only private knowledge: one remembered patch, its bearing, and how good it was last time. Rule 1, an unemployed bee watches a dance and copies its bearing with an angular error. Rule 2, a returning forager dances with a duration proportional to the nectar quality it found, so richer patches recruit more followers per return. Rule 3, a forager abandons a patch when its yield falls below a threshold. Rule 4, bees follow the odour gradient over the last 30 m. Rule 5, a bee fans above its own temperature threshold and clusters below it. Nothing in the code compares patches, computes a colony average, or issues an order. Colony-level quantities are read out but never written to. Patch share and hive temperature must therefore be measured from the agents, never imposed on them, or the whole demonstration is a fraud.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Individual rules	Multi-select	Follow a dance / Dance if rich / Abandon if poor / Follow odour / Fan or cluster	all on	—	Which behaviours exist in every bee; switching one off removes it colony-wide
Colony size	Slider	50–2000	600	bees	How many agents run; the emergent behaviours weaken then vanish as it falls
Patch count	Stepper	1–6	4	patches	How many flower patches exist in the field; patches can also be dragged
Richness of selected patch	Slider	0–100	70	% bloom	Nectar concentration in the patch currently selected
Nectar regrowth	Slider	0–20	6	% per min	How fast a depleted patch refills, which sets whether a choice stays stable
Dance angular error	Slider	0–30	8	°	Precision with which a follower copies a dance bearing
Outside temperature	Slider	0–45	22	°C	Drives fanning and clustering, and whether bees can fly at all
Follow one bee	Toggle	On / Off	Off	—	Locks the camera to a single agent and shows only what it knows
Rain event	Toggle	On / Off	Off	—	Grounds all foragers for the duration
Time compression	Dial	1×–120×	30×	—	Simulated minutes per real second

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Four patches, one best	patches=4; richness A=90, B=45, C=30, D=20; rules=all on; colony=600	Does the colony end up mostly at the best patch? Now inspect ten bees. How many of them compared the patches?
S2	Silence the dance	rules: Dance if rich OFF; patches and richness as S1	Every bee still forages perfectly well. What exactly has the colony lost the ability to do?
S3	The best patch runs dry	richness A drops to 5 at minute 20; regrowth=0; rules=all on	How many minutes does the colony take to switch, and what individual behaviour makes the switch happen?
S4	Cold snap in the almond bloom	outside=4; rules: Fan or cluster ON; colony=1200	Hive core temperature holds near 34 °C. Which bee measured it, and what is each bee actually responding to?

STUDENT ACTIVITIES

1. Run S1 for 30 simulated minutes and record the share of foragers at each patch every 5 minutes.
2. Click five individual bees and write down, for each, the number of patches it knows about. Compare that to the number the colony is exploiting.
3. Switch off "Dance if rich", re-run S1, and record final patch share. State in one sentence what disappeared.
4. Run S3 and record the time from the patch drying out to the moment more than half the foragers are elsewhere.
5. Drop colony size to 60 with all rules on. Record whether the colony still converges, and give the smallest size at which it still works.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Patch share	Line graph	% of foragers	Four stacked traces over time, updated every 10 simulated seconds
Colony choice accuracy	Live numeric	%	Share of foragers at whichever patch is objectively richest right now
Total nectar stored	Line graph	g	Colony-level stock over the run
Individual knowledge	Data table	count	For the inspected bee: patches known, last yield, bearing held, dances watched
Hive core temperature	Line graph	°C	Measured from the agent field, with outside temperature as a second trace
Decision time	Timer	min	Time from a change in patch richness to the colony's majority moving
Emergence check	Pass-fail badge	—	Green when a colony-level property holds while no individual holds it
Run log	Data table	mixed	Exportable CSV: agent states, patch stocks, colony readouts per simulated minute

WHAT THE STUDENT SHOULD REALISE

Students assume that organised behaviour needs someone organising it: the queen must be deciding, or some bee must be keeping score. Inspecting any bee shows it knows one patch and one bearing, and switching off a single rule destroys a colony-level ability while leaving every bee intact. The whole has properties the parts do not. The student should be able to say: *"The colony chooses, and not one bee is choosing."*

A1.5 · Systems across scales, from a cell to a planet

Experiment name: From Chloroplast to Coastline

Render mode: 2.5D Layered

Simulation engine: Data-driven model (coupled multi-scale) + Agent-based

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: Supporting — MS-LS2-1

THEME & SCENE

A continuous vertical column of Monterey Bay water, sun shafts raking down through a swaying kelp canopy, particles drifting in the green light. On the left, a logarithmic scale slider runs from 1 μm to 1000 km with a live scale bar that redraws at every step. At the deepest setting the view is inside one kelp cell: jade chloroplasts, thylakoid stacks, a pale cell wall. Pull outward and the scene cross-dissolves through blade, whole plant, forest patch with urchins and otters, the whole bay, and finally the California coast from orbit with a chlorophyll satellite wash. Each layer keeps its own small HUD in the same position, so the numbers change while the frame does not. Control panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Chloroplast	Structure	5 μm lens, jade, internal thylakoid stacks as ribbed discs, starch grain inclusions; 60 per cell, streaming slowly around the vacuole	No
2	Kelp cell	Structure	30 μm rectangular cell, translucent wall 0.8 μm thick, large central vacuole, plasmodesmata pits at the walls	No
3	Sieve tube element	Structure	Elongated conducting cell with a perforated end plate; carries sugar down the stipe at 10 cm/min, drawn as a moving stripe	No
4	Blade and pneumatocyst	Structure	40 cm corrugated blade with a 3 cm gas bladder at its base; bladder holds carbon monoxide and buoys the blade upward	Yes: place
5	Stipe and holdfast	Structure	Stipe up to 30 m, holdfast a tangle of 200 haptera gripping a granite boulder; sways on a 6 s swell period	No
6	Giant kelp individual	Actor	Whole plant agent; growth up to 30 cm/day when nitrate and light allow; senesces and detaches in storms	Yes: place
7	Kelp forest patch	Environment	One hectare, 180 individuals, canopy fraction 0–100 %, understory light shaded accordingly	Yes: resize
8	Purple urchin agents	Actor	6 cm test with moving spines, 0–40 per m^2 ; graze holdfasts; switch to a dormant "barren" state when food is gone	Yes: place
9	Sea otter agents	Actor	1.2 m agents rafting at the surface, each eating 25 % of body mass daily; visibly crack urchins on their chests	Yes: place
10	Sunflower star and sheephead	Actor	Second and third urchin predators, added or removed to change the food web without touching kelp directly	Yes: place
11	Water column layers	Environment	Light attenuating with depth at $k=0.12 \text{ m}^{-1}$, nitrate stock rising with depth, thermocline drawn as a shimmer band	No
12	Scale slider and scale bar	UI-Probe	Log slider 1 μm to 1000 km with a bar that relabels itself: 10 μm , 1 cm, 1 m, 100 m, 10 km, 500 km	Yes: drag
13	Cross-scale link arrows	Overlay	Arrows showing which quantity at one level feeds the next: photosynthesis per chloroplast to sugar per blade to biomass per plant to tonnes per hectare	No
14	Carbon accountant	Instrument	Invisible integrator summing fixed carbon upward through every level and reporting it in that level's natural units	No
15	Perturbation tool	UI-Probe	Drop a change at any level: block light on one chloroplast, cut a blade, remove otters, warm the bay; marks the injection point with a pin	Yes: place

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Satellite chlorophyll layer	Overlay	Coast-scale false-colour wash from MODIS-style data, 1 km pixels, with the bay outlined	No

HOW IT WORKS — THE MODEL

Six nested models run at once, each with its own units and its own clock, coupled only upward through explicit aggregation. At the chloroplast level, photosynthesis follows a light-response curve saturating near 400 $\mu\text{mol photons/m}^2/\text{s}$. Cell-level fixation is chloroplast rate multiplied by count. Blade fixation sums cells over area; plant growth sums blades and subtracts respiration; patch biomass sums plants and subtracts urchin grazing; bay production sums patches. Grazing is agent-based: urchin density rises when predators are removed, and above about 9 urchins/ m^2 a patch flips to a barren state that persists even after grazing falls, which is the hysteresis the coast really shows. Characteristic times differ by level: chloroplast seconds, plant days, forest years. The honesty rule: a perturbation at one level must propagate only through the actual coupling arrows, so blocking light on a single chloroplast genuinely changes nothing measurable at the bay.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Scale	Slider (log)	1 μm – 1000 km	10 m	m	Which nested level is on screen and which HUD units are shown
Perturbation site	Dropdown	Chloroplast / Cell / Blade / Whole plant / Forest patch / Whole bay	Forest patch	—	Level at which the next change is injected
Perturbation type	Dropdown	Block light / Cut nutrients / Remove otters / Warm water +3 °C / Storm swell / Harvest canopy	Remove otters	—	What the injected change actually does
Otter population	Stepper	0–40	18	otters	Predation pressure on urchins in the modelled patch
Upwelling nitrate	Slider	0–30	18	$\mu\text{mol/L}$	Nutrient supply to help growth, the Central California seasonal driver
Surface light	Slider	0–2000	1200	$\mu\text{mol photons/m}^2/\text{s}$	Light at the sea surface before attenuation with depth
Sea temperature	Slider	10–24	13	°C	Growth rate and, above 18 °C, nutrient starvation in surface water
Time span	Timeline scrubber	0–20	0	years	Position within a 20-year run, scrubbable both ways
Cross-scale arrows	Toggle	On / Off	On	—	Draws the aggregation links between adjacent levels
Level clock	Radio	Seconds / Hours / Days / Years	Days	—	Which characteristic time the current level is stepped in

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Climb the ladder	perturbation=none; light=1200; nitrate=18; temp=13	Read carbon fixed at all six levels. The number changes by twelve orders of magnitude. What stays the same?
S2	Break it at the bottom	site=Chloroplast; type=Block light; span=0–5 years	One chloroplast is shaded. Measure the effect at cell, plant, patch and bay. At which level does it vanish into the noise?

#	Scenario	Preset	The question the student answers
S3	Break it in the middle	site=Forest patch; type=Remove otters; otters=0; span=0–20 years	The forest collapses to a barren. Has photosynthesis per chloroplast changed? What has changed?
S4	The 2014 warm blob	temp=19; nitrate=3; otters=6; type=Warm water +3 °C; span=0–20 years	This is what the California coast actually experienced. Which level did the heat act on, and which level showed it first?

STUDENT ACTIVITIES

1. Set the scale slider to each of the six levels in turn and record carbon fixed with its units at each. Note where the unit changes from femtograms to tonnes.
2. Inject "Block light" at the chloroplast, then read the bay-level output. Record the change to three significant figures.
3. Inject "Remove otters" at the patch level and scrub the timeline to year 20. Record urchin density and canopy area at years 1, 5 and 20.
4. With the barren established, set otters back to 18 and scrub forward five more years. Record whether the forest returns and how long it takes.
5. Run S4 and write one sentence naming the level the disturbance entered and the level at which you first measured it.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Carbon fixed at current level	Live numeric	fg/s, mg/day, kg/day or t/year	Rescales automatically to the level on screen; the unit change is the lesson
Scale ladder table	Data table	mixed	All six levels side by side: object, size, characteristic time, output, unit
Canopy area	Line graph	ha	Forest patch canopy over the 20-year timeline
Urchin density	Line graph	urchins/m ²	Plotted with the 9 per m ² barren threshold marked
Otter count	Live numeric	otters	Current predator population in the modelled patch
Effect size by level	Bar chart	% change	How much the injected perturbation moved each of the six levels
Recovery time	Timer	years	Time from restoring predators to canopy passing 50 % of its former area
Run log	Data table	mixed	Exportable CSV: every level's state at every timestep

WHAT THE STUDENT SHOULD REALISE

Students think of a system as one fixed size, usually the size they can see. Here the same kelp is a system at six sizes at once, and the units, the clock and the answerable questions all change with the level. A change that is catastrophic at one level is invisible at another. The student should be able to say: *"Whether something counts as a big change depends on which level I chose to look at."*

A2 · Boundaries, inputs and outputs

A2.1 · Drawing a system's boundary

Experiment name: Where You Draw the Line

Render mode: Hybrid 2D+3D

Simulation engine: Data-driven model (flow network) + State machine

Interaction level: Investigate

Session length: 12–18 min

NGSS anchor: Supporting — MS-ESS3-3

THEME & SCENE

An isometric cutaway of a Sierra-foothills middle school at ten in the morning, drawn like a clean architectural model: four classroom blocks with their roofs lifted off, a cafeteria with a working kitchen, a rooftop solar array, a bus loop with two buses, a garden with three compost bins, a trash and recycling enclosure, and the utility connections at the street edge. Flows move as coloured ribbons across the model: blue water, yellow electricity, green food, grey waste, white people. A dashed magenta boundary polygon lies over the site with grab handles at each vertex; everything inside is rendered in full colour, everything outside falls to 40 % grey. The ledger panel on the right lists Inputs, Outputs and Internal transfers and rewrites itself the instant a handle moves.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Site terrain plate	Environment	180 m × 140 m isometric base with asphalt, playing field, oak shade and a street edge; sun angle set by the day clock	No
2	Classroom blocks	Structure	Four single-storey blocks, roofs lifted 3 m to show lighting, HVAC ducts and occupancy; each draws a metered electrical load	Yes: place
3	Cafeteria and kitchen	Actor	Serving line, two ovens, a walk-in chiller, three sinks; converts delivered food into meals, food waste and grey water on a lunch-hour schedule	Yes: place
4	Rooftop solar array	Actor	60 kW of panels on the gym roof, tilt 20°, output following a daily curve with a visible inverter cabinet	Yes: place
5	Grid connection and meter	Instrument	Pole transformer and a bidirectional meter at the street; reads import and export separately in kWh	No
6	Water main and meter	Instrument	100 mm main under the street with a pit-mounted meter reading cumulative m ³	No
7	Sewer lateral and flow meter	Instrument	150 mm lateral leaving the site with an inline flow meter; carries kitchen and washroom outflow	No
8	Food delivery truck	Actor	Arrives 07:30, unloads 240 kg of food into the kitchen store, departs; the ribbon it carries is drawn only while it moves	Yes: place
9	School buses	Actor	Two 12 m buses on the loop, arriving 07:50 and 15:10; consume diesel, carry students in and out	Yes: place
10	Garden and compost bins	Actor	Three 1 m ³ bins plus 12 raised beds; takes kitchen scraps and returns soil and a small vegetable yield to the kitchen	Yes: place
11	Trash and recycling enclosure	Actor	Six wheeled bins behind a screen wall; a hauler empties them twice a week and drives off site	Yes: place
12	Flow ribbons	Overlay	Bezier ribbons between nodes, width ∝ rate, colour by type, animated dashes showing direction	No
13	Boundary polygon tool	UI-Probe	Dashed magenta closed polygon, 3–12 draggable vertices, snapping to building corners and fence lines	Yes: drag
14	Crossing detector	Instrument	Invisible geometry test: every ribbon-polygon intersection is registered with its type, direction and rate	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Ledger panel	Overlay	Three live columns, Inputs, Outputs and Internal transfers, each row naming the flow, its rate and its unit	No
16	Balance check badge	Instrument	Tests inputs minus outputs minus accumulation for each flow type and reports whether the books close	No

HOW IT WORKS — THE MODEL

The campus is a directed flow network: 14 nodes, 26 edges, each edge carrying a typed flow with a rate schedule across the school day. Water enters at the main, splits to washrooms, kitchen and irrigation, and leaves by sewer or by evaporation. Electricity enters from the grid, is generated on the roof, and is consumed by lighting, HVAC and kitchen. Food enters by truck, becomes meals, scraps and packaging. Nothing about these flows depends on the boundary. The boundary is purely a geometric classifier evaluated every frame: an edge with both endpoints inside is an internal transfer, an edge crossing inward is an input, an edge crossing outward is an output. Dragging a vertex therefore reclassifies flows without changing a single rate, and the ledger animates the reclassification. The failure to avoid: the sim must never alter a flow rate when the boundary moves, because the entire idea is that the boundary is an accounting choice, not a wall.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Boundary preset	Dropdown	Whole campus / Cafeteria only / Buildings only / Campus plus buses and supplier / Custom	Whole campus	—	Jumps the polygon to a saved shape and rewrites the ledger
Boundary vertices	Drag-handle	3–12 handles, snapping to corners and fences	6 handles	—	The actual shape of the system boundary
Flow types shown	Multi-select	Water / Electricity / Food / Waste / People	all	—	Which ribbons are drawn and counted in the ledger
Time of day	Timeline scrubber	06:00–18:00	12:00	h:mm	Every flow rate follows its own daily schedule
Solar output	Slider	0–60	42	kW	Rooftop generation, which can turn grid import into export
Enrolment	Slider	200–900	520	students	Scales water use, food, waste and bus trips together
Composting	Toggle	On / Off	On	—	Whether kitchen scraps go to the garden or to the hauler
Irrigation	Slider	0–40	12	m ³ /day	Water sent to the field and garden rather than to the sewer
Show crossings only	Toggle	On / Off	Off	—	Hides internal ribbons so only boundary crossings remain visible
Aggregation window	Radio	Per minute / Per day / Per school year	Per day	—	Units in which the ledger reports every flow

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	The whole campus	preset=Whole campus; time=12:00; flows=all; window=Per day	List every input and every output. How many flows cross the line, and how many are purely internal?
S2	Just the cafeteria	preset=Cafeteria only; composting=On	The compost ribbon was internal a moment ago. Which column is it in now, and what physically changed?
S3	Solar left outside	preset=Buildings only; solar=42	The array is 8 m away, on the gym roof, outside the line. Why does the ledger now show electricity as an input?

#	Scenario	Preset	The question the student answers
S4	Draw it wide	preset=Campus plus buses and supplier; window=Per school year	Bus diesel and the supplier depot are now inside. Does the total crossing count go up or down, and why?

STUDENT ACTIVITIES

1. Load S1 and copy the full ledger into your table: name, type, direction and rate for every crossing.
2. Drag one vertex so the garden falls outside the line. Record which two flows changed column, and confirm that neither rate changed.
3. Switch the preset to Cafeteria only and count crossings. Compare with the whole-campus count and explain the difference.
4. Set composting off, then on, at the whole-campus boundary. Record what happens to the waste output and why the total mass leaving the site changes.
5. Draw a custom boundary that makes water an internal transfer rather than an input. Record whether that is possible on this site, and say why.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Crossing list	Data table	mixed	Every boundary crossing: name, type, direction, rate, unit; rewritten on every vertex drag
Inputs vs outputs by type	Bar chart	m ³ , kWh, kg	Paired bars per flow type for the current boundary and window
Internal transfers	Counter	count	How many flows are wholly inside the line right now
Reclassified by last drag	Counter	count	Flows that changed column when the boundary last moved
Accumulation	Live numeric	m ³ , kWh, kg	What is piling up inside the boundary, per flow type
Balance check	Pass-fail badge	—	Green when inputs minus outputs equals accumulation for every type
Ledger export	Data table	mixed	Exportable CSV of the ledger for up to six saved boundaries side by side

WHAT THE STUDENT SHOULD REALISE

Students treat a boundary as a real wall that decides what is in the system. Dragging one vertex turns an internal compost flow into an output, and turns the school's own solar electricity into an import, while no physical rate changes. The boundary is a line the investigator draws, not a thing the world contains. The student should be able to say: *"Nothing moved. I moved the line, and that changed what counts as going in and out."*

A2.2 · Open vs closed systems

Experiment name: Three Jars, One Lamp

Render mode: 3D Scene

Simulation engine: Data-driven model (mass and energy balance) + Fluid/thermal

Interaction level: Investigate

Session length: 12–18 min

NGSS anchor: Supporting — MS-PS1-5

THEME & SCENE

A dark lab bench under one adjustable LED grow bar on a rail. Three identical two-litre wide-mouth jars stand in a row, each on its own digital balance whose display reads to 0.01 g. Jar A is open, its mouth bare. Jar B wears a silicone-gasket lid clamped down. Jar C is sealed the same way but sits inside a mirrored vacuum-flask jacket with a small shutter over its light window. Inside each jar: three centimetres of moist dark soil, four duckweed fronds on a water film, a moss cushion, one snail, and condensation beading on the cold upper glass. Grey-green, clinical, high contrast, with the lamp the only warm thing in frame. Behind the jars, three vertical accounting strips fill live: mass, energy, matter species. Control panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Bench and lamp rail	Environment	1.2 m matte-black bench, anodised rail 40 cm above it, lamp head slides along the rail and can be aimed	Yes: drag
2	Jar shells	Structure	Three 2 L borosilicate jars, 14 cm tall, 11 cm wide, 3 mm wall, faint mould seam; identical in every respect	No
3	Gasket lids	Structure	Silicone-sealed screw lids with a visible red gasket ring; a lid can be removed, replaced, or fitted with a 1 mm pinhole	Yes: swap
4	Vacuum-flask jacket and shutter	Structure	Double-walled mirrored sleeve around Jar C with a 30 mm sliding shutter over its light window; blocks conduction, radiation and convection when closed	Yes: place
5	Digital balances	Instrument	Three 0.01 g platforms with backlit displays and a tare button; drift-free, reading total jar mass continuously	No
6	Soil layer and microbes	Structure	3 cm of dark loam, visible aggregate texture; carries an invisible microbe stock that respire and decomposes	Yes: resize
7	Duckweed and moss	Actor	Four 4 mm fronds floating plus one 3 cm moss cushion; photosynthesis follows a light-response curve, fronds yellow and sink when starved	Yes: place
8	Snail	Actor	12 mm garden snail agent, grazes moss, respire continuously, retracts and slows below 12 °C	Yes: place
9	Water in three states	Field + Particle	One water stock split between soil water, free liquid, and vapour; condensation droplets render on the inside glass and run back down	No
10	Gas stocks	Field	Per-jar stocks of O ₂ , CO ₂ , N ₂ and water vapour in millimoles, drawn as four segmented bars beside each jar	No
11	LED grow lamp	Actor	30 cm bar, 0–400 μmol photons/m ² /s, 4000 K, on a daily timer; casts a hard rectangle of light on the bench	Yes: swap
12	Thermal probe and gas sensor	Instrument	Thin stainless probe and a miniature O ₂ /CO ₂ sensor through each lid; live per-jar readouts on stalk displays	No
13	Energy flux arrows	Overlay	Yellow arrows for light in, red arrows for infrared out, drawn crossing or stopping at each jar wall; Jar C's arrows visibly terminate at the jacket	No
14	Mass ledger strip	Overlay	Vertical strip per jar showing starting mass, current mass, and cumulative mass crossing the wall	No
15	Tagged carbon tracer	Particle	20 carbon atoms rendered as magenta spheres cycling CO ₂ to sugar to tissue to CO ₂ ; toggleable, counted but never created or destroyed	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Time compression clock	UI-Probe	Bench-mounted clock face reading elapsed simulated days, with a scrub handle	Yes: drag

HOW IT WORKS — THE MODEL

Three parallel balance models, one per jar, each conserving mass and energy explicitly. Photosynthesis converts CO₂ and water to sugar and O₂ at a rate set by the light-response curve. Respiration by snail, plants and soil microbes runs the reaction backwards day and night. Water evaporates and condenses between the three water stores according to temperature and the vapour deficit. Wall exchange is what differs: Jar A exchanges both matter and energy, Jar B exchanges energy only (light in, infrared out through glass), Jar C exchanges neither while the shutter is closed. Every gram and every joule that crosses a wall is logged. Nothing is permitted to appear or vanish, so a sealed jar's total mass must stay flat to the last decimal even as the water cycles between liquid, soil and vapour. The classification badge is computed from the crossing counters, never hard-coded, so a lid with a pinhole is correctly reported as an open system.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Jar A lid	Dropdown	Open / Sealed / Sealed with 1 mm pinhole	Open	—	Whether matter can cross Jar A's wall, and how fast
Jar B lid	Dropdown	Open / Sealed / Sealed with 1 mm pinhole	Sealed	—	Whether matter can cross Jar B's wall
Jar C jacket	Toggle	Fitted / Removed	Fitted	—	Whether energy can cross Jar C's wall by conduction and radiation
Jar C light shutter	Toggle	Open / Closed	Closed	—	The last energy pathway into Jar C; closing it makes the jar isolated
Lamp intensity	Slider	0–400	220	μmol photons/m ² /s	Energy input to Jars A and B, and to C when the shutter is open
Light hours	Slider	0–24	12	h/day	Length of the daily photosynthesis window
Room temperature	Slider	10–35	21	°C	Drives evaporation, respiration rate and the heat flow through each wall
Living things per jar	Stepper	0–6	3	organisms	How many plants and animals are inside each jar at the start
Tagged carbon tracer	Toggle	On / Off	Off	—	Renders 20 labelled carbon atoms so they can be counted as they cycle
Time compression	Dial	1×–20000×	5000×	—	Simulated days per real second, up to a 90-day run

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Thirty days side by side	A=Open; B=Sealed; C=Fitted with shutter Open; lamp=220; hours=12	Which jar loses mass, and which jar's mass never changes at all? Which jars are still alive on day 30?
S2	Sealed and dark	B=Sealed; lamp=0 for Jar B only via shutter and lamp aim	Matter is fully conserved and the jar still dies. What was crossing the wall before, and what is not crossing now?
S3	Truly isolated	C=Jacket Fitted; shutter=Closed; time=20000×	Neither matter nor energy crosses. Record how long anything stays alive and what the gas bars do at the end.
S4	The pinhole test	A=Sealed with 1 mm pinhole; room=30	The lid is on and it looks sealed. What does the balance say after 30 days, and what does the badge classify it as?

STUDENT ACTIVITIES

1. Tare all three balances, run 30 simulated days, and record mass for each jar on days 0, 10, 20 and 30.
2. Record the energy crossing counter for Jar B. State whether a closed system can still receive anything, and what.

3. Close Jar C's shutter and run again. Record the day on which CO₂ stops rising and explain what has run out.
4. Fit the pinhole lid to Jar A, run 30 days, and record the total mass lost. Predict first, then compare.
5. Switch on the carbon tracer for Jar B and count the magenta atoms on day 0 and day 30. Record where each one is at both times.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Jar mass	Line graph	g	Three traces to 0.01 g over the run; the sealed trace is a flat line
Gas concentrations	Line graph	mmol	O ₂ and CO ₂ for the selected jar, with day and night bands shaded
Energy crossing wall	Live numeric	W	In and out separately for each jar, updated every simulated hour
Matter crossing wall	Counter	g cumulative	Total mass that has crossed each wall since the run started
System classification	Pass-fail badge	—	Reads Open, Closed or Isolated per jar, computed from the two counters
Survival status	Live numeric	days	Days since the last living organism in that jar stopped respiring
Carbon atom census	Data table	count	Where each tagged atom sits: gas, water, plant tissue, snail, soil
Run log	Data table	mixed	Exportable CSV of all three jars, every stock and crossing, per simulated day

WHAT THE STUDENT SHOULD REALISE

Students hear "closed system" and picture a box that nothing gets into at all. Jar B is sealed, its mass never moves, and it thrives for a month because light keeps pouring through the glass. Jar C, which really does exclude everything, dies. Closed means matter cannot cross; energy still can, and usually must. The student should be able to say: *"Sealed to matter is not sealed to energy, and that difference is what keeps the jar alive."*

A2.3 · Inputs and outputs

Experiment name: On the Dyno: In One End, Out the Other

Render mode: Hybrid 2D+3D

Simulation engine: Data-driven model (process balance) + State machine

Interaction level: Investigate

Session length: 12–18 min

NGSS anchor: Supporting — MS-PS3-5

THEME & SCENE

A school engineering test cell seen through a thick observation window, everything clearly labelled as a virtual rig. Centre stage, a single-cylinder 250 cm³ engine bolted to a dynamometer, cut away down its axis so the piston, valves, spark plug and crankshaft run visibly through their four strokes. Fuel arrives from a graduated glass burette on the wall whose level drops as you watch. Air comes in through a bell mouth with a mass-flow meter. A coolant loop leaves for a radiator and returns. The exhaust runs into a gas analyser cabinet. Five ribbons wrap the rig: amber fuel, pale blue air, green shaft work, red heat, grey exhaust. A Sankey bar across the bottom splits the fuel energy live. Control panel right, atom inventory left.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Engine block cutaway	Structure	250 cm ³ single cylinder, 70 mm bore, 65 mm stroke, alloy fins, sectioned on the centre plane with cut faces in hatched grey	No
2	Piston, rings, conrod, crank	Actor	Three rings modelled separately, conrod journal and counterweight; runs 600–4000 rpm with correct phase against the valves	No
3	Intake and exhaust valves with camshaft	Actor	Two 30 mm poppet valves on springs, single overhead cam with visible lobes; opening events shown on a small timing wheel	No
4	Spark plug and spark	Actor	14 mm plug, electrode gap 0.8 mm, blue spark sprite once per two revolutions, advance angle adjustable	Yes: swap
5	Fuel injector and burette	Actor + Instrument	Injector spraying a visible cone into the intake; 100 mL glass burette on the wall with 1 mL graduations and a falling meniscus	No
6	Air intake and mass-flow meter	Instrument	Bell mouth with a hot-wire meter reading g/s to one decimal; intake air drawn as pale blue streamlines	No
7	Coolant loop	Actor + Instrument	Pump, hoses, radiator with a fan, and two thermometers; heat removed computed as $\text{flow} \times \Delta T \times \text{specific heat}$	Yes: connect
8	Exhaust pipe and gas analyser	Instrument	38 mm pipe into a cabinet reading CO ₂ , H ₂ O, O ₂ , N ₂ and unburnt hydrocarbons as percentages	No
9	Dynamometer brake and load cell	Instrument	Water brake with a 0.5 m torque arm on a load cell; reads N·m, and with rpm gives shaft power in kW	Yes: swap
10	Fuel and air molecule tokens	Particle	C ₈ H ₁₈ tokens with charcoal C and white H spheres, O ₂ as paired red spheres, N ₂ as paired blue; recombine into CO ₂ and H ₂ O at combustion	No
11	Atom inventory panel	Overlay	Live count of C, H, O and N atoms entering and leaving per second, before and after combustion, with a difference column that must read zero	No
12	Energy Sankey ribbon	Overlay	One wide amber input band splitting into green shaft work, red coolant heat, grey exhaust heat and a thin friction band; widths $\propto$ kW	No
13	Control volume frame	Overlay	Dashed white box around the counted region; snaps to engine only, engine plus radiator, or engine plus radiator plus fuel tank	Yes: resize
14	Input disconnect valves	Actor	Four physical shutoffs on the fuel line, air duct, ignition lead and coolant hose; each can be closed by hand	Yes: connect

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Extraction hood and hazard badge	Environment	Stainless hood over the rig, running fan, plus a persistent "simulated rig" badge and interlock light	No
16	Data logger	Instrument	Rack-mounted unit sampling every channel at 10 Hz with a run marker and an export button	No

HOW IT WORKS — THE MODEL

A steady-state process balance with a four-stroke state machine driving the animation. Fuel energy in equals mass flow $\times$ 44 MJ/kg. Shaft power out equals torque $\times$ angular speed. Coolant heat out equals flow $\times$ 4.18 kJ/kg·K $\times$ ΔT . Exhaust heat out equals exhaust mass flow $\times$ specific heat $\times$ temperature rise. Friction takes what remains, and by construction the four outputs sum exactly to the input, so the Sankey always closes to 100 %. Combustion stoichiometry is explicit: two C_8H_{18} plus twenty-five O_2 gives sixteen CO_2 plus eighteen H_2O , so the atom inventory balances at every air-fuel ratio, with unburnt hydrocarbons appearing in the exhaust when the mixture is rich. Blocking an output is modelled honestly: with coolant stopped, heat accumulates in the block's thermal mass, temperature climbs, and the engine derates then seizes. Efficiency must never be allowed to exceed roughly 30 %, because the wasted heat is the point.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Throttle	Slider	0–100	40	% open	Air and fuel mass flow into the cylinder
Dyno load	Slider	0–20	8	N·m	Braking torque the engine must overcome, which sets rpm
Air-fuel ratio	Slider	10:1–18:1	14.7:1	mass ratio	Completeness of combustion and what appears in the exhaust
Spark timing	Slider	–5 to +35	20	° before top dead centre	How much of the fuel energy becomes work rather than exhaust heat
Coolant flow	Slider	0–30	12	L/min	Rate of the heat output pathway through the radiator
Disconnect an input	Multi-select	Fuel / Air / Ignition / Coolant	none	—	Physically closes that supply, which is a structural change to the rig
Control volume	Dropdown	Engine only / Engine plus radiator / Engine plus radiator plus fuel tank	Engine only	—	Which components are inside the counted box, and therefore what counts as a crossing
Run duration	Stepper	10–120	60	s	Length of a logged dyno run
Ledger units	Radio	kW / % of fuel energy / g per second	% of fuel energy	—	Units for the Sankey and the input-output table
Slow-motion cycle	Toggle	On / Off	Off	—	Drops to 1/40 speed so the four strokes can be counted individually

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Steady cruise	throttle=40; load=8; ratio=14.7:1; timing=20; coolant=12	List every input and every output with its rate. What fraction of the fuel energy leaves as useful work, and where does the rest go?
S2	Starve one input	disconnect=Air; everything else as S1	Cut one input and every output collapses. Which output keeps flowing for a while afterwards, and why?
S3	Rich mixture	ratio=10:1; throttle=60	Unburnt fuel appears in the exhaust. Check the atom inventory. Did any atoms go missing, and if not, where did they go?
S4	Block an output	coolant=0; throttle=60; load=12; duration=120	Every input is still connected. Trace what happens to block temperature and power when heat has nowhere to leave.

STUDENT ACTIVITIES

1. Run S1 for 60 s and record fuel flow, air flow, shaft power, coolant heat and exhaust heat. Add the outputs and compare with the input.
2. Set the ledger to "% of fuel energy" and copy the four Sankey widths into your table. State which output is the largest.
3. Disconnect the air supply and record the time to stall and the coolant heat still leaving 20 s later.
4. Set the air-fuel ratio to 10:1 and record the carbon atom count in and out. Then say what the difference column reads.
5. Set coolant flow to zero and record block temperature every 20 s until the engine derates. Record the temperature at which it stops.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Input table	Data table	g/s and kW	Fuel, air and coolant flow, each with its energy equivalent
Output table	Data table	g/s and kW	Shaft work, coolant heat, exhaust heat, friction and exhaust mass
Energy split	Bar chart	% of fuel energy	Live Sankey widths as numbers; must sum to 100 %
Atom inventory	Data table	atoms/s	C, H, O and N in, out and difference; the difference column must read zero
Shaft power and torque	Live numeric	kW and N·m	Updated at 10 Hz with rpm alongside
Block temperature	Line graph	°C	Over the run, with derate and seizure thresholds marked
Thermal efficiency	Live numeric	%	Shaft work divided by fuel energy
Run log	Data table	mixed	Exportable CSV at 10 Hz for every channel

WHAT THE STUDENT SHOULD REALISE

Students list a machine's inputs, assuming the outputs are just the useful one. This rig shows three quarters of the fuel energy leaving as heat, and shows that closing the radiator, an output, kills the engine as surely as closing the fuel line. Meanwhile the atom count in equals the atom count out. The student should be able to say: *"A system needs its outputs open as much as its inputs, and the energy that leaves is still all there."*

A2.4 · Tracing matter and energy through a system

Experiment name: Follow One Drop, Follow One Joule

Render mode: 2.5D Layered

Simulation engine: Agent-based (tracer particles on a flow network) + Data-driven model

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: Supporting — MS-ESS2-4

THEME & SCENE

A long cutaway landscape scrolling left to right, from Sierra granite down to San Francisco Bay, at twelve times vertical exaggeration so every drop of elevation is legible. Left, a blue-white snowfield above a reservoir held by a concrete arch dam. Then a penstock plunging to a powerhouse, an aqueduct running the valley, a treatment plant of rectangular basins, a pump station, a hilltop tank, a school with its plumbing exposed, a sewer, a wastewater plant, and an outfall into the bay. Above everything, faint evaporation arrows curl up to a cloud that drifts back east. The tracers are the only saturated colour on screen: a glowing magenta droplet with a tag number, and a small orange dot labelled 1 J. Control panel docks bottom-right; a path log scrolls left.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Terrain profile plate	Environment	260 km horizontal at 1:400 000, vertical exaggeration ×12, elevation ticks every 200 m from 2600 m to sea level, granite to alluvium textures	No
2	Snowpack and melt front	Actor	Snowfield with a retreating edge, melt rate driven by air temperature; meltwater renders as thin braided threads	Yes: resize
3	Reservoir and dam	Structure	Arch dam 90 m high, reservoir stock in millions of m ³ drawn as a filling wedge with a visible bathtub ring at low level	Yes: resize
4	Penstock, turbine, generator	Actor	3 m steel pipe dropping 300 m, Francis runner spinning, generator housing with a live kW display; converts gravitational energy to electrical	Yes: connect
5	Aqueduct and inverted siphon	Structure	4 m concrete channel on piers with a siphon crossing a canyon; flow velocity 1.2 m/s shown by drifting surface texture	Yes: connect
6	Treatment plant nodes	Actor	Four sequential nodes: bar screen, flocculation basin with paddle mixers, sand filter bed, chlorine contact tank; each has its own residence time	Yes: place
7	Pump station	Actor	Three centrifugal pumps in a shed, kWh meter on the wall; lifts water 140 m and adds energy that shows on the joule tracer	Yes: connect
8	Hilltop storage tank	Structure	8 million litre welded steel tank on a ridge, level gauge on the side; the buffer that decouples supply from demand	No
9	School plumbing and fountain	Actor	Service line, meter, riser, and a drinking fountain in cutaway; consumption follows the school-day schedule	Yes: place
10	Student body node	Actor	Simple body outline showing where drunk water goes: blood, sweat, breath vapour, urine; splits the tracer's onward path	No
11	Sewer and wastewater plant	Actor	Gravity sewer, then screening, aeration basins with churning surface, clarifier with a rotating arm and a returning sludge line	Yes: place
12	Outfall, bay and cloud	Environment	Diffuser pipe on the bay floor, then evaporation arrows to a drifting cloud that closes the loop back to the snowfield	No
13	Water tracer particle	Particle	Numbered magenta droplet, 6 px, leaves a fading trail; obeys the same flow field as the bulk water, with a random draw at every branch	Yes: place
14	Energy tracer particle	Particle	Orange token labelled 1 J; at every node a slice of it turns dull red and drifts away as waste heat, so the token visibly shrinks and dims	Yes: place
15	Path log	Instrument	Scrolling record: node entered, arrival time, residence time, transformation applied, energy remaining	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Node inspector	UI-Probe	Click any node to open its stock, inflow, outflow, mean residence time and current tracer count	Yes: drag

HOW IT WORKS — THE MODEL

A directed flow network of 18 nodes, each a stock with an inflow, an outflow and a computed residence time equal to stock divided by outflow. Bulk flows are solved first, so the reservoir genuinely holds water for years while the pipe holds it for minutes. Tracers are then advected as agents: at each node a tracer waits a time drawn from that node's residence distribution, then takes an exit branch with probability proportional to that branch's flow. Matter tracers are never destroyed, only moved, and the loop through evaporation and snowfall closes, so a water tracer can return to its starting node. Energy tracers behave differently by design: at every node a fixed fraction of the token is converted to low-grade heat and removed to the environment, and the token cannot be recycled back into the network. That asymmetry is the entire lesson, so the engine must never let an energy tracer complete a loop.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Tracer type	Radio	Water molecule / Joule of energy / Both	Water molecule	—	Which kind of tagged packet is released and logged
Release point	Dropdown	Snowpack / Reservoir / Aqueduct / Treatment plant / School fountain / Sewer / Bay	Snowpack	—	Where the tracer enters the network
Number of tracers	Stepper	1–200	1	tracers	One tells a story; two hundred reveal the spread of paths and times
Snowmelt rate	Slider	0–120	35	m ³ /s	Inflow to the reservoir and therefore reservoir residence time
Pump station power	Slider	0–2500	900	kW	Energy added to the water, visible as a step on the joule tracer
School demand	Slider	0–60	18	m ³ /day	Outflow rate at the school and how often a tracer is drawn through it
Aqueduct leak	Slider	0–25	0	% of flow	Diverts a share of flow to a soil node, adding a branch the tracer can take
Show residence times	Toggle	On / Off	Off	—	Prints each node's mean residence time on the scene
Energy quality shading	Toggle	On / Off	On	—	Fades the joule token from bright orange to dull red as it degrades
Time compression	Dial	1×–100000×	20000×	—	Simulated hours per real second, enough to run multi-year reservoir storage

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	One drop, whole journey	tracer=Water molecule; release=Snowpack; count=1; time=20000×	List every node the drop passes through in order. How many days from snow to bay, and where did it wait longest?
S2	Two hundred drops	tracer=Water molecule; count=200; release=Snowpack	The drops started together. Why do their arrival times at the school differ by years rather than hours?
S3	One joule, same route	tracer=Joule of energy; release=Snowpack; shading=On; pump=900	Follow the token to the outfall. How much of the original joule is still useful, and what is the rest now?

#	Scenario	Preset	The question the student answers
S4	The leaky aqueduct	leak=20; tracer=Both; count=50	A fifth of the flow is lost. Where do the lost tracers actually end up, and does any water disappear?

STUDENT ACTIVITIES

1. Release one water tracer at the snowpack and copy the path log: every node, its arrival time and its residence time.
2. Release 200 tracers at once and record the earliest and latest arrival at the school fountain. Sketch the spread.
3. Switch to an energy tracer from the same release point and record the token's remaining value at the turbine, the pump station, the school and the outfall.
4. Set the aqueduct leak to 20 % and run 50 tracers. Record how many reach the school and where the others finished.
5. Run a water tracer for a full simulated cycle and record whether it can return to the snowpack. Repeat with an energy tracer and record the result.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Path log	Data table	mixed	Node, arrival time, residence time, transformation, energy remaining, for the selected tracer
Arrival time distribution	Bar chart	days	Histogram of when a batch of tracers reaches the chosen node
Node residence times	Data table	h or years	Stock divided by outflow for all 18 nodes, updating with the flow settings
Energy remaining	Line graph	J	The tracer token's value against distance along the network; a staircase down
Waste heat released	Counter	J cumulative	Total energy shed to the environment so far on this run
Tracer census	Live numeric	count	How many tracers are currently in each node
Loop closed	Pass-fail badge	—	Green for a matter tracer that returns to its start; permanently grey for an energy tracer
Run log	Data table	mixed	Exportable CSV of every tracer's full path and timing

WHAT THE STUDENT SHOULD REALISE

Students say energy is "used up", and they picture water as vanishing once it goes down a drain. Tagging one droplet shows it circling from snow to bay to cloud to snow again with nothing lost, while the tagged joule dwindles at every step into heat that never comes back. Matter cycles; energy flows through once and degrades. The student should be able to say: *"The same water comes back around, and the same energy never does."*

A2.5 · Choosing a boundary for a purpose

Experiment name: The Zero-Emission Bus Argument

Render mode: Data Dashboard

Simulation engine: Data-driven model + Agent-based

Interaction level: Argue-from-data

Session length: 18–25 min

NGSS anchor: Supporting — MS-ESS3-3

THEME & SCENE

A working analyst's dashboard in cool slate and paper white, alive rather than static. Down the left, a narrow live map strip of a Fresno school bus route: 22 km, fourteen stops, three electric buses and one diesel bus crawling through a compressed school day. Across the centre, a horizontal lifecycle chain of seven boxes from lithium mine to battery recycling, with animated flow arrows whose widths breathe as the day runs. Over the chain lies a resizable dashed boundary frame with corner handles; boxes inside are crisp and counted, boxes outside desaturate to grey and their flows become unlabelled arrows entering the frame. Top centre, a question card. Right, the ledger and the claim builder. A hard-edged fit badge sits beside the question, red, amber or green.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Route map strip	Environment	22 km loop drawn schematically over a Fresno street grid, 14 stops as pins, an elevation ribbon underneath, a moving time-of-day marker	No
2	Electric bus agents	Actor	Three buses, 155 kWh pack each, 1.1 kWh/km consumption, animated along the route; state of charge shown as a filling bar over each icon	Yes: place
3	Diesel bus agent	Actor	One comparison bus, 32 L/100 km, animated grey tailpipe plume that thickens on the uphill segment	Yes: swap
4	Depot charger and meter	Actor	150 kW charger with a schedulable window and a kWh meter that spins only while charging	Yes: connect
5	Grid mix ribbon	Overlay	Hourly California generation stack: solar, wind, hydro, gas, imports; drives a live gCO ₂ per kWh figure that dips at midday and spikes after sunset	No
6	Lifecycle chain nodes	Structure	Seven boxes: lithium mine, cell factory, bus assembly, depot charger, route operation, battery second life, recycling; each carries energy, CO ₂ , cost and pollutant values with uncertainty bands	Yes: place
7	Flow arrows between nodes	Overlay	Arrow width $\propto$ mass or energy moved; arrows crossing the boundary frame render as bold entering or leaving arrows with a crossing tag	No
8	Boundary frame	UI-Probe	Dashed white rectangle with four corner handles, snapping to node edges; can enclose any contiguous run of the chain	Yes: resize
9	Question card deck	UI-Probe	Six cards, each stating one question and carrying a hidden list of the quantities needed to answer it; a card is dragged onto the board to arm the run	Yes: drag
10	Fit badge	Instrument	Compares the question's required quantities against what the current boundary encloses; green if all inside, amber if partly, red if a required quantity is outside	No
11	Ledger panel	Overlay	Totals inside the current boundary: energy, CO ₂ , running cost, NO _x and PM _{2.5} , each with an uncertainty range	No
12	Claim builder	UI-Probe	Three slots: claim, evidence rows dragged from the ledger, and the boundary used; scores whether the evidence can support the claim under that boundary	Yes: drag
13	Neighbourhood air probe	Instrument	A sensor pin dropped beside the school gate reading PM _{2.5} in $\mu\text{g}/\text{m}^3$ as buses arrive and depart	Yes: place
14	Time-of-day scrubber	UI-Probe	00:00 to 23:00 handle that drives the grid mix, the route and the charging window together	Yes: drag

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Uncertainty band overlay	Overlay	Grey whiskers on every ledger figure showing the plausible range, so a difference smaller than the band is flagged as not a difference	No
16	Boundary comparison tray	Instrument	Holds up to four saved boundary and question pairs side by side for direct comparison	Yes: place

HOW IT WORKS — THE MODEL

Two coupled models feed one accounting engine. The route model is agent-based: each bus consumes energy per kilometre with a gradient and stop-frequency correction, drains its pack, and returns to the depot to charge in the chosen window. The lifecycle model is a static inventory per node, in kWh, kg CO₂, dollars and grams of pollutant per bus-year, each with an uncertainty band. The accounting engine sums only what the boundary frame encloses, exactly as in a real life-cycle assessment. The grid mix is a real hourly California profile, so charging at 13:00 draws mostly solar and charging at 19:00 draws mostly gas, and the same boundary yields different answers at different hours. The fit badge is the heart of it: every question card declares the quantities it needs, and the badge is computed, not scripted. A claim scored on a red boundary is rejected however true it happens to be.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Question card	Dropdown	Air outside our school / Total CO ₂ over the year / CO ₂ in the first year / Cheaper to run / Where does the pollution move to / Is the electricity clean when we charge	Air outside our school	—	Which question the run must answer, and which quantities the badge demands
Boundary preset	Dropdown	Tailpipe only / Bus and charger / Bus, charger and power plants / Whole life cycle / Custom	Tailpipe only	—	Jumps the frame to a saved span of the chain
Boundary frame	Drag-handle	Any contiguous run of the 7 chain nodes	Route operation only	—	Which nodes are counted; everything outside greys out and becomes a crossing
Charging window	Timeline scrubber	00:00–23:00	22:00	h:mm	Which hourly grid mix the depot draws, which sets gCO ₂ per kWh
Grid scenario	Dropdown	California average / Sunny midday / Evening peak / Rooftop solar depot	California average	—	The generation mix behind every kWh the buses use
Fleet size	Stepper	1–40	12	buses	Scales every ledger figure and the depot load together
Bus service life	Slider	4–18	12	years	How far the manufacturing burden is spread, which moves the break-even year
Comparison vehicle	Dropdown	2008 diesel / 2020 diesel / Compressed natural gas / Electric	2020 diesel	—	The baseline the electric fleet is measured against
Ledger metrics	Multi-select	CO ₂ / Energy / Running cost / NO _x / PM _{2.5}	CO ₂ , PM _{2.5}	—	Which totals the ledger reports for the enclosed nodes
Uncertainty bands	Toggle	On / Off	On	—	Shows the plausible range on every figure and flags false differences

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	The right small boundary	question=Air outside our school; boundary=Tailpipe only; probe placed at the gate	The badge is green and the answer is zero. Why is a tiny boundary the correct one for this question?
S2	The same boundary, a new question	question=Total CO ₂ over the year; boundary=Tailpipe only	The badge turns red. Name the quantity the question needs that this boundary leaves outside.
S3	The whole life cycle	question=CO ₂ in the first year; boundary=Whole life cycle; life=12	The answer flips to no. Move the service life slider and find the year in which the electric fleet breaks even.
S4	Charge at noon	question=Is the electricity clean when we charge; boundary=Bus, charger and power plants; charging=13:00 then 19:00	The boundary never moves. Why does the answer change by a factor of three?

STUDENT ACTIVITIES

1. Drag the "Air outside our school" card onto the board with the tailpipe boundary. Record the badge colour, the PM2.5 reading and your answer.
2. Keep the boundary and swap to the "Total CO₂" card. Record the badge colour and write down exactly which node sits outside the frame.
3. Widen the frame node by node until the badge turns green. Record the smallest boundary that can answer the question.
4. Set the whole-life-cycle boundary and step the service life from 4 to 18 years. Record the break-even year against the 2020 diesel baseline.
5. Build a claim in the claim builder with three evidence rows and the boundary you used. Record the score and the reason given if it is rejected.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Ledger totals	Data table	kg CO ₂ , kWh, \$, g NOx, g PM2.5	Totals for the enclosed nodes only, each with an uncertainty range
Boundary fit	Pass-fail badge	—	Green, amber or red, computed from the question's required quantities against the frame
Nodes inside vs outside	Counter	count	How many of the seven chain nodes the frame currently encloses
Gate air quality	Live numeric	µg/m ³	PM2.5 at the school gate probe as buses arrive and leave
Grid carbon intensity	Line graph	gCO ₂ /kWh	Hourly California profile with the charging window shaded
Break-even year	Live numeric	years	When cumulative electric emissions fall below the comparison vehicle's
Claim score	Pass-fail badge	—	Whether the selected evidence can support the claim under the stated boundary
Comparison tray	Data table	mixed	Up to four saved question and boundary pairs side by side, exportable as CSV

WHAT THE STUDENT SHOULD REALISE

Students look for the one correct boundary. Here the tailpipe boundary is exactly right for air at the school gate and exactly wrong for yearly carbon, and both facts are true at once. A boundary is fit for a purpose, not true or false, and every claim carries the boundary that produced it. The student should be able to say: *"Before I answer, I have to say where I drew the line and why that line fits this question."*

A3 · Models of systems

A3.1 · Why scientists build models

Experiment name: The Experiment You Cannot Run

Render mode: Hybrid 2D+3D

Simulation engine: Data-driven model + State machine

Interaction level: Investigate

Session length: 12–18 min

NGSS anchor: Supporting — MS-ETS1-4

THEME & SCENE

A dim field station at dusk. Along the back wall stand four armoured cases holding systems the student may watch but never touch: a Sierra snowpack under a frosted dome, a magma chamber glowing beneath a sectioned block of Long Valley, a redwood seedling in a pot with a 90 m adult ghosted behind it, and a near-Earth asteroid turning slowly on a wire. Red lamps above each case read TOO SLOW, TOO BIG, TOO DANGEROUS, TOO FAR, whichever apply. Facing them, warmly lit and open to the hands, is the model bench: a turntable, a glass tank, a whiteboard and a laptop. One clock hangs between the halves with two faces, real time on the left, model time on the right. Control panel docks right, cost meter above it.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Reality case	Structure	1.2 m armoured vitrine, 40 mm laminated glass, brushed steel frame, interlocked door that never opens; four instanced along the back wall	No
2	Sierra snowpack specimen	Actor	2 m × 2 m slab of layered snow on granite, 6 visible strata from November crust to April firn, melts at the true seasonal rate, meltwater trickles into a weir	No
3	Long Valley magma chamber	Actor	Sectioned rock block 3 m tall, rhyolite melt lens glowing 780 °C orange at 6–8 km depth, crystal mush rim, convection cells drifting at 1 cm/year	No
4	Coast redwood pair	Actor	Potted 30 cm seedling in the foreground, 90 m adult ghosted at 20 % opacity behind, growth ring counter reading 0 and 1,240 years	No
5	Near-Earth asteroid	Actor	340 m irregular basalt body on a gimbal, 4.2 h rotation, orbit trace drawn as a faint ellipse crossing Earth's	No
6	Barrier lamp cluster	Instrument	Four red lamps per case labelled TOO SLOW / TOO BIG / TOO DANGEROUS / TOO FAR; lit lamps dim one by one as a model removes each barrier	No
7	Ground-truth engine	Field	Invisible state model per case, run once, irreversibly, at 1× real time; supplies the answer the student is eventually graded against	No
8	Reality sensor set	Instrument	Surface-only probes: snow stake, satellite pass every 16 days, thermal camera, one seismometer, one telescope; each reads a single number, never a cross-section	Yes: place
9	Scale-model turntable and tank	Structure	60 cm turntable and 40 L glass tank on the model bench, holds sand, ice, dyed water or wax depending on the case	Yes: swap
10	Computational model laptop	Instrument	Open laptop showing the case's equations as editable blocks; runs the same state model with adjustable process count and speed	Yes: swap
11	Mathematical model whiteboard	Instrument	Whiteboard carrying two or three relations for the case, for example melt = degree-day factor × positive degrees; solvable in one step	Yes: swap
12	Analogue model tray	Structure	Tray of stand-ins: flour and a marble for impact, a heat lamp over corn syrup for convection, a fan over a sand ridge for drift	Yes: swap
13	Dual-face clock	Instrument	Wall clock with two dials, real time fixed at 1×, model time driven by the compression dial, both showing elapsed days	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
14	Rewind lever	UI-Probe	Brass lever on the model bench, throws freely; the identical lever on the reality wall is welded solid and rings when pulled	Yes: drag
15	Cross-section knife	UI-Probe	Cutting plane that slices any model object open; refuses to cross the vitrine glass	Yes: drag
16	Prediction card and error meter	Overlay	Index card the student writes a numeric prediction on before running, then a needle gauge showing prediction minus outcome	Yes: place

HOW IT WORKS — THE MODEL

Both tracks run the same governing state model for the selected case, so the comparison is honest. The reality track runs at 1× with process noise, exposes only the sensors the student has placed, and can be run exactly once. The model track runs the same equations with three levers: a process count from 1 to 10, which switches terms on in order of importance; a time compression dial; and a repeat count, which reruns with fresh noise draws to give a spread rather than a single answer. Each medium adds its own systematic bias term: scale models carry a distortion offset, mathematical models carry a truncation error, analogue models carry a material mismatch. Error therefore falls with more processes but never reaches zero. The failure to avoid is a model that is simply reality at high speed. It must visibly get things wrong, and the student must be able to say which barrier it removed in exchange.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Reality case	Dropdown	Sierra snowpack / Long Valley magma / Coast redwood / Near-Earth asteroid	Sierra snowpack	—	Which system is behind the glass, which barrier lamps light, and which ground truth runs
Model medium	Dropdown	None / Scale physical / Computational / Mathematical / Analogue	None	—	What appears on the model bench and which bias term applies
Processes included	Slider	1–10	4	count	How many terms of the state model the model track computes
Model time compression	Dial	1×–1,000,000×	1,000×	—	Simulated days per real second on the model track only
Repeat runs	Stepper	1–50	1	runs	How many times the model is rerun with new noise; reality stays locked at one
Reality sensors	Multi-select	Snow stake / Satellite pass / Thermal camera / Seismometer / Telescope	Snow stake	—	Which single numbers the sealed case will give up
Rewind	Toggle	On / Off	Off	—	Enables rewind on the model track; stays locked off for reality
Cut open	Toggle	On / Off	Off	—	Slices the model object to expose its interior
Lab time allowed	Slider	1–72	8	h	Budget the run consumes; long real-time waits exhaust it

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Ask reality	case=Sierra snowpack; medium=None; sensors=Snow stake; repeat=1; rewind=Off	How long must you wait for the melt-out date, and how many times can you repeat the season to check it?
S2	Speed it up	case=Sierra snowpack; medium=Computational; processes=6; compression=1,000,000×; repeat=30	Thirty melt seasons finish in two minutes. What is the spread of melt-out dates, and what did one run hide?
S3	Cut it open	case=Long Valley magma; medium=Scale physical; cut open=On; repeat=5	What can you see inside the model that no instrument on the real chamber can show, and what does the model get wrong?

#	Scenario	Preset	The question the student answers
S4	Never touchable	case=Near-Earth asteroid; medium=Mathematical; sensors=Telescope; processes=3; repeat=50	You cannot experiment on this object at all. How can two equations still give a miss distance you would bet on?

STUDENT ACTIVITIES

1. Run S1 with no model and record the elapsed real time on the left clock when the melt-out date finally arrives, plus how many repeats you got.
2. Write a melt-out date on the prediction card, then run the computational model 30 times and record the earliest, latest and most common date.
3. Swap the medium to Scale physical, cut the model open, and list two things you can now measure that the sensors on the sealed case cannot reach.
4. Pull the rewind lever on the reality wall, then on the model bench, and record what each one does.
5. For each of the four cases, record which barrier lamps the model switched off and which one it could not.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Prediction error	Live numeric	days or km	Prediction minus the ground-truth outcome, shown once reality resolves
Answer spread	Bar chart	count per bin	Histogram of results across repeat runs; a single run has no spread at all
Time to answer	Timer	h	Real-track and model-track elapsed time side by side
Barriers removed	Counter	count	How many of the four red lamps this model configuration switched off
Bias contribution	Data table	days or km	Systematic error added by the chosen medium at the chosen process count
Run log	Data table	mixed	Exportable CSV: case, medium, processes, repeats, prediction, outcome, error

WHAT THE STUDENT SHOULD REALISE

Students think a model is a small copy built to show what something looks like. Here the real snowpack melts once, at its own pace, and yields only surface readings; the model runs thirty times, a million times faster, and can be cut in half. A model exists to let you do what reality forbids. The student should be able to say: *"We model what we cannot rewind, reach, or risk."*

A3.2 · Diagrams and flowcharts as models

Experiment name: The Diagram That Runs

Render mode: Hybrid 2D+3D

Simulation engine: State machine + Agent-based

Interaction level: Design

Session length: 18–25 min

NGSS anchor: Supporting — MS-ESS3-3

THEME & SCENE

The screen splits down the middle. Left is a small, warm, working plant: the drinking-water treatment works of a Sierra foothills town of 12,000, drawn as a low three-quarter cutaway with the canal intake, the alum doser ticking, paddles turning in the flocc basin, sludge settling brown in the sedimentation tank, and a sand filter whose bed you can see through. Right is a cold blue drafting grid with a palette of empty shapes down its edge: rounded terminators, rectangles, diamonds, parallelograms, tanks, arrows. The student's diagram lives there. When Run is pressed a small white token enters the diagram and walks it, one shape at a time, while the real plant runs the same minute beside it. Control panel sits along the bottom.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Canal intake and bar screen	Actor	Concrete headworks with a 20 mm steel bar rack, weed and leaf debris catching on it and raking off every 40 s	No
2	Coagulant doser	Actor	200 L alum tank with a peristaltic pump, dose rate 5–90 mg/L, drip visible in a sight glass, level falls as it runs	Yes: swap
3	Flocculation basin and paddles	Actor	8 m × 4 m basin, three paddle banks at 3–15 rpm, floc particles growing from specks to 2 mm snow over 20 min	No
4	Sedimentation tank and sludge blanket	Structure	25 m rectangular tank, inclined plate pack, brown blanket thickening on the floor, scraper crawling at 2 cm/s	No
5	Rapid sand filter bed	Actor	1 m of graded sand over gravel in a glass-walled box; a headloss column beside it rises from 0.2 m to 2.5 m as the bed loads	No
6	Backwash pump and waste line	Actor	Reversing pump and a grey waste pipe; fires only if the diagram contains a loop back to the filter	Yes: connect
7	Chlorine contact tank	Structure	Serpentine baffled tank giving 30 min contact, dosing point with a residual probe at the outlet	No
8	Clearwell and distribution pumps	Structure	Buried 4 ML clearwell with a level gauge and two pumps feeding the town main	No
9	Water parcel tokens	Particle	1,000 L parcels, colour-coded by turbidity from mud brown to clear blue; carry their own state through every step	No
10	Diagram canvas	Environment	Snap grid, 24 px pitch, cold blue, holds the student's shapes and arrows	Yes: resize
11	Shape palette	UI-Probe	Terminator, process box, decision diamond, input/output parallelogram, stock tank, off-page connector; drag out to instantiate	Yes: place
12	Flow arrows	Structure	Directed connectors that snap to shape ports; decision diamonds require two labelled exits or they refuse to close	Yes: connect
13	Token walker	Actor	Execution head that highlights the shape it is in, pauses 0.4 s, and takes the branch the diagram tells it to	No
14	Reality event log	Instrument	Ground-truth trace from the plant: a timestamped list of every real step and decision the water actually went through	No
15	Trace diff overlay	Overlay	Draws the reality trace beside the diagram trace, green where they match, amber for a missing step, red for a wrong branch	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
16	Notation switcher	UI-Probe	Redraws the same plant as flowchart, cycle diagram, cross-section schematic or single black box	Yes: swap

HOW IT WORKS — THE MODEL

The diagram is not a picture of the model, it is the model. Shapes compile to states, arrows to transitions, diamonds to conditional branches evaluated on the token's own state: turbidity in NTU, chlorine residual in mg/L, filter headloss in metres. Each token entering a process box has that box's real transfer function applied, for example sedimentation removing 60–90 % of turbidity as a function of floc size and residence time. The plant on the left runs the same physics from its own hard-wired process order, and every token it passes writes a line to the reality event log. On Run, the two traces are aligned step by step and scored. A diagram missing the headloss diamond keeps pushing water through a clogged bed until throughput collapses; a diagram missing the backwash loop can never recover. Nothing is scored on neatness. It is scored on whether the arrows predict what the plant does.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Notation	Dropdown	Flowchart / Cycle diagram / Cross-section schematic / Black box	Flowchart	—	Which kind of model the canvas builds and which questions it can answer
Available shapes	Multi-select	Terminator / Process / Decision / Input-output / Stock tank / Loop-back arrow	All six	—	Which symbols the palette offers; removing one makes some systems undrawable
Raw water turbidity	Slider	0.5–120	3.0	NTU	How dirty the incoming canal water is, and which branches must exist
Town demand	Slider	1–40	12	ML/day	Token release rate and how fast the clearwell drains
Run mode	Radio	Step one token / Continuous / Fast 24 h	Step one token	—	Whether the walker moves on click, in real time, or over a compressed day
Fault injection	Dropdown	None / Filter clogged / Doser empty / Power cut / Algal bloom	None	—	Injects a real plant fault that only a correctly branched diagram survives
Show reality trace	Toggle	On / Off	On	—	Displays the plant's own event log beside the diagram trace
Auto-check arrows	Toggle	On / Off	Off	—	Flags dangling arrows and unlabelled diamonds before the run
Diagram time compression	Dial	1×–2,000×	60×	—	Simulated minutes per real second for both panes

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Draw the works	turbidity=3.0; demand=12; run mode=Step one token; show reality trace=On	Draw the plant as a flowchart, run one token, and report what percentage of your steps match the reality trace.
S2	Muddy river day	turbidity=90; demand=12; fault=None	The raw water goes brown. Which decision diamond does your diagram need, and what reaches the town without it?
S3	The missing loop	fault=Filter clogged; available shapes=all except Loop-back arrow	Throughput falls to zero and stays there. Which single shape would fix it, and why is a straight-line diagram wrong here?
S4	Four ways to draw one plant	notation=Cycle diagram, then Cross-section schematic, then Black box	Ask "where does the sludge go?" of each notation. Which ones can answer it, and what did each drop to stay readable?

STUDENT ACTIVITIES

1. Drag out shapes and connect them into a flowchart of the plant, then run one token and record your trace match percentage.
2. Set turbidity to 90 NTU, run again, and record the outlet turbidity before and after you add a coagulant decision diamond.
3. Inject the filter-clogged fault and record throughput each minute for ten minutes, first without a loop-back arrow and then with one.
4. Redraw the same plant as a cycle diagram and as a black box; record for each which of three questions it can answer.
5. Delete one arrow of your choice, predict which token step will break, run, and record whether you were right.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Trace match	Live numeric	%	Fraction of reality-log steps that the diagram's token path reproduced in order
Water delivered	Line graph	ML/day	Throughput to the town over the run, per diagram version
Outlet turbidity	Live numeric	NTU	Final token turbidity, red above 1.0 NTU
Unmatched steps	Data table	—	Named list of missing steps and wrong branches, one row each
Deadlock check	Pass-fail badge	—	Green if every token reaches a terminator, red if any token stalls or loops forever
Diagram complexity	Counter	shapes	Number of shapes and arrows used, so a working simple diagram beats a working cluttered one

WHAT THE STUDENT SHOULD REALISE

Students treat a diagram as a picture that decorates an explanation. Here the arrows are executed, so a missing decision diamond sends untreated water into the town and a missing loop-back arrow leaves the filter clogged forever. The layout is a claim about order and dependency, and it can be wrong. The student should be able to say: *"A flowchart is a model, and its arrows are predictions I can test."*

A3.3 · Physical and digital models

Experiment name: Two Bays: The Warehouse and the Solver

Render mode: Hybrid 2D+3D

Simulation engine: Fluid/thermal + Data-driven model

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: Supporting — MS-ETS1-4

THEME & SCENE

A vast Sausalito warehouse, skylit and echoing. Underfoot, 1.5 acres of concrete San Francisco Bay: the Golden Gate, the two bay lobes, Carquinez Strait, Suisun Bay and the Delta out to Stockton, all filled with real water that glints under the roof lights. Horizontal scale is 1:1000 and vertical scale 1:100, so the vertical exaggeration is 10× and the shallows look like canyons. Thousands of copper roughness strips bristle from the shallows. At the seaward end a tide machine sighs once every 14.9 minutes, one full lunar day. A steel walkway crosses the model; a monitor on a trolley shows the same Bay as a grid of coloured cells. Controls dock right; a dye bottle follows the cursor.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Concrete bay basin	Environment	320 modelled bathymetry panels forming the Bay and Delta, 1:1000 horizontal, 1:100 vertical, painted grey-green, water depth 1–12 cm	No
2	Tide machine	Actor	Hydraulic plunger at the seaward end, stroke adjustable, one cycle every 14.9 min representing a 24.8 h lunar day, audible sigh at each turn	No
3	Copper roughness strips	Structure	250,000 vertical copper tabs 15 mm high, instanced across the shallows, tuning friction so the model's tidal timing matches the real Bay	Yes: place
4	Delta inflow header	Actor	Pump manifold at Stockton and Sacramento ends delivering scaled Sacramento and San Joaquin river flow, flowmeter dial visible	No
5	Dye and dye ports	Particle	40 brass ports; green fluorescein for tracing, red rhodamine for a second release; plumes stretch and fold with the tide	Yes: place
6	Salinity probe gantry	Instrument	Rolling gantry with five conductivity probes lowering on cables at Golden Gate, Richmond, Carquinez, Antioch and Rio Vista	Yes: drag
7	Reber Plan barrier blocks	Structure	Two placeable earth-fill dam blocks, one across the north bay and one across the south, each with a lock and a highway deck moulded on	Yes: place
8	Dredged channel tool	Structure	Removable insert cutting a 15 m deep, 200 m wide shipping channel through Suisun Bay	Yes: swap
9	Digital grid field	Field	Finite-volume mesh over the same Bay, cell size 50–2000 m, each cell carrying depth, velocity, salinity; drawn as a false-colour skin on the monitor	Yes: resize
10	Digital solver panel	Instrument	Shows equations being stepped, cell count, time step and wall-clock cost per simulated day	No
11	Digital tracer field	Field	Numerical dye concentration advected on the grid, plotted as contours to compare with the physical plume	No
12	Field observation dataset	Instrument	Real station records for the five sites: tidal range, salinity, slack-water timing; the arbiter both models are judged against	No
13	Scale-effect overlay	Overlay	Amber wash marking where the physical model misleads: surface tension at millimetre depths, 10× vertical exaggeration, warm indoor water	No
14	Comparison chart panel	Overlay	Three-series plot, physical vs digital vs field, one panel per station	No
15	Surprise log	Overlay	Auto-logged list of behaviours seen in the water that no line of the digital model produces, for example a standing eddy behind a barrier	No
16	Observation walkway	Structure	Steel grating bridge with handrail crossing the model at Carquinez; the camera can be parked on it	Yes: drag

HOW IT WORKS — THE MODEL

The physical basin is real shallow-water flow at 1:1000, tuned with copper roughness until its tidal timing matches the field record; it therefore produces eddies, salt wedges and sediment fans that nobody wrote down. The digital model solves depth-averaged continuity and momentum plus a salt advection-diffusion equation on the chosen mesh, stepping at a stability-limited time step, so its cost rises steeply as cells shrink. Both are compared against the same field dataset. The physical model carries scale distortions that are stated openly: with 10× vertical exaggeration, slopes are ten times too steep, and at millimetre depths surface tension and viscosity matter far more than they do in the real Bay. The digital model has no such distortion but contains only what was programmed. The whole point is that neither is "the real Bay", and each fails in its own characteristic direction.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Model in use	Radio	Physical / Digital / Side-by-side	Side-by-side	—	Which basin, monitor, or both are driving the run
Delta inflow	Slider	100–3,000	700	m ³ /s	Combined Sacramento and San Joaquin freshwater flow, scaled to the basin
Tide amplitude	Slider	0.0–2.0	1.2	m	Tide machine stroke and the digital boundary forcing
Barriers and channels	Multi-select	None / Reber north dam / Reber south dam / Dredged 15 m channel	None	—	Places physical blocks and edits the digital bathymetry to match
Dye release point	Drag-handle	Any of 40 ports	Golden Gate	—	Where the tracer plume starts in both models
Digital cell size	Slider	50–2,000	200	m	Mesh resolution, solver cost, and whether Carquinez Strait exists as more than one cell
Run length	Slider	1–30	14	tidal days	How long both models are run before the salt front is read
Scale-effect overlay	Toggle	On / Off	Off	—	Marks where the physical model's distortions dominate
Digital speed	Dial	1×–100,000×	5,000×	—	Simulated seconds per real second on the solver only

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Same question, two models	model=Side-by-side; inflow=700; tide=1.2; run length=14	Where does the salt front sit in each model after 14 tidal days, and which is closer to the field stations?
S2	Test the Reber Plan	barriers=Reber north dam, Reber south dam; inflow=700	Two dams are meant to make fresh-water lakes. What happens to flushing and salinity behind them?
S3	Cheap and coarse	model=Digital; cell size=2,000; inflow=200	The strait is now narrower than one cell. How far does the salt front move, and can you trust the number?
S4	Where the water lies	model=Physical; scale-effect overlay=On; inflow=100	Which observed behaviours are real Bay physics, and which are artefacts of 10× exaggeration and centimetre depths?

STUDENT ACTIVITIES

- Run S1 and record salt front position and salinity at all five stations for the physical model, the digital model and the field dataset.
- Place both Reber dams, run 14 tidal days, and record salinity and dye clearance time behind each dam.
- Step the digital cell size through 50, 200, 800 and 2,000 m and record the salt front position and the solver cost at each.
- Release dye at the Golden Gate in both models and record every behaviour listed in the surprise log that the digital model did not reproduce.

5. Turn on the scale-effect overlay and name two measurements you would refuse to take from the physical model, with a reason for each.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Salt front position	Live numeric	km from Golden Gate	Location of the 2 PSU contour in each model, updated each tidal cycle
Salinity at five stations	Line graph	PSU	Three series per station: physical, digital, field observation
Tracer clearance time	Live numeric	h	Time for peak dye concentration to fall by 90 % at the release point
Agreement chart	Bar chart	PSU error	Absolute error against field data for each model at each station
Surprises	Counter	count	Behaviours logged in the physical model with no counterpart in the digital output
Cost to answer	Data table	mixed	Solver wall-clock seconds, cells, and physical model run time per scenario

WHAT THE STUDENT SHOULD REALISE

Students rank models as more or less realistic, with the computer at the top. Here the concrete Bay throws eddies nobody programmed but lies about slopes and surface tension, while the solver is undistorted yet contains only what was typed in. Both miss, in opposite directions. The student should be able to say: *"A physical model can surprise me and a digital model cannot, but only the digital one lets me change the Bay before lunch."*

A3.4 · What a model leaves out on purpose

Experiment name: The Model That Flattened the Sierra

Render mode: Hybrid 2D+3D

Simulation engine: Data-driven model + Fluid/thermal

Interaction level: Argue-from-data

Session length: 18–25 min

NGSS anchor: Supporting — MS-ESS2-6

THEME & SCENE

One landscape, drawn twice, with a draggable wipe between them. On the right is reality: California from Sacramento to Reno rendered from a 10 m elevation model, oak savanna rising into pine, granite crest, then the grey sagebrush drop into the Great Basin, with five instrument huts on their real sites. On the left is the model's world, the same landscape built from blocks whose size the student sets: at 200 km the Sierra is a single grey slab, at 4 km the crest and lee appear. A cold westerly stream of vapour parcels blows in over both. Below runs a modelling console: an omission ledger of process cards with checkboxes, a compute budget bar, and a scatter plot with a 1:1 line waiting to be filled.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Reality terrain	Environment	10 m DEM strip 260 km west to east, vertical exaggeration 3×, real land cover textures, unchangeable by the student	No
2	Model terrain blocks	Environment	The same strip rebuilt as flat-topped blocks at the chosen cell size, each block one averaged elevation; visibly rebuilds when the slider moves	Yes: resize
3	Wipe divider	UI-Probe	Vertical draggable seam with a grab handle, sweeping model over reality anywhere across the scene	Yes: drag
4	Vapour parcel stream	Particle	2,000 translucent parcels entering from the west, each carrying water vapour in kg, speed set by the wind dial	No
5	Orographic lift field	Field	Vertical velocity field computed from wind crossing terrain slope; drawn as pale upward arrows when the process card is on	No
6	Condensation and cloud layer	Particle	Parcels turn opaque white above their lifting condensation level, thickening as they climb	No
7	Precipitation cells	Particle	Falling streaks, blue for rain, white six-point flakes for snow, chosen against the freezing level	No
8	Freezing level plane	Overlay	Horizontal translucent cyan sheet at the set altitude, cutting the terrain visibly at the snow line	Yes: drag
9	Rain gauge huts	Instrument	Five huts at Sacramento 8 m, Blue Canyon 1,610 m, Donner Summit 2,100 m, Truckee 1,800 m and Reno 1,340 m, each with a tipping bucket and a running total	Yes: place
10	Process cards	UI-Probe	Seven cards: terrain detail, orographic lift, rain-snow phase, valley evaporation, cloud microphysics, canopy interception, wind drift; each shows its compute cost	Yes: place
11	Omission ledger	Overlay	Two-column board, IN and LEFT OUT, with a reason field for each omitted card and its measured error cost	Yes: swap
12	Compute budget bar	Instrument	Horizontal bar of 100 units filling as cards and resolution are added; turns red past the limit and refuses the run	No
13	Model-vs-reality scatter	Overlay	Five points, modelled against observed annual precipitation, with a dashed 1:1 line and error bars	No
14	Error map	Overlay	False-colour ribbon over the terrain, blue where the model is too dry, red where too wet, grey within 10 %	No
15	Run timer	Instrument	Stopwatch showing solver seconds for the current configuration, so cheapness is visible	No

HOW IT WORKS — THE MODEL

A one-dimensional west-to-east moisture budget on the chosen grid. Each parcel carries vapour; where terrain rises, forced ascent cools it at 6 °C per km until saturation, and the excess falls out as precipitation, so the windward slope is wet by construction. Descending on the lee side, the parcel warms and dries and nothing falls, producing the rain shadow. Every process card switched off removes a term: no terrain detail averages elevation over the whole cell, so a 200 km cell has no crest to lift anything; no phase rule puts rain where snow belongs; no valley evaporation starves the parcels of the moisture the Central Valley returns. Ground truth is the 1991–2020 gauge normals, roughly 470 mm at Sacramento, 1,500 mm at Blue Canyon and 190 mm at Reno. The honest part is the compute budget: it is never large enough for everything, so omission is forced and must be argued for, not apologised for.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Grid cell size	Slider	1–200	25	km	How coarsely the terrain is rebuilt, and whether the crest exists at all
Processes included	Multi-select	Terrain detail / Orographic lift / Rain-snow phase / Valley evaporation / Cloud microphysics / Canopy interception / Wind drift	Terrain detail, Orographic lift, Rain-snow phase	—	Which terms the moisture budget computes; each has a compute cost
Compute budget	Slider	10–200	100	units	Total cost allowed; cards and fine grids must fit inside it
Storm strength	Slider	100–1,200	400	kg/m/s	Incoming vapour transport, the moisture available to fall out
Wind direction	Dial	180–360	250	°	Angle of approach, and therefore which slopes are windward
Freezing level	Slider	500–4,000	1,800	m	Altitude of the rain-to-snow switch and the visible snow line
Wipe position	Drag-handle	0–100	50	% of width	Where the seam between model world and real world sits
Compare against	Dropdown	1991–2020 normals / Feb 2017 storm / Dec 2021 storm	1991–2020 normals	—	Which observed dataset the scatter plot is scored on
Terrain exaggeration	Slider	1–10	3	×	Vertical stretch of both terrains for legibility only

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	A flat California	cell size=200; processes=none; storm=400	With no mountains in the model, how much rain does it give Blue Canyon and Reno, and what do the gauges say?
S2	Put the mountains back	cell size=4; processes=Terrain detail, Orographic lift, Rain-snow phase	What appears at Blue Canyon and disappears at Reno, and what is the windward to leeward ratio now?
S3	Spend the budget	budget=100; choose any four cards; cell size=10	You cannot afford all seven processes. Which four omissions cost you the least error, and why?
S4	Storm of record	compare=Feb 2017 storm; storm=900; freezing level=2,700	The model puts snow where the gauges recorded rain. Which omitted process explains the miss?

STUDENT ACTIVITIES

1. Set the cell size to 200 km with all process cards off, run, and record modelled and observed precipitation at all five gauges.
2. Turn on terrain detail and orographic lift, drop the cell size to 4 km, re-run, and record the change in mean absolute error.
3. Fill the omission ledger: for every card you left out, write the reason and record the error cost the ledger reports for it.
4. Hold the budget at 100 units and find the cheapest configuration that keeps error under 150 mm; record the cards you kept.
5. Drag the wipe across the crest and describe, in one sentence, exactly what the model's world is missing there.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Precipitation at five gauges	Bar chart	mm/year	Paired modelled and observed bars for each station
Model vs reality scatter	Line graph	mm vs mm	Five points against a dashed 1:1 line; distance from the line is the error
Mean absolute error	Live numeric	mm	Averaged across the five stations, updated at the end of each run
Rain-shadow ratio	Live numeric	—	Blue Canyon divided by Reno precipitation, observed value close to 8
Omission ledger	Data table	mm and units	One row per omitted process: reason given, compute saved, error added
Run time	Timer	s	Solver seconds for the current configuration

WHAT THE STUDENT SHOULD REALISE

Students assume a better model is one that includes more, and that leaving things out is cheating. Here the budget forbids everything, and a model with four well-chosen processes beats one with seven crammed onto a coarse grid. Omission is a design decision with a stated price. The student should be able to say: *"Every model leaves things out on purpose, and I should be able to name what mine left out and what that cost me."*

A3.5 · Building and revising a model of a system

Experiment name: Three Versions of a Kelp Forest

Render mode: Hybrid 2D+3D

Simulation engine: Data-driven model (stock-and-flow) + Agent-based

Interaction level: Design

Session length: 18–25 min

NGSS anchor: Supporting — MS-LS2-4

THEME & SCENE

The left half of the screen is underwater at Point Lobos: a giant kelp forest lit by green shafts, blades rocking on the swell, sardine flicker in the gloom, and off to one side a pale pink patch of bare rock crawling with purple urchins. Otters raft at the surface, silhouetted against the light. The right half is a pale drafting canvas holding nothing yet, with a block palette down its edge: square stocks, valved flow pipes, small circular converters, thin connectors and curved feedback links with a plus or minus at the head. Along the bottom runs a ribbon of ground truth, thirty-five years of measured canopy area with a violent trough after 2013. Controls dock right; the fit score sits above the canvas.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Giant kelp plant	Actor	20 m stipes with 40 blades and gas-filled floats, jade to olive, growing up to 0.4 m/day; canopy thins to bare stipes as biomass falls	No
2	Holdfast and rocky reef	Structure	Granite reef at 12 m with 25 cm branching holdfasts; a detached holdfast leaves a pale scar the model counts as lost habitat	No
3	Purple urchin agents	Actor	6 cm spiny hemispheres, up to 4,000 instanced, crawl at 1 cm/s toward kelp, mass into a visible barren front when food runs out	Yes: place
4	Sea otter agents	Actor	1.2 m rafting bodies, 6–60 instanced, dive on a 90 s cycle, surface with an urchin and a rock, tool-tapping animation	Yes: place
5	Sunflower sea star	Actor	60 cm twenty-armed star, ochre, creeping at 0.2 cm/s; removed from the scene entirely by the wasting shock	Yes: place
6	Sea temperature field	Field	Invisible scalar over the reef, 9–20 °C, drives kelp growth through a nutrient proxy; drawn as a thermal wash when the overlay is on	No
7	Model canvas	Environment	Snap grid, warm white, holds the student's blocks; blocks refuse to connect if the units on the two ends disagree	Yes: resize
8	Stock block	Structure	Rectangular tank with a fill level and a units label, for example kelp biomass in tonnes or urchins in thousands	Yes: place
9	Flow block	Structure	Pipe with a valve and a rate label, always joining a stock to a stock or to a cloud; valve aperture animates with the live rate	Yes: connect
10	Converter block	Structure	Small circle holding a constant, a data series or a formula, for example sea temperature or grazing per urchin	Yes: place
11	Feedback link	Structure	Curved arrow carrying a plus or minus at its head; closed loops are auto-detected and shaded, reinforcing in red, balancing in blue	Yes: connect
12	Source and sink clouds	Structure	Cloud glyphs marking where matter enters or leaves the model boundary; a model with no sink cannot ever fall	Yes: place
13	Ground-truth ribbon	Instrument	Measured kelp canopy area for this stretch of coast, 1985–2020, one value per year, with the 2014–2016 collapse to near zero	No
14	Fit overlay and residual band	Overlay	Model curve drawn over the ground-truth ribbon with a shaded residual band; the worst three years are pinned and labelled	No
15	Version stack	UI-Probe	Card stack of saved versions v1 to v8, each showing its block count, loop count and score; two cards can be opened side by side	Yes: swap
16	Structure diff panel	Overlay	Highlights blocks and links added, removed or rewired between two versions, with the score change attached	No

HOW IT WORKS — THE MODEL

Whatever the student assembles is what runs. Blocks compile to a stock-and-flow system integrated once per simulated day: each stock changes by inflows minus outflows, each flow reads only the blocks wired into it. A minimal kelp model grows logistically toward a carrying capacity and can never crash. Wire in an urchin stock and a grazing flow and the system gains a balancing loop; wire otter predation onto the urchin stock and it gains a second. The observed 2014 collapse cannot be produced by any parameter setting of those loops, because it was driven by a temperature shock and by the loss of a predator, so the fit only improves when structure is added. Auto-tune therefore searches parameters only, which is the honest and important limitation: it will grind to a floor and stop, and that floor is the evidence that something is structurally missing.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Block palette	Multi-select	Stock / Flow / Converter / Connector / Feedback link / Cloud	Stock, Flow	—	Which blocks can be placed on the canvas, and therefore what kinds of model are buildable
Kelp growth rate	Slider	0.00–0.25	0.08	per day	Speed of canopy regrowth after any loss
Grazing per urchin	Slider	0.000–0.050	0.012	kg kelp/urchin/day	How fast urchins strip the reef and how quickly a barren forms
Otter predation rate	Slider	0–40	22	urchins/otter/day	Strength of the top-down balancing loop
Temperature forcing	Dropdown	Constant 13 °C / Observed 1985–2020 / Observed plus 2 °C	Constant 13 °C	—	Whether the converter reads a flat value or the real ocean record
Shock events	Multi-select	None / Sea-star wasting 2013 / Marine heatwave 2014–2016 / El Niño storms 1998	None	—	Injects real disturbances at their real dates
Auto-tune parameters	Toggle	On / Off	Off	—	Searches numeric values only; it can never add a block
Run span	Slider	5–35	35	years	Length of the simulated record compared against the data
Model version	Stepper	v1–v8	v1	—	Loads a saved structure for editing or comparison
Playback speed	Dial	0.25–5	1	years/s	Simulated years per real second

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Two boxes and a pipe	palette=Stock, Flow; growth=0.08; temperature=Constant 13 °C; shocks=None	Build kelp with growth alone. Your curve climbs and flattens. What can a model with no consumer never reproduce?
S2	Add the grazers	palette=all; grazing=0.012; predation=0; shocks=None	Add an urchin stock and a grazing flow. Does the reef now crash, and does it ever come back?
S3	The auto-tune wall	structure from S2; auto-tune=On; temperature=Observed 1985–2020	Tune every number as hard as the machine can. What is the lowest score you reach, and which years still refuse to fit?
S4	The real record	palette=all; temperature=Observed 1985–2020; shocks=Sea-star wasting 2013, Marine heatwave 2014–2016	Add otters and a temperature converter. Which single added block cut the error most?

STUDENT ACTIVITIES

1. Build v1 from one stock and one flow, run the full 35 years, and record the fit score and the first year the model and the data disagree by more than half.
2. Add an urchin stock, a grazing flow and the feedback link that closes the loop. Save as v2 and record the new score and the loop count.
3. Turn on auto-tune with v2's structure and record the best score it can reach, then write one sentence saying why it stops there.
4. Add the temperature converter, the otter stock and the two shocks. Save as v3, record the score, and open the diff against v2.
5. Rank your three versions by score and state, for each, the one behaviour in the data it still cannot produce.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Fit score	Live numeric	km ² canopy	Root-mean-square error between model and measured canopy area, updated at run end
Model vs data	Line graph	km ² vs year	Model curve over the ground-truth ribbon for the whole span
Residuals	Line graph	km ²	Model minus data per year; systematic humps reveal missing structure
Loop inventory	Counter	count	Reinforcing and balancing loops detected in the current wiring
Barren area	Live numeric	% of reef	Fraction of reef in urchin barren, model value beside the surveyed value
Version diff	Data table	mixed	Blocks and links added or removed between two versions, with the score change for each

WHAT THE STUDENT SHOULD REALISE

Students think a model that misses the data has the wrong numbers in it. Auto-tune here searches every number and still cannot reproduce the 2014 collapse, because no amount of tuning invents a predator or a heatwave. The fix was a new connection, not a new value. The student should be able to say: *"When my model fails, I check what I left out of it before I change what is in it."*

A4 · Earth as a system

A4.1 · The geosphere

Experiment name: Peel the Planet: Reading the Rock Shells

Render mode: 3D Scene

Simulation engine: Procedural geology + Ray/wave

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: Supporting — MS-ESS2-1

THEME & SCENE

Earth hangs in black, turning once every twenty seconds, drawn true to scale with no vertical exaggeration on the globe itself. A quarter wedge has been lifted out over the North Pacific, so the planet is a cut fruit: buff crust barely thicker than a drawn line, then banded mantle in deepening ochres, then the outer core glowing a dull sodium orange, then a hard white inner core at the centre. Fine white contour rings mark 410, 660 and 2,891 km. A blue push-pin sits on the San Andreas near Parkfield; a ring of small seismograph huts is scattered over the globe, each with a paper drum turning. A depth-profile column stands to the left; the control panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Globe body and cut wedge	Environment	6,371 km radius sphere with a removable 90° wedge; the cut face is the only place layers can be seen edge-on	Yes: drag
2	Continental crust	Structure	Buff granite shell 30–50 km thick, thickening to 55 km under the Sierra root, speckled feldspar texture, density 2.7 g/cm ³	No
3	Oceanic crust	Structure	Dark basalt shell only 7 km thick with pillow-lava texture and a thin sediment drape, density 3.0 g/cm ³	No
4	Lithospheric mantle and plates	Structure	Rigid peridotite shell to about 100 km, olive green, scored into 15 plate outlines that creep 2–7 cm/year when the timeline runs	Yes: place
5	Asthenosphere	Structure	100–410 km, slightly translucent and softly deforming, 1–2 % partial melt shown as a faint shimmer	No
6	Transition zone	Structure	410–660 km, two sharp bands where minerals repack; seismic rays visibly kink at both boundaries	No
7	Lower mantle	Structure	660–2,891 km, dense ochre, slow convection cells drawn as pale streamlines when the plate timeline runs	No
8	Outer core	Structure	2,891–5,150 km, liquid iron and nickel, glowing orange, swirling at centimetres per second; transmits P waves and stops S waves	No
9	Inner core	Structure	5,150–6,371 km, solid iron, white-hot at about 5,200 °C under 360 GPa, rotating fractionally faster than the mantle	No
10	Moho and core-mantle boundary	Overlay	Two labelled interfaces drawn as thin cyan lines on the cut face, with their depths and the velocity jump across each	No
11	Earthquake source	Actor	Placeable focus with a depth handle 0–700 km; releases a pulse and an expanding wavefront shell on trigger	Yes: place
12	P and S ray bundles	Field	180 traced rays per event, P in yellow and S in magenta, refracting at every velocity boundary and reflecting off the core	No
13	Seismograph stations	Instrument	Placeable huts with drum recorders; each prints P arrival, S arrival, and a flat trace where a wave never came	Yes: place
14	Shadow-zone band	Overlay	Translucent grey band painted on the globe from 103° to 142° for P and beyond 103° for S, drawn only after a run	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	California drill core	Instrument	Draggable rig producing a scaled core column: Central Valley alluvium 0–600 m, marine sediment, then Sierra granodiorite or Franciscan mélange, with a 20× exaggerated surface inset	Yes: drag
16	Depth profile column	Overlay	Vertical strip plotting temperature, pressure and density against depth, with a slider cursor tied to the peel depth	No

HOW IT WORKS — THE MODEL

Layers are defined by their seismic velocities, and the rays are what the student actually measures. Each ray is traced through a radial velocity model, bending by Snell's law at every boundary: P velocity climbs from 6 km/s in crust to 13.7 km/s at the base of the mantle, then drops abruptly to 8 km/s entering the liquid outer core, which is what bends P waves aside and creates the 103° to 142° shadow band. S waves are shear waves and simply cannot travel through liquid, so they stop dead at the core-mantle boundary and no station beyond 103° ever records one. Ray travel is played at real speed by default: about 13 minutes for P to cross the planet. Changing the core-state hypothesis rebuilds the velocity model, so a solid-core Earth genuinely delivers S waves everywhere, and the station record then contradicts the real data.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Peel depth	Slider	0–6,371	0	km	How far the shells are stripped back on the cut face, and where the profile cursor sits
Shells shown	Multi-select	Crust / Lithospheric mantle / Asthenosphere / Transition zone / Lower mantle / Outer core / Inner core	All seven	—	Which shells are drawn; hidden shells still transmit waves
Earthquake site	Drag-handle	Any surface point, focus depth 0–700	Parkfield, 8 km	km depth	Where the pulse starts and therefore which stations sit in shadow
Magnitude	Slider	4.0–8.5	6.5	Mw	Wave amplitude on the drums and how far a readable arrival travels
Wave type	Radio	P only / S only / P and S / Surface waves	P and S	—	Which ray bundles are traced and drawn
Seismograph stations	Stepper	4–24	12	count	How densely the globe is instrumented, and whether the shadow edge is resolvable
Core-state hypothesis	Dropdown	Liquid outer core / Solid all the way through / Uniform rock, no core	Liquid outer core	—	Rebuilds the velocity model, so a wrong Earth gives visibly wrong arrivals
Drill site	Drag-handle	Central Valley / Sierra crest / Coast Ranges / Mojave	Central Valley	—	Where the core column is taken and what rock sequence comes up
Surface inset exaggeration	Slider	1–50	20	×	Vertical stretch of the drill-core inset only; the globe stays true scale
Playback speed	Dial	0.25×–20×	1×	—	Ray travel speed, from real time to a fast sweep

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	One quake, twelve drums	site=Parkfield; magnitude=6.5; wave type=P and S; stations=12	Which stations record both waves, which record only P, and where on the globe are they?
S2	Find the shadow	stations=24; wave type=S only; magnitude=7.5	Mark every station with no S arrival. What angle does the silent zone begin at, and what does that silence prove?
S3	A wrong Earth	core state=Solid all the way through; stations=24; wave type=P and S	Predict the drums first, then run. Which stations now record S waves, and how does that clash with the real record?
S4	Drill California	drill site=Central Valley, then Sierra crest; exaggeration=20×	How deep is bedrock at each site, and after ten kilometres of drilling which shell are you still inside?

STUDENT ACTIVITIES

1. Place twelve stations evenly around the globe, fire the Parkfield event, and record P and S arrival times for each in a table.
2. Set wave type to S only, raise the station count to 24, and record the angular distance at which S arrivals stop.
3. Switch the hypothesis to a solid core, predict what each drum will show, run, and record which predictions failed.
4. Drag the peel slider from 0 to 6,371 km and record the temperature, pressure and density at 35, 660, 2,891 and 5,150 km.
5. Drill at the Central Valley site and the Sierra crest and record the layer sequence and bedrock depth for each.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Arrival table	Data table	s	P and S arrival time per station, with a dash where no arrival came
Travel-time curve	Line graph	s vs degrees	Arrival time against angular distance, with the shadow band showing as a gap
Shadow-zone map	Heat map	—	Globe painted by which waves reached each point
Layer profile	Line graph	°C, GPa, g/cm ³	Temperature, pressure and density against depth on the cut face
Hypothesis check	Pass-fail badge	—	Green when the chosen core state reproduces the real observed shadow zones
Drill core log	Data table	m	Rock unit, thickness and depth for the chosen California site

WHAT THE STUDENT SHOULD REALISE

Students picture Earth's interior as molten rock all the way down, and believe the layers were seen by digging. Nobody has drilled past 12 km. The layers are inferred from waves: a band of stations that never records an S wave is the evidence that the outer core is liquid. The student should be able to say: *"We know the inside of the Earth from what its waves do, not from having looked."*

A4.2 · The hydrosphere

Experiment name: Ninety-Seven, Two, and a Splash

Render mode: Hybrid 2D+3D

Simulation engine: Data-driven model + Particle system

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: Supporting — MS-ESS2-4

THEME & SCENE

The same slowly turning Earth, but the rock is dimmed to a grey ghost and only water is lit. Ocean basins glow in depth bands from pale shelf turquoise to near-black abyss, with the Mariana trench a dark scar at 10,935 m. Antarctic and Greenland ice sit as thick blue-white shells. Under the continents a subsurface amber haze marks groundwater; rivers thread the surface as fine silver lines; a soft haze of vapour hangs above everything. Floating beside the planet are three matched spheres, drawn to scale: all Earth's water at 1,385 km across, all fresh water much smaller, and all liquid fresh surface water a dot. A California inset shows the Sierra snowpack, the Central Valley and the Delta. Panel right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Ocean water body	Structure	1.338 billion km ³ shell, hypsometric colour ramp from 0 to 11,000 m, animated surface currents as faint drifting streaks	No
2	Ice sheets and glaciers	Structure	Antarctic shell to 4.8 km thick, Greenland to 3.2 km, plus 200,000 small mountain glaciers as bright specks; 24.06 million km ³ total	Yes: swap
3	Groundwater body	Structure	Amber subsurface layer to 2 km depth with a visible water table surface that can be dragged down; 23.4 million km ³ total	Yes: resize
4	Surface fresh water	Structure	Lakes as flat discs and rivers as 3 px silver threads; Lake Tahoe, Mono Lake, the Sacramento and San Joaquin individually named	No
5	Soil moisture film	Structure	Thin dark film in the top 2 m of land, dries visibly to pale tan under drought settings	No
6	Atmospheric vapour haze	Particle	Translucent white haze, densest over the warm tropics, holding only 12,900 km ³ at any instant	No
7	Comparison spheres	Structure	Three to-scale spheres, 1,385 km, 434 km and 56 km across, held beside the globe with labels and volumes	No
8	Tagged water tracers	Particle	Numbered glowing droplets, up to 500, moving between reservoirs by residence-time probability, leaving a fading trail	Yes: place
9	Tracer passport	Overlay	Card per tracer stamped with each reservoir entered, arrival year and time spent there	No
10	Central Valley aquifer cutaway	Structure	Vertical section from Sacramento to Bakersfield showing Corcoran clay, upper and lower aquifers, well screens and the water table	Yes: drag
11	Subsidence bench mark	Instrument	Survey pole near Mendota with 1925, 1977 and today marks; ground surface drops as pumping continues, up to 8.5 m	No
12	Sierra snowpack wedge	Actor	White wedge across the range whose thickness scales with April snow water equivalent; melts on a spring schedule into the rivers	Yes: resize
13	Sea-level ruler	Instrument	Graduated pole planted on the San Francisco waterfront reading metres above today's mean sea level	Yes: place
14	Reservoir stock bars	Overlay	Horizontal bar panel, one bar per reservoir, in both km ³ and percentage of all water; a log toggle makes the small ones visible	No
15	Salinity overlay	Overlay	False colour from 0 to 40 PSU across every water body, making the fresh fraction visibly tiny	No
16	Reservoir probe	UI-Probe	Cursor-following pointer reading volume, salinity and residence time wherever it is held	Yes: drag

HOW IT WORKS — THE MODEL

A reservoir-and-transfer model of the whole hydrosphere. Eight stocks hold real volumes: ocean 1.338 billion km³, ice 24.06 million, groundwater 23.4 million, lakes 176,400, soil moisture 16,500, atmosphere 12,900, rivers 2,120. Transfers run at measured global rates, evaporation 505,000 km³ per year, precipitation the same, river discharge 45,000. Each tracer is a random walker: on entering a reservoir it draws an exit time from that reservoir's residence-time distribution, nine days for the atmosphere, months for a river, 3,200 years for the deep ocean, centuries to ten millennia for groundwater. Pumping adds an extra outflow from the Central Valley aquifer that no natural inflow replaces, so the stock falls and the bench mark sinks, which is the honest lesson: not every reservoir refills on a human timescale.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Water bodies shown	Multi-select	Ocean / Ice / Groundwater / Lakes and rivers / Soil moisture / Atmosphere	All six	—	Which reservoirs are drawn and included in the stock bars
Tracer release site	Drag-handle	Any point on the globe	Pacific surface, off Monterey	—	Where tagged droplets enter the system
Tracer count	Stepper	1–500	100	droplets	How many walkers run, and whether one path or a distribution is visible
Time compression	Dial	1 day/s – 1,000 years/s	10 years/s	—	Simulated time per real second
Central Valley pumping	Slider	0–20	8	km ³ /year	Extra groundwater outflow; drives water-table fall and land subsidence
Sierra snowpack	Slider	0–200	100	% of average	April snow water equivalent, and therefore spring river flow
Ice scenario	Dropdown	Today / Mountain glaciers melt / Greenland melts / All ice melts	Today	—	Moves ice volume into the ocean and raises the sea-level ruler
Salinity overlay	Toggle	On / Off	Off	—	Colours every water body by salt content
View mode	Radio	Globe / Comparison spheres / Stock bars	Globe	—	Switches between the planet, the to-scale spheres and the accounting panel

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Count the water	view mode=Comparison spheres, then Stock bars; salinity overlay=On	Of all the water on Earth, what percentage is liquid, fresh, and above ground where you could drink it?
S2	Follow one droplet	release site=Pacific surface; tracers=200; compression=100 years/s	Over 5,000 simulated years, which reservoirs do the droplets visit, and in which one do they spend the longest?
S3	Drought pumping	snowpack=40; pumping=16; compression=1 year/s; run 10 years	How far does the water table fall, how far does the bench mark sink, and does the aquifer refill when pumping stops?
S4	If all the ice went	ice scenario=All ice melts; view mode=Globe; salinity overlay=On	How much does the sea-level ruler rise, which California coastline is lost, and does the fresh fraction go up or down?

STUDENT ACTIVITIES

1. Switch to comparison spheres and record the diameter and volume of each of the three spheres, then convert the smallest to a percentage of the largest.
2. Release 200 tracers in the Pacific, run 5,000 years, and record from the passports the mean time spent in the ocean, the atmosphere and groundwater.

3. Set pumping to 16 km³/year for ten years, record water-table depth and subsidence each year, then set pumping to 0 and record how much recovers in ten more.
4. Step the snowpack slider through 40, 100 and 160 % and record peak river discharge and summer soil moisture for each.
5. Run each ice scenario and record the sea-level ruler reading and the change in the ice stock bar.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Reservoir stocks	Bar chart	km ³ and %	Live volume in each of the eight reservoirs, with a log scale toggle
Residence times	Data table	days to years	Measured mean residence time per reservoir from the tracers actually run
Tracer passport	Data table	mixed	Per droplet: every reservoir entered, the year it arrived and how long it stayed
Water-table depth and subsidence	Line graph	m	Two traces against time for the Central Valley section
Sea level	Live numeric	m	Ruler reading above today's mean sea level for the current ice scenario
Fresh and accessible fraction	Live numeric	%	Share of all water that is liquid, fresh and at or near the surface

WHAT THE STUDENT SHOULD REALISE

Students believe fresh water is abundant and endlessly renewed, and that groundwater is an underground river. The accounting refuses both: 96.5 % of the water is salt, most of the rest is locked in ice, and the aquifer that sinks the valley floor takes centuries to refill. The student should be able to say: *"Almost all Earth's water is unusable, and the small usable part sits in reservoirs that refill at very different speeds."*

A4.3 · The atmosphere

Experiment name: Ride the Balloon to the Edge of Air

Render mode: 3D Scene

Simulation engine: Fluid/thermal + Data-driven model

Interaction level: Investigate

Session length: 12–18 min

NGSS anchor: Supporting — MS-ESS2-5

THEME & SCENE

Dawn on a stubble field outside Sacramento. A 2 m latex balloon strains on its line above a small styrofoam gondola bristling with sensors and a parachute folded beneath. Release, and the camera rides the payload: the valley falls away, the Sierra crest slides past on the right, and the sky bleeds from pale blue to indigo to black while the horizon curves. Translucent shells slide past like the panes of a lantern, each labelled with its altitude and named at the boundary where the temperature turns around. The balloon swells visibly as pressure falls. On the left a profile chart draws itself in real time, three traces climbing with the payload. Control panel docks right, altimeter top-centre.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Latex balloon envelope	Actor	2.0 m sphere at launch, inflating continuously to about 10 m as external pressure falls, skin thinning and going translucent, then bursting into ribbons	Yes: resize
2	Instrument gondola	Actor	25 cm styrofoam box on a 15 m tether, spinning slowly, with sensor booms and a camera port; mass set by the payload control	Yes: swap
3	Parachute	Structure	1.2 m orange canopy stowed above the gondola, deploying on burst and giving a 5 m/s descent	No
4	Troposphere shell	Structure	Ground to 11 km over mid-latitude California, hazy blue, holding all the weather, temperature falling 6.5 °C per km	No
5	Stratosphere shell	Structure	11–50 km, clear and dark, temperature flat then rising to about –3 °C at the top; layered with the ozone band	No
6	Ozone band	Field	Concentration maximum near 25 km at about 8 parts per million, drawn as a faint violet glow that brightens under the UV overlay	No
7	Mesosphere shell	Structure	50–85 km, near-black, coldest place in the atmosphere at about –90 °C; meteor streaks burn up inside it	No
8	Thermosphere shell	Structure	85–600 km, molecules drawn far apart, temperature reading over 1,000 °C while the density readout is almost zero	No
9	Air molecule field	Particle	N ₂ blue dumbbells, O ₂ red dumbbells, Ar violet spheres, CO ₂ and H ₂ O tagged; instanced density falls exponentially with height	No
10	Altitude landmarks	Structure	To-scale markers: cumulus base 2 km, Mt Whitney 4,421 m, airliner 11 km, ozone peak 25 km, noctilucent cloud 82 km, Kármán line 100 km, ISS 410 km	No
11	Jet stream ribbon	Field	Fast west-to-east flow near 10 km, up to 60 m/s, drawn as a translucent ribbon that visibly shoves the balloon downwind	No
12	Cumulative mass column	Overlay	Vertical bar beside the profile filling as the balloon climbs, showing the fraction of atmospheric mass now below the payload	No
13	Composition probe	UI-Probe	Draggable altitude marker returning a live composition pie and a molecule count per cubic centimetre	Yes: drag
14	Smoke and aerosol layer	Particle	Optional grey-brown plume from a Sierra fire, lofting to 6–12 km and spreading laterally under the tropopause	Yes: place
15	Profile chart	Overlay	Pressure on a log axis, temperature and ozone against altitude, drawn live and kept between runs for comparison	No
16	Data logger	Instrument	Onboard recorder writing one row per 100 m of ascent, downloadable after the flight	No

HOW IT WORKS — THE MODEL

Pressure follows the barometric relation $p = p_0 \cdot \exp(-z/H)$ with a scale height of about 8.5 km, so half the atmosphere's mass lies below 5.5 km and 90 % below 16 km. Temperature follows the standard atmosphere: falling at 6.5 °C per km through the troposphere to -56.5 °C at 11 km, flat to 20 km, rising through the stratosphere to about -3 °C at 50 km because ozone absorbs ultraviolet, falling again to -90 °C at the mesopause, then rising in the thermosphere where the few remaining molecules move very fast. Balloon volume scales inversely with pressure, so it bursts when its skin reaches its stretch limit, typically 30–34 km. The lesson the sim must protect is that the thermosphere is hot and would not feel hot: temperature and heat are separated on screen by putting the density readout beside the thermometer.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Payload mass	Slider	0.5–5.0	1.2	kg	Ascent rate and burst altitude
Helium fill	Slider	1.0–8.0	3.5	m ³	Lift, initial ascent rate and how soon the skin reaches its limit
Launch site	Dropdown	Sacramento 8 m / Owens Valley 1,150 m / Mt Whitney 4,421 m / San Diego coast 20 m	Sacramento 8 m	—	Starting altitude, pressure and how much atmosphere is already below you
Instrument package	Multi-select	Pressure / Temperature / Ozone / CO ₂ / Humidity / UV / Particle counter / Camera	Pressure, Temperature, Ozone	—	Which traces the profile chart draws and which columns the logger writes
Layer shells	Toggle	On / Off	On	—	Draws the labelled translucent shells and their boundary names
Air mass	Dropdown	Standard day / Winter storm / Summer heat / Marine layer / Wildfire smoke plume	Standard day	—	Reshapes the temperature profile and adds inversions or a smoke layer
Composition probe altitude	Drag-handle	0–120	25	km	Where the pie chart and molecule count are sampled
Ascent rate target	Slider	1–8	5	m/s	Climb speed, and therefore how finely the profile is sampled
Time compression	Dial	1×–200×	60×	—	Real seconds per simulated second of flight

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Straight up	payload=1.2; fill=3.5; site=Sacramento 8 m; package=Pressure, Temperature, Ozone	At what altitude does the temperature stop falling and start rising, and what is that boundary called?
S2	Half the air	probe altitude=5.5, then 16, then 48; layer shells=On	At what altitude is half of all the air below you, and how does that compare with the height of Mt Whitney?
S3	Smoke over the valley	air mass=Wildfire smoke plume; package adds Particle counter; site=Sacramento 8 m	How high does the smoke rise, what stops it going higher, and what does the particle count read at 3 km and 15 km?
S4	Hot and empty	probe altitude=110; package adds Camera	The thermometer reads over 1,000 °C. Why would a hand held out there freeze rather than burn?

STUDENT ACTIVITIES

1. Launch the default flight and record temperature and pressure every 2 km up to burst, then mark on your chart where each trace changes direction.
2. Drag the composition probe to 0, 25 and 110 km and record the molecule count per cubic centimetre and the percentage of nitrogen at each.

3. Set the air mass to the wildfire smoke plume and record the top of the smoke layer and the particle count above and below it.
4. Run two flights with payloads of 1.2 kg and 5.0 kg and record burst altitude and total flight time for each.
5. Read the cumulative mass column at 5.5 km and 16 km and write down what fraction of the atmosphere lies above 16 km.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Pressure profile	Line graph	hPa vs km	Log-scaled pressure against altitude, drawn live during ascent
Temperature profile	Line graph	°C vs km	The double reversal at the tropopause and stratopause is the shape to notice
Ozone profile	Line graph	ppm vs km	Concentration against altitude, peaking near 25 km
Cumulative mass below	Live numeric	%	Share of atmospheric mass beneath the payload at the current altitude
Composition at probe	Bar chart	% by volume	N ₂ , O ₂ , Ar, CO ₂ and water vapour at the probe altitude
Burst altitude and flight log	Data table	m and mixed	Burst height plus one logged row per 100 m for the whole flight

WHAT THE STUDENT SHOULD REALISE

Students picture the atmosphere as a uniform blanket that thins gradually to space, and read a high thermosphere temperature as scalding. The flight shows a layered structure defined by where energy is absorbed, temperature reversing twice, and half of all the air lying below the height of a Sierra peak. The student should be able to say:

"The air is layered and astonishingly shallow, and its layers are set by what absorbs the Sun's energy."

A4.4 · The biosphere

Experiment name: The Living Skin: Painting Life onto Bare Rock

Render mode: Hybrid 2D+3D

Simulation engine: Agent-based + Data-driven model

Interaction level: Design

Session length: 18–25 min

NGSS anchor: Supporting — MS-LS2-3, MS-ESS2-1

THEME & SCENE

The planet loads dead. Tan regolith, grey scree slopes, blue ocean, not one leaf, turning slowly under a hard sun. Along the bottom edge sits a biome tray of loaded brushes: coast redwood, chaparral, Central Valley grassland, Mojave scrub, kelp bed, open-ocean plankton, Arctic tundra. Paint one onto the globe and colour floods outward as the vegetation grows in over simulated decades. Hinged out on the left is a one square metre core patch in cutaway at three times vertical exaggeration: canopy, litter, dark leaf mould, crumb-textured topsoil, mottled subsoil, grit-and-clay saprolite, then fractured granodiorite with pale root tips wedged into every crack. Three glass bins across the top, labelled air, life and rock, are joined by pipes whose bore swells with whatever is flowing. Controls dock right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Bare globe base	Environment	6,371 km sphere at true scale, unvegetated: tan regolith, grey talus, hypsometric ocean; terminator sweeps once per minute; per-cell albedo recomputed whenever cover changes	Yes: drag
2	Biome patch	Structure	Paintable surface cell about 400 km across, carrying its own leaf area index, rooting depth and soil age; grows in over 5–80 simulated years rather than appearing instantly	Yes: place
3	Coast redwood stand	Actor	90 m tapered trunks in fog-belt cells, 40–60 instanced per patch, flat needle sprays, fog drip animating down the boles at 2 cm/s	Yes: place
4	Valley grassland sward	Actor	60 cm bunchgrass tussocks, 12,000 instanced, green in winter and straw-gold by June, root mass drawn to 1.8 m in the cutaway	Yes: place
5	Chaparral shrub	Actor	2 m chamise and manzanita domes, waxy grey-green, dense lignotuber knot at the base for resprouting	Yes: place
6	Mojave creosote and Joshua tree	Actor	Creosote at 1.5 m spaced 4 m apart on bare grit, Joshua tree at 6 m with bayonet leaves; 5 % ground cover only	Yes: place
7	Marine producers	Actor	Kelp blades on the shelf and a fine emerald phytoplankton haze offshore, bloom density pulsing on a spring and autumn cycle	Yes: place
8	Decomposer community	Actor	White fungal hyphae threads, bacterial stipple, 3 mm springtails and 10 cm earthworms in the litter and A horizon; population scales with litter mass	Yes: place
9	Herbivore agents	Actor	Tule elk, pocket gophers and grasshoppers as small walkers, 0–60 per km ² , cropping leaf area and dropping dung pellets that re-enter the litter	Yes: place
10	Root network and rhizosphere	Structure	Branching roots to species depth, translucent halo of exuded acid around every tip, visible crack propagation into granite at the deepest tips	Yes: resize
11	Soil column cutaway	Structure	One square metre column at 3× vertical exaggeration with O, A, B and C horizons drawn separately, plus a graduated stake reading depth in millimetres	Yes: drag
12	Bedrock and saprolite	Structure	Sierra granodiorite below, grading up through 40 cm of crumbly saprolite; the weathering front advances upward as a visibly ragged boundary	No
13	Gas exchange particles	Particle	CO ₂ as red-and-charcoal triads, O ₂ as red dimers, up to 4,000 per patch, flowing both directions at once, day and night streams drawn separately	No
14	Transpiration plume	Particle	Faint vapour column rising from the canopy, opacity scaled to millimetres per day; condenses into fog over the redwood belt	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Sphere flux ledger	Overlay	Three glass bins, air, life and rock, joined by six pipes whose bore is proportional to live flux, each pipe carrying its own live number	No
16	Flux chamber probe	Instrument	Clear 50 cm dome the student clamps over any patch; seals, then reads CO ₂ in and out, O ₂ in and out, and water vapour for 60 simulated seconds	Yes: place

HOW IT WORKS — THE MODEL

Every painted cell runs two flows, not one. Gross photosynthesis fixes carbon at a rate set by biome, light, temperature, rainfall and CO₂: about 1,300 g C per square metre per year under coast redwood, 550 under valley grassland, 60 under Mojave scrub, 130 in open ocean. Plant respiration burns roughly half of that straight back, and decomposers in the litter and soil return nearly all the remainder, so net storage is the small difference between two large opposing flows. Oxygen tracks carbon at 2.67 g O₂ per gram of carbon fixed, and is consumed again at the same ratio. Roots and soil microbes push soil CO₂ to 0.5–3 %, ten to seventy times the air above, which acidifies pore water and roughly triples chemical weathering of the bedrock beneath, so soil thickens at 0.02–0.1 mm per year. The sim must never draw photosynthesis as one-way oxygen production; both arrows stay live at all times.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Biome brush	Multi-select	Coast redwood / Chaparral / Valley grassland / Mojave scrub / Kelp bed / Open-ocean plankton / Arctic tundra / Bare rock eraser	Valley grassland	—	Which biome the brush paints, and therefore which organisms exist in the scene at all
Patch coverage	Slider	0–100	35	% of land surface	How much of the globe carries living cover; drives every global total
Herbivore stocking	Slider	0–60	12	animals/km ²	Grazing pressure, leaf area removed, and how much carbon takes the dung route
Decomposer activity	Slider	0–200	100	% of normal	Speed litter is returned to CO ₂ , and therefore how much carbon stays in the soil
Air temperature	Slider	–10–40	15	°C	Photosynthesis and respiration respond differently, so the net flip point moves
Rainfall	Slider	0–3,000	500	mm/year	Water available; a biome painted outside its range thins and dies back
Atmospheric CO ₂	Slider	180–800	420	ppm	Photosynthesis rate, soil CO ₂ , and weathering of the bedrock below
Time compression	Dial	1 day/s – 500 years/s	1 year/s	—	Simulated time per real second, from single days to soil-building millennia
View	Radio	Globe / Core patch / Flux ledger	Globe	—	Switches between the planet, the one square metre cutaway and the accounting bins
Flux chamber site	Drag-handle	Any painted patch	Central Valley grassland	—	Where the sealed dome takes its 60-second measurement

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	A world with nothing living	Biome brush=Bare rock eraser; Patch coverage=0; Time compression=500 years/s	With no life anywhere, do the pipes between air and rock stop? Does soil ever form on the core patch?
S2	Paint California	Biome brush=Coast redwood, Valley grassland, Mojave scrub; Patch coverage=35; Flux chamber site=Central Valley grassland	Which of the three moves the most carbon per square metre, and which moves the most in total once you count its area?
S3	The warm night shift	Air temperature=30; Decomposer activity=200; Patch coverage=60; View=Flux ledger	Photosynthesis is still running at full rate. Why has the net oxygen pipe dropped to zero and then reversed?
S4	Ten thousand years of soil	Biome brush=Coast redwood; Patch coverage=100; Time compression=500 years/s; View=Core patch	How deep is the soil after 10,000 years, and how many millimetres of granite were consumed to make it?

STUDENT ACTIVITIES

1. Erase all life, run 1,000 years at full compression, and record soil depth, weathering rate and every one of the six pipe values.
2. Paint redwood, grassland and Mojave scrub onto their real California positions, then clamp the flux chamber on each in turn and record gross photosynthesis, respiration and net carbon for all three.
3. Set decomposer activity to 200 % and air temperature to 30 °C, and record the moment the net oxygen flux crosses zero.
4. Raise herbivore stocking from 0 to 60 in four steps, recording leaf area index and net carbon storage at each step.
5. Run the core patch for 10,000 years under redwood, recording soil depth every 1,000 years, then plot depth against time and describe the shape.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Sphere exchange ledger	Data table	g C/m ² /day	All six flows between air, life and rock, in both directions, updated every tick
Gross versus net production	Bar chart	g C/m ² /year	Photosynthesis, plant respiration, decomposer respiration and the net remainder side by side
Net CO ₂ and O ₂ flux	Live numeric	g/m ² /day	Signed values from the sealed flux chamber, positive out of the patch
Soil depth	Line graph	mm vs years	Depth of O, A and B horizons on the core patch as it builds
Weathering rate	Live numeric	mm/1,000 years	Advance of the weathering front into fresh granodiorite, with the bare-rock value shown beside it
Transpiration	Live numeric	mm/day	Water moved from soil to air by the current cover
Global cover map	Heat map	% cover	Painted globe coloured by leaf area index, with total land area vegetated

WHAT THE STUDENT SHOULD REALISE

Students treat the biosphere as a layer of living things sitting on top of the Earth, and picture plants as oxygen factories that only give. The ledger refuses both. Photosynthesis and respiration run at once, and any net gain is the small difference between them, while the same roots that feed the plant are dissolving the granite that becomes soil at a tenth of a millimetre a year. The student should be able to say: *"Life is not on the Earth, it is one of the flows that builds it."*

A4.5 · Interactions among Earth's four spheres

Experiment name: Pull One Thread: The Four-Sphere Web

Render mode: 2.5D Layered

Simulation engine: Data-driven model + Field/vector

Interaction level: Investigate

Session length: 12–18 min

NGSS anchor: Supporting — MS-ESS2-2, MS-ESS2-6

THEME & SCENE

Four translucent planes hang one above another in a dark studio, each carrying the same 700 km slice of California drawn from the Pacific through the Coast Ranges, the Central Valley and the Sierra crest down into the Owens Valley. Lowest is the geosphere plane: granite batholith, valley fill, the San Andreas trace. Above it the hydrosphere plane, with ocean, fog bank, snowpack, silver rivers and an amber aquifer. Above that the atmosphere plane, onshore flow on the left and rain shadow on the right. On top the biosphere plane: redwood belt, oak savanna, irrigated cropland, mixed conifer, sagebrush. Twelve arrows run vertically between the planes, each named, each with a small clock at its midpoint. A loaded injector sits lower left; controls dock right; the cause log runs along the bottom.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Transect frame	Environment	700 km west-to-east slice at 20× vertical exaggeration, four parallel planes 120 px apart with parallax on drag; exaggeration factor printed on the frame	Yes: drag
2	Geosphere plane	Structure	Franciscan mélange under the Coast Ranges, 800 m of Central Valley alluvium, Sierra granodiorite batholith, Owens fault scarp; state variables are uplift, erosion rate and soil depth	No
3	Hydrosphere plane	Structure	Ocean with a cold upwelling tongue off Monterey, coastal fog bank, Sierra snowpack wedge, Sacramento and San Joaquin threads, aquifer glow, two reservoir discs	No
4	Atmosphere plane	Structure	Marine layer at 400 m, onshore westerly arrows, orographic cloud on the west slope, dry descending air east of the crest, jet stream ribbon at 10 km	No
5	Biosphere plane	Structure	Redwood belt on the coast, oak savanna on the foothills, irrigated cropland on the valley floor, mixed conifer to 2,600 m, sagebrush beyond; colour tracks leaf area	No
6	Sphere hub node	Instrument	Four labelled glass discs, one per plane, each carrying four live state gauges shown as fractions of baseline with a coloured deviation ring	No
7	Causal link arrow	Structure	Twelve directed arrows between hubs, thickness proportional to the flux carried, blue for a damping effect and amber for an amplifying one, name printed along the shaft	Yes: connect
8	Link delay clock	Instrument	Small dial at each arrow midpoint showing that link's lag, from one week to fifty years, counting down while an effect is in transit	No
9	Perturbation injector	Instrument	Dart tool the student aims at any hub; barrel shows the chosen perturbation and its size, and fires a visible pulse into that plane	Yes: drag
10	Travelling pulse	Particle	Bright bead running along an arrow at a speed set by that link's lag, splitting at every junction so branching chains are visible	No
11	Feedback loop halo	Overlay	Closed chains auto-detected and outlined, red for reinforcing and blue for balancing, with the loop's gain printed beside it	No
12	Human works	Actor	Placeable dams, the California Aqueduct, groundwater wells and irrigated fields; each adds or reroutes named links when placed	Yes: place
13	Cause log	Overlay	Ribbon along the bottom writing one plain sentence per hop, timestamped, in the order the engine actually fired them	No
14	Cut-link scissors	UI-Probe	Cursor tool that disables a chosen arrow, greying it out; the engine then runs with that pathway genuinely absent	Yes: connect

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Cross-section probe	UI-Probe	Vertical pin the student drops at any point along the transect, reading all sixteen state variables in that column	Yes: drag
16	Response order board	Instrument	Four-row scoreboard stamping the simulated time at which each sphere first moves more than 5 % from baseline	No

HOW IT WORKS — THE MODEL

Each hub holds four state variables expressed as fractions of their baseline, twelve directed links join the hubs, and every link carries a gain and a lag. On injection the engine integrates the coupled set once per simulated month: each state changes by the summed contributions of its incoming links, each contribution delayed by that link's own lag. Real couplings set the numbers. A 1 °C atmospheric warming lifts the Sierra snow line about 150 m and shifts peak runoff two to three weeks earlier, with a lag of one to three years. Clearing forest raises surface albedo and roughly doubles hillslope erosion within a single wet season. Ocean warming off Monterey weakens upwelling and thins kelp inside a year. Gains are linear and lags fixed, which is honest only for small kicks, and the panel says so on screen. The failure to avoid is a one-way hierarchy with life on top: every link has a return link, and cutting one must visibly break the chain.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Target sphere	Radio	Geosphere / Hydrosphere / Atmosphere / Biosphere	Atmosphere	—	Which hub the injector fires into, and therefore where the chain starts
Perturbation	Dropdown	Ash eruption / Fault uplift pulse / Ocean warms / Aqueduct diversion / CO ₂ step / Wind shift / Forest cleared / Bark beetle outbreak	CO ₂ step	—	The specific kick delivered; the list filters to match the target sphere
Perturbation size	Slider	0–100	25	% of baseline	Magnitude of the kick, and how far downstream it stays above the 5 % threshold
Links active	Multi-select	All twelve named arrows	All twelve	—	Which pathways the engine may use; disabling one removes that route entirely
Human works	Multi-select	Dams / Aqueduct / Groundwater wells / Irrigated cropland / None	Dams, Aqueduct, Irrigated cropland	—	Adds engineered links that reroute water between spheres
Delay scaling	Slider	0.1–10	1	×	Stretches or compresses every lag together, so slow chains can be seen inside a lesson
Run span	Slider	1–200	50	years	How long the coupled system is integrated after the kick
Arrow display	Radio	Sign only / Thickness by flux / Hidden	Thickness by flux	—	How much of the causal web is drawn, so a student can predict before seeing
Loop highlight	Toggle	On / Off	On	—	Outlines detected reinforcing and balancing loops and prints their gain
Playback speed	Dial	0.25–20	2	years/s	Simulated years per real second

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	One kick, four answers	Target sphere=Atmosphere; Perturbation=CO ₂ step; Perturbation size=25; Links active=All twelve; Run span=50	In what order do the other three spheres first move, and how many years does each take to get there?
S2	Cut the thread	Same as S1, but Links active removes Atmosphere to hydrosphere	Which downstream changes disappear entirely, and what does that prove that link was carrying?
S3	Kick the rock instead	Target sphere=Geosphere; Perturbation=Ash eruption; Perturbation size=60; Run span=20	Starting in the geosphere, does the biosphere respond faster or slower than it did in S1, and by which route?
S4	California plumbing	Target sphere=Hydrosphere; Perturbation=Aqueduct diversion; Perturbation size=40; Human works=Dams, Aqueduct, Groundwater wells, Irrigated cropland	Which sphere absorbs the water shortfall first, which one shows it last, and does the geosphere respond at all?

STUDENT ACTIVITIES

1. Set arrow display to Sign only, predict on paper which sphere will move second after a CO₂ step, then run S1 and record the response order board against your prediction.
2. Cut the atmosphere to hydrosphere link, re-run the identical kick, and record which of the four hubs still move and by how much.
3. Fire the same 25 % kick into each of the four spheres in turn, recording for each run how many hubs ended more than 5 % from baseline.
4. Place dams, wells and irrigated cropland, run the aqueduct diversion, and copy out the cause log as a numbered chain of sentences.
5. Turn loop highlight on, list every closed loop the engine finds, and mark each as reinforcing or balancing.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Sphere state gauges	Live numeric	% of baseline	Sixteen state variables, four per hub, updated every simulated month
Response order and lag	Data table	years	First time each sphere crosses 5 % from baseline, in the order it happened
Link flux traces	Line graph	% of baseline vs years	One trace per active arrow, so the delayed arrivals are visible as offset humps
Cause chain	Data table	mixed	The cause log as ordered rows: source, link, effect, delay, size
Loop inventory	Counter	count	Reinforcing and balancing loops detected in the currently active wiring
Entry-point sensitivity	Bar chart	hubs moved	Number of spheres pushed past 5 % per unit of kick, one bar per entry sphere

WHAT THE STUDENT SHOULD REALISE

Students imagine four spheres stacked like floors of a building, with effects running only downward and life sitting on top. Kicking the geosphere and watching the atmosphere answer first, then cutting a single arrow and seeing a distant effect vanish, makes that untenable. Every link has a return link, and every link has a delay. The student should be able to say: *"The spheres are not layers, they are one machine, and a push anywhere comes back changed."*

A4.6 · Modeling an Earth-system event

Experiment name: One Spark, Sixty Years: A Sierra Watershed

Render mode: Hybrid 2D+3D

Simulation engine: State machine + Data-driven model + Particle system

Interaction level: Argue-from-data

Session length: 18–25 min

NGSS anchor: Supporting — MS-ESS3-2

THEME & SCENE

An oblique aerial over twelve kilometres of the American River headwaters: granite domes and mixed conifer running from 1,400 m at the reservoir up to a ridge at 2,600 m, with a two-lane highway threading the canyon and a small town on the far bench. Late August light, the air already tan. A timeline ribbon spans the full width of the screen, its ticks spaced logarithmically, hours across the first week, then months, then decades out to year sixty, with a chunky scrubber handle the student drags. Everything redraws to whatever moment the handle sits on: terrain, sky, water, forest. Four stacked sphere cards on the right each carry a sparkline. Controls dock right beneath them.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Watershed terrain	Environment	12 × 12 km elevation model, 1,400–2,600 m, 1.5× vertical exaggeration, granite outcrop and glacial bench, full drainage network to the reservoir	Yes: drag
2	Mixed conifer stand	Actor	80,000 instanced trees: ponderosa, sugar pine, white fir, incense cedar, 20–50 m; per-tree state of green, scorched, torched or fallen	Yes: place
3	Understory and chaparral	Actor	Manzanita and chamise domes on south aspects, 1–2 m, with lignotuber knots that resprout within one season of burning	Yes: place
4	Fuel load field	Field	Invisible grid in tonnes per hectare, 8 t/ha after recent fire rising to 90 t/ha after ninety years of exclusion; drawn as a straw-to-red wash when the overlay is on	Yes: resize
5	Fire front	Actor	Advancing perimeter polygon with a flame band 3–40 m high, spotting embers thrown up to 1.5 km ahead, perimeter timestamped every hour	No
6	Smoke plume	Particle	Grey-brown column lofting to 6–12 km then flattening under the tropopause, drifting downwind, with a PM2.5 concentration field trailing to Sacramento and Reno	No
7	Soil column and hydrophobic layer	Structure	1 m profile per cell; under high severity a waxy water-repellent band bakes in 2–4 cm down, drawn as a dark seal that rain visibly beads on	No
8	Snowpack patch	Structure	White wedge above 1,900 m, thickness set by snow water equivalent; ash fall darkens it from albedo 0.80 to about 0.50, and melt-out date shifts visibly	Yes: resize
9	Reservoir and river reach	Structure	Reservoir disc with a water treatment intake, plus the river reach above it; both take a turbidity colour ramp from clear green to milk-chocolate	No
10	Rill and debris flow	Actor	Erosion channels incising the burned hillslope, coalescing into a debris lobe that runs the canyon and buries a stretch of highway	No
11	Storm driver	Environment	Scheduled winter storm with intensity in millimetres per hour, rain-on-snow flag, and an atmospheric river ribbon drawn offshore when that event is selected	Yes: swap
12	Wildlife and insect agents	Actor	Mule deer browsing new shrubs, black-backed woodpeckers arriving in year one on standing snags, spotted owl territory outlines, bark beetle population on stressed survivors	No
13	Regrowth succession stack	Structure	Per-cell state machine drawn as real vegetation: bare, herbs by year one, shrub field years three to fifteen, pine sapling to closed canopy by year sixty	No
14	Sensor kit	Instrument	Placeable air quality monitor, stream gauge, snow pillow, soil moisture probe and reservoir turbidity meter, each logging its own series for the whole run	Yes: place

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Timeline scrubber	UI-Probe	Logarithmic ribbon from hour zero to year sixty with labelled event pins; dragging it moves the scene and every readout together	Yes: drag
16	Four-sphere impact dashboard	Overlay	Four stacked cards, geosphere, hydrosphere, atmosphere, biosphere, each with a 0–100 impact index, a sparkline and a recovery clock	No

HOW IT WORKS — THE MODEL

Three stages run off one clock. Spread uses a simplified Rothermel rule: rate of spread scales with fuel load, roughly doubles for every 20 km/h of wind, doubles again per 20° of upslope, and rises sharply once dead fuel moisture falls below 8 %, at which point the fire climbs into the crowns. Burn severity per cell then sets the post-fire state: high severity bakes a water-repellent layer 2–4 cm down and the runoff coefficient jumps from about 0.1 to 0.6. Erosion follows rainfall intensity, so above roughly 24 mm in an hour on a steep high-severity slope the cell fails as a debris flow, and sediment yield decays with a three-year e-folding time as cover returns. Ash on snow drops albedo from 0.80 to about 0.50 and melts the pack two to three weeks early. Regrowth is a state machine, herbs to shrubs to pines, over sixty years. Timeline ticks are logarithmic on purpose: the whole point is that the four clocks differ.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Event	Dropdown	Sierra wildfire / Atmospheric river / Multi-year drought / Coastal upwelling collapse	Sierra wildfire	—	Loads a different driver, a different cascade and a relabelled panel; the scene, objects and sensors change with it
Ignition point	Drag-handle	Any terrain cell, 1,400–2,600 m	Ridge at 1,850 m	m elevation	Where the fire starts, and therefore what slope and aspect it runs on
Wind speed	Slider	0–90	35	km/h	Rate of spread, spotting distance and which way the plume carries
Fuel moisture	Slider	3–30	6	%	Flammability, and whether the fire stays on the ground or reaches the crowns
Years since last fire	Slider	2–120	90	years	Fuel load on every cell, which sets how severe the burn can become
First-winter storm	Slider	0–150	60	mm in 24 h	Intensity of the December storm, and whether debris flows trigger at all
Post-fire action	Multi-select	None / Mulch treatment / Replanting / Salvage logging / Prescribed burn beforehand	None	—	Adds or removes objects on the hillslope and changes which recovery path the state machine takes
Timeline	Timeline scrubber	Hour 0 – Year 60	Hour 0	hours to years	Moves the scene and every readout to that moment, forward or back
Sphere focus	Multi-select	Geosphere / Hydrosphere / Atmosphere / Biosphere	All four	—	Which impact cards and overlays are drawn
Playback speed	Dial	0.5×–50×	1×	—	How fast the clock advances when the timeline is left to run

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	The baseline fire	Event=Sierra wildfire; Ignition point=Ridge at 1,850 m; Wind speed=35; Fuel moisture=6; Years since last fire=90; First-winter storm=60; Post-fire action=None	Scrub the whole timeline. At what moment is each of the four spheres worst hit, and how far apart are those moments?
S2	The winter after	Same as S1, but First-winter storm=140	Did the greatest hydrosphere and geosphere damage come from the fire itself, or from rain that fell three months after it was out?

#	Scenario	Preset	The question the student answers
S3	Prepared ground	Years since last fire=8; Post-fire action=Prescribed burn beforehand; Wind speed=35; Fuel moisture=6	With the same wind and the same dryness, how do burn severity, sediment yield and the biosphere recovery clock change?
S4	A different event entirely	Event=Atmospheric river; First-winter storm=150; Post-fire action=None; Sphere focus=All four	Which spheres respond in a different order this time, and which one is still not back to baseline at year sixty?

STUDENT ACTIVITIES

1. Run the baseline fire, then drag the timeline to hour 6, day 21, December, April and year five, recording all four impact indices at each stop.
2. Place the air monitor in Sacramento, the stream gauge below the burn and the snow pillow at 2,100 m, then record each sensor's peak value and the date it occurred.
3. Set the first-winter storm to 140 mm, re-run, and record sediment yield and reservoir turbidity against the 60 mm run.
4. Compare the prepared-ground run with the baseline and record burn severity, debris flow count and years to 90 % canopy recovery for both.
5. Rank the four spheres by recovery time, then write one sentence naming the sphere that recovered fastest and the one that had not recovered by year sixty.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Four-sphere impact index	Bar chart	0–100	Live severity per sphere at the scrubber's current position, with the run's peak marked on each bar
Recovery clocks	Data table	days and years	Time for each sphere to return to 90 % of its pre-event value, one row per sphere
Air quality at Sacramento	Line graph	µg/m ³ PM2.5	Concentration against time on the log axis, with the healthy threshold drawn across it
Sediment yield and turbidity	Line graph	t/ha/year and NTU	Hillslope loss and reservoir turbidity on shared time axis, peaking in the first wet season
Snowmelt timing	Live numeric	days earlier	Shift in melt-out date at the snow pillow caused by ash darkening the pack
Carbon ledger	Line graph	t C/ha	Carbon released during the burn against carbon regained by regrowth, over the full sixty years
Burn severity map	Heat map	severity class	Watershed painted unburned, low, moderate and high, with the area of each class
Event chronology	Data table	mixed	Every modelled event with its timestamp, sphere and magnitude, exportable as CSV

WHAT THE STUDENT SHOULD REALISE

Students think a disaster is one event, in one place, that ends when the news does. Here the smoke has cleared in three weeks, the worst hydrosphere and geosphere damage arrives in December on rain that falls after the fire is out, the snow melts early the following April, and the forest is still recovering at year sixty. The student should be able to say: *"One event, four spheres, four different clocks, and the biggest damage did not arrive first."*

A5 · Investigation, measurement and evidence

A5.1 · Lab safety and working like a scientist

Experiment name: The Bench That Bites

Render mode: 3D Scene

Simulation engine: State machine + Fluid/thermal

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: Supporting — MS-PS1-2

THEME & SCENE

A school laboratory bench at the student's own eye height, close enough that two gloved hands reach in from below the frame. Grey epoxy worktop, a gas tap, a sink with an eyewash arm folded above it, a hot plate whose coil glows dull orange, a spirit burner with a lazy blue flame, a rack of glassware holding pale blue copper sulfate solution. Lighting is cool and flat like a real lab, except that every liquid carries a diagonal SIMULATION watermark and every accident plays as a clean graphic effect: colour, a card, never gore. Behind the bench hangs a peg wall of goggles, aprons and gloves. The fume cupboard sash slides at the left. Control panel docks right; a risk overlay can wash the bench amber.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Bench worktop and services	Environment	120×60 cm grey epoxy top, upstand at the rear, gas tap, twin sockets, sink with swan neck; surfaces carry a wet/dry state and a contamination map	No
2	Gloved hands avatar	Actor	Two rigged forearms entering from the frame bottom; sleeve length swappable short/long; skin and glove show a soft red glow at a contact site, never a wound	Yes: drag
3	PPE peg wall	Structure	Wall rail holding splash goggles, side-shield spectacles, cotton apron, nitrile gloves, hair tie, closed shoes; each item drags onto the avatar and snaps into a worn slot	Yes: place
4	Hot plate and thermal field	Actor	18 cm ceramic top on a steel body, coil glow ramps black → dull red → orange with setpoint; invisible temperature field extends 4 cm above the plate and 2 cm past its rim	Yes: place
5	Spirit burner and flame	Actor	60 mL glass reservoir, wick, cap; 6 cm blue flame with a 3 cm near-invisible upper cone; flame tilts to a draught and relights from any surface above 250 °C	Yes: place
6	Glassware rack	Structure	250 mL beaker, 100 mL conical flask, six 16 mm test tubes in a wooden rack, watch glass, stirring rod; each has a fill level, a hot/cold state and a topple threshold	Yes: swap
7	Reagent bottles	Structure	Four 250 mL amber bottles with real GHS pictogram labels: 0.1 M hydrochloric acid, copper sulfate solution, limewater, ethanol; ground-glass stoppers lift off and can be left off	Yes: swap
8	Splash droplet particles	Particle	60–400 droplets per pour, radius 0.5–2 mm, tinted to the reagent, launched at exit speed from the pour lip; each droplet reports where it lands	No
9	Spill puddle	Particle	Shallow fluid sheet spreading at 3 cm/s to an area set by volume; darkens the worktop, wets paper, and becomes a slip hazard on the floor	No
10	Emergency station	Instrument	Eyewash arm with twin aerated heads, drench shower pull-ring, fire blanket in a red pouch, 2 kg CO ₂ extinguisher on a bracket; each has a reach distance from the bench and an activation time	Yes: place
11	Fume cupboard sash	Structure	Toughened glass sash on a counterweight, 0–60 cm travel, airflow arrows drawn in the aperture; above 40 cm the containment arrows break up and vapour escapes	Yes: drag
12	Bench clutter	Structure	Rucksack, exercise book, phone, coat, coiled cable; each is a topple, ignition or trip object with its own flammability	Yes: drag

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
13	Broken glass and sharps bin	Structure	Shard set generated on any topple from above 15 cm, dustpan and brush, blue sharps bin; shards cut only on a bare-hand grab	Yes: place
14	Hazard state machine	Instrument	Invisible per-object channel table (thermal, chemical, mechanical, electrical) plus the contact-event queue and the countdown timer for each correct response	No
15	Risk heat overlay	Overlay	Toggleable amber-to-red wash over every bench zone, intensity $\propto$ current hazard magnitude; grey where a hazard is present but currently inert	No
16	Shake table	Environment	The whole bench sits on a hidden platform that can run a 6 s, 3 cm horizontal sway; unsecured glassware slides toward the nearest edge	No

HOW IT WORKS — THE MODEL

A hazard state machine runs beside the physics. Every object carries hazard channels: thermal (surface temperature), chemical (corrosive, irritant, flammable), mechanical (sharp, heavy, unstable) and electrical. Every student action generates a contact event carrying which channel, what magnitude, which body part and for how long. PPE is a filter on that event: goggles block splash to the eye entirely, gloves cut chemical contact to a fraction and cut thermal contact not at all above 120 °C, an apron intercepts torso splash. Severity is magnitude minus what the PPE absorbed, and severity selects the outcome card and the seconds allowed for a correct response. Splashes are real particles: pour height sets exit speed, exit speed sets spray radius, so a 25 cm pour genuinely reaches the face. The failure to avoid is random punishment. Every incident replays as a traceable chain from the first unsafe choice, and every hazard is visible before it fires.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
PPE worn	Multi-select	Splash goggles / Apron / Nitrile gloves / Hair tied / Closed shoes	Splash goggles	—	Which channels the consequence engine can still reach the avatar through
Hot plate setpoint	Slider	20–350	180	°C	Coil glow, plate surface temperature, burn severity, ignition of vapour above it
Reagent in use	Dropdown	Water / 0.1 M hydrochloric acid / Copper sulfate solution / Limewater / Ethanol	Copper sulfate solution	—	Splash consequence, flammability, the pictograms on the label
Pour height	Slider	1–30	5	cm	Droplet exit speed, spray radius, how far splash travels toward the face
Bench clutter	Stepper	0–6	3	items	How many loose objects can topple, catch fire or snag a sleeve
Fume cupboard sash	Drag-handle	0–60	20	cm open	Vapour containment; above 40 cm ethanol vapour reaches the room
Bench layout	Drag-handle	Free placement of hot plate, burner, glassware, bottles and bag within 120×60 cm	Standard layout	—	Where every item sits, and therefore what a reach passes over
Reach path	Radio	Across the hot plate / Around the front	Across the hot plate	—	Whether the sleeve and forearm cross the thermal field
Earthquake drill	Toggle	On / Off	Off	—	Runs a 6 s shake that slides unsecured glassware toward the bench edge
Step through incidents	Toggle	On / Off	On	—	Freezes at the instant of contact and draws the cause chain

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Suited up	PPE=Splash goggles, Apron, Nitrile gloves, Hair tied, Closed shoes; reagent=Copper sulfate solution; pour height=5; clutter=0	Run the full pour-and-heat sequence. How many incidents does the log record, and which single item of PPE absorbed the one splash that still happened?
S2	Bare-eyed	PPE=Apron; reagent=0.1 M hydrochloric acid; pour height=25; clutter=3	Where exactly did the droplets land, and how many seconds did the timer give you to reach the eyewash?
S3	Three things at once	reagent=Ethanol; hot plate setpoint=300; sash=55; reach path=Across the hot plate; clutter=4	The vapour ignites. Name the three separate conditions that all had to be true, and remove the single cheapest one.
S4	Shake-table drill	earthquake=On; hot plate setpoint=180; bench layout=glassware at the front edge; PPE=all	Which items moved and how far? Move them 15 cm inboard, re-run the shake, and record what changed.

STUDENT ACTIVITIES

1. Drag PPE onto the avatar one item at a time, running the same acid pour after each addition. Record which incident disappears at each step.
2. Set pour height to 5 cm, then 15 cm, then 25 cm with goggles off. Measure the maximum splash distance from the beaker in each case and record all three.
3. Place the hot plate, burner, ethanol bottle and rucksack yourself, then run the heating sequence. Record the risk overlay's peak value for your layout and for one improved layout.
4. Trigger the ethanol ignition deliberately, then use step-through to write down the cause chain in order, from the first choice to the flame.
5. Run the earthquake drill twice, once with glassware at the edge and once secured inboard, and record the number of breakages each time.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Incident log	Data table	mixed	One row per contact event: time, channel, magnitude, PPE absorbed, severity, response taken
Safety score	Live numeric	0–100	Starts at 100, deducts by severity, restores partially for a correct response inside the timer
Exposure counter	Counter	count	Eye, skin, inhalation and cut exposures, tallied separately
Response time	Timer	s	Seconds from incident to eyewash, shower, blanket or extinguisher activation
Risk map	Heat map	—	Bench-zone hazard intensity right now, updated every frame
PPE compliance	Pass-fail badge	—	Green only when every PPE item required by the current reagent and heat source is worn before the first action
Cause chain	Data table	—	The ordered list of choices leading to each incident, exportable as CSV

WHAT THE STUDENT SHOULD REALISE

Students believe lab safety is a list of rules to memorise and obey. Here the rules are never stated first: a 25 cm pour genuinely throws droplets to eye height, and goggles genuinely stop them. Safety turns out to be prediction, done before you act, about what a hazard can reach. The student should be able to say: *"Before I touch anything, I work out what could reach me, and I put something in the way."*

A5.2 · Variables and fair tests

Experiment name: Four Chambers, One Question

Render mode: Hybrid 2D+3D

Simulation engine: Data-driven model + State machine

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: Supporting — MS-LS1-5

THEME & SCENE

Four sealed growth chambers stand in a row on a steel rack in a dim grow room, each a clear acrylic box the size of a shoebox with its own LED bar, drip line and heating plate. Inside each, lettuce seedlings push up from peat plugs against a black backdrop, so pale green reads hard against dark. A window at the right end of the rack leaks daylight across chamber four. The camera sits level with the plugs, close enough to see cotyledons open. Left of the rack hangs the variable-locking panel: seven rows, each with a padlock, a value and a live difference reading. Below, a strip chart draws four height traces. Time runs fast; leaves visibly unfold.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Chamber rack	Environment	1.6 m steel shelf with four bays at 40 cm spacing, cable trays beneath; bays can be emptied or filled	Yes: place
2	Growth chamber shell	Structure	30×20×25 cm clear acrylic box with a sealed lid, hygrometer window, vent grommet; four instances, each independently labelled	Yes: resize
3	LED grow bar	Actor	28 cm strip of 24 white-plus-red diodes above each chamber, dimmable, on a per-chamber timer; casts a real pool of light with visible falloff	Yes: swap
4	Drip emitter and water meter	Instrument	4 mm line ending in a dripper over each plug, with a graduated 100 mL reservoir; drips are visible and the reservoir level falls	Yes: connect
5	Peat plug and substrate	Structure	40 mm cylindrical plug, three substrate options with different water-holding capacity, colour darkens when wet	Yes: swap
6	Lettuce seedling	Actor	Procedural plant: hypocotyl, two cotyledons, then true leaves at 3 day intervals; height, leaf count, leaf area and colour all driven by the model; six per chamber by default	Yes: place
7	Heating plate and temperature field	Field	Thin plate under each chamber with a setpoint; invisible scalar field inside the box, 0.5 °C hysteresis, 40 min thermal lag	No
8	Ruler stake	Instrument	150 mm stake beside each plug with 1 mm graduations, read automatically each simulated midnight	No
9	Variable lock panel	UI-Probe	Seven rows: light hours, water, temperature, substrate, seed variety, pot size, rack position. Each row has a padlock; locked rows force all chambers identical and shake when you try to change one chamber alone	Yes: drag
10	Confound detector	Instrument	Invisible comparator counting how many variables differ between any two chambers; feeds the attribution meter	No
11	Attribution meter	Overlay	Vertical gauge, 0–100 %, green at 100 when exactly one variable differs, falling to 0 when two or more do	No
12	Natural variation generator	Field	Per-seed vigour drawn from a distribution whose spread is slider-set; the reason two identical chambers still differ	No
13	Ghost curve overlay	Overlay	When two variables are unlocked, draws the translucent curve each one alone would have produced; the two overlap the measured curve and cannot be separated	No
14	Chamber label cards	Structure	Magnetic cards on each bay front showing that chamber's full setting list, with any differing value highlighted amber	Yes: swap

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Time-lapse camera and strip chart	Instrument	Fixed camera per chamber producing a daily frame; strip chart below the rack plots four height traces on shared axes	No
16	Randomiser shuffle tray	Instrument	A mechanism that swaps chamber positions on the rack each simulated day, breaking the window-side light and warmth gradient	Yes: drag

HOW IT WORKS — THE MODEL

Growth is a rate model run once per simulated hour. Each seedling accumulates biomass at a rate proportional to the product of four limiting factors: light (saturating above 14 h/day), water (falling off sharply below 10 mL/day and above 50 mL/day through waterlogging), temperature (an optimum near 20 °C with a steep drop past 30 °C) and substrate nutrient supply. Height and leaf area follow biomass with a fixed allometry. Each seed carries a vigour multiplier drawn from a spread the student sets, so two identically treated chambers never match exactly. The rack imposes a genuine nuisance gradient: chamber four receives 900 lx of extra window light and sits 2 °C warmer. The engine is honest about attribution: the confound detector counts differing variables directly, and when the count exceeds one the ghost overlay draws both candidate explanations sitting on top of each other, because the data genuinely cannot separate them.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Independent variable	Dropdown	Light hours / Water per day / Temperature / Substrate / Seed variety	Light hours	—	Which row the lock panel releases, and which chambers are allowed to differ in
Locked variables	Multi-select	Light hours / Water / Temperature / Substrate / Seed variety / Pot size / Rack position	Water, Temperature, Substrate, Seed variety, Pot size, Rack position	—	Structural: a locked variable is forced identical across all four chambers
Light hours per chamber	Slider (one per chamber)	0–20	6 / 10 / 14 / 18	h/day	LED on-time in each chamber, and the pool of light you can see
Water per chamber	Slider (one per chamber)	0–60	20 / 20 / 20 / 20	mL/day	Dripper volume, substrate wetness, waterlogging above 50
Temperature per chamber	Slider (one per chamber)	8–35	22 / 22 / 22 / 22	°C	Plate setpoint and the internal air temperature field
Seedlings per chamber	Stepper	1–8	6	count	Replicates; how much a single unlucky seed can shift the mean
Natural seed variation	Slider	0–40	15	% spread	How different identically treated plants are from each other
Rack position effect	Toggle	On / Off	On	—	Whether the window gradient adds light and warmth to the right-hand bay
Randomise positions daily	Toggle	On / Off	Off	—	Shuffles which chamber occupies which bay, averaging the gradient away
Time compression	Dial	100×–20000×	5000×	—	Simulated hours per real second

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	A fair test of light	Independent variable=Light hours; locked=Water, Temperature, Substrate, Seed variety, Pot size, Rack position; light=6/10/14/18; seedlings=6	Which chamber grows tallest by day 21, and what exactly lets you say the light caused it?
S2	Two things at once	Unlock Water; light=6/10/14/18; water=10/20/40/60; seedlings=6	Chamber four wins by 34 mm. Was it the light or the water? Point at the thing on screen that proves you cannot tell.
S3	One seed each	seedlings=1; natural variation=35; independent variable=Light hours; light=6/10/14/18	Run the identical setup three times. Does the same chamber win every time? What does that do to your conclusion?
S4	Central Valley water trial	Independent variable=Water per day; water=5/15/30/60; temperature locked at 30; seedlings=8; randomise=On	Below what daily volume does growth collapse, and what would you say to a grower who tested only one plant?

STUDENT ACTIVITIES

1. Set the lock panel so only light hours is unlocked, run to day 21, and record final mean height for all four chambers.
2. Unlock water as well, set water to 10/20/40/60, re-run, and record the attribution meter reading and what the ghost curves show.
3. Drop seedlings to 1 and raise natural variation to 35 %. Run the same light test three times and record the winning chamber each time.
4. Turn rack position effect on with randomise off, run, then turn randomise on and re-run. Record the difference between chamber four and chamber one in both.
5. Design and run your own fair test of temperature: choose the four values, lock everything else, and record your prediction before pressing Run.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Height traces	Line graph	mm vs day	Four traces on shared axes, one per chamber, drawn from the midnight ruler reading
Mean height with range	Bar chart	mm	One bar per chamber, whisker showing the spread across that chamber's seedlings
Attribution meter	Live numeric	%	100 when one variable differs, 0 when two or more do; updated whenever a control moves
Differences table	Data table	mixed	Every variable, chamber by chamber, with differing cells highlighted
Fair-test verdict	Pass-fail badge	—	Green only when exactly one variable differs and replicates are at least three
Confound list	Data table	—	Named list of every variable currently free to differ, with its size of difference
Run export	Data table	mixed	CSV of every seedling's daily height, leaf count and treatment

WHAT THE STUDENT SHOULD REALISE

Students believe a fair test means being careful and tidy. The rig makes it mechanical: unlock a second variable and the simulation itself reports that it cannot say which change caused the difference, and the two ghost curves lie on top of each other to prove it. Fairness is not an attitude, it is a countable property of the setup. The student should be able to say: *"If two things differ, the result cannot tell me which one did it."*

A5.3 · SI units and measurement

Experiment name: Read It Right: The Metrology Bench

Render mode: Hybrid 2D+3D

Simulation engine: Rigid-body + State machine

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: Supporting — MS-PS1-2

THEME & SCENE

A dark rubber-matted bench under one warm pool of light, shot from just above the instruments so their scales face the student. Along the mat: a triple-beam balance with its three notched riders, a 100 mL graduated cylinder half full of tinted water, a vernier caliper lying open, a steel metre rule, an alcohol thermometer in a clip, a stopwatch. A tray of specimens sits at the front: a river pebble, a steel bolt, a 20 mm aluminium cube, a wooden block, an irregular brass nut. Top right, a magnifier panel blows up whatever scale the cursor touches, so the divisions are always readable. Left, a small eye icon on a vertical track shows exactly where the student is looking from. Control panel docks right.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Bench mat and levelling feet	Environment	90×45 cm black rubber mat on three adjustable feet with a spirit-level bubble at the rear; tilt shifts the bubble and biases the balance	Yes: drag
2	Triple-beam balance	Instrument	32 cm beam assembly, 12 cm stainless pan, three beams reading 0–500 g in 100 g notches, 0–100 g in 10 g notches, 0–10 g continuous with 0.1 g graduations; pointer swings with damped oscillation and settles in 4 s	Yes: place
3	Balance riders and zero screw	Structure	Three brass sliders that drag along their beams and snap into notches on the two rear beams, plus a knurled zero screw under the pan giving ± 0.5 g of trim; beam torque is computed from real rider positions, not from a lookup	Yes: drag
4	Digital balance	Instrument	0.01 g resolution, tare key, 2 s settling, drifts 0.02 g when the bench is tilted; provided as a contrast to the beam balance	Yes: swap
5	Graduated cylinder set	Instrument	10 mL with 0.1 mL divisions, 100 mL with 1 mL divisions, 250 mL with 5 mL divisions; borosilicate look with a real refractive water column	Yes: swap
6	Meniscus surface	Structure	Curved liquid top rendered with a concave dip and a bright edge highlight; its apparent position on the scale shifts with the eye icon's height	No
7	Vernier caliper	Instrument	150 mm stainless caliper: main scale in 1 mm, vernier scale of 20 divisions giving 0.05 mm, external jaws, internal jaws, depth rod, thumbwheel and lock screw; jaws close on the specimen and stop at its true width	Yes: drag
8	Rules	Instrument	1 m steel rule and a 30 cm plastic rule, both graduated in mm; the plastic rule has a visibly worn end so the zero mark is not at the edge	Yes: swap
9	Thermometer	Instrument	–10 to 110 °C alcohol column in glass, 1 °C divisions, 20 s response time; reads air unless the bulb is fully immersed	Yes: place
10	Stopwatch	Instrument	0.01 s display with start, stop and lap; adds a 0.15–0.25 s human reaction offset to every manual start and stop	Yes: place
11	Specimen tray	Actor	River pebble 24.3 g, steel bolt 11.85 mm shank, aluminium cube 20.0 mm, wooden block 45×30×20 mm, brass nut of irregular shape, 50 mL water sample; each carries hidden true values to four significant figures	Yes: swap
12	Displacement kit	Structure	250 mL beaker, overflow can with spout, small catch cylinder, thread hook; used for the volume of an irregular solid	Yes: place
13	Eye-level marker	UI-Probe	Eye icon on a vertical track beside the cylinder, ± 15 cm about the meniscus; drives the parallax offset applied to the apparent reading	Yes: drag
14	Magnifier scale panel	UI-Probe	Fixed panel at top right showing a 6× view of whichever scale the cursor is over, with the nearest graduation labelled	Yes: drag

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Reading entry and tolerance checker	Instrument	Numeric field plus a unit dropdown; checks the value against the true value within half the smallest division, and separately checks the number of digits claimed	No
16	Unit conversion ladder	Overlay	Vertical ladder for km–m–cm–mm, kg–g–mg, L–mL–cm ³ , with animated ×10 and ÷10 rungs; lights the rungs the current conversion crosses	No

HOW IT WORKS — THE MODEL

Every instrument is mechanical, not a lookup. The balance sums real torques: pan load times arm length against each rider's mass times its distance, so the pointer settles only when the sum is zero, and the zero-adjust trim and the bench tilt both enter that sum. The cylinder reading is a geometric projection: apparent scale position equals true meniscus height plus the eye offset times a parallax factor set by the viewing distance, which is why the same 45 mL reads 47 mL from above. The caliper computes vernier alignment directly, so exactly one vernier line coincides with a main-scale line. Every instrument declares a smallest division, and the checker demands a reading estimated one digit beyond it, with an uncertainty of half a division. The failure to avoid is the calculator answer: no reading may ever appear pre-formatted, and no derived quantity may carry more significant figures than its least precise input.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Instrument on the bench	Dropdown	Triple-beam balance / Digital balance / Graduated cylinder / Vernier caliper / Rule / Thermometer / Stopwatch	Triple-beam balance	—	Structural: which instrument is live, magnified and accepting a reading
Specimen	Dropdown	River pebble / Steel bolt / Aluminium cube / Wooden block / Brass nut / Water sample	River pebble	—	The hidden true mass, volume and dimensions the instruments must reveal
Rider positions	Drag-handle	Rear beam 0–500 g in 100 g notches, middle beam 0–100 g in 10 g notches, front beam 0.0–10.0 g continuous	0 / 0 / 0.0	g	Beam torque, pointer position and how long the pointer takes to settle
Zero-adjust trim	Dial	–0.50 to +0.50	0.00	g	Systematic offset on every mass reading until the student zeroes it
Bench tilt	Dial	–2.0 to +2.0	0.0	°	Spirit bubble position and a proportional bias on both balances
Cylinder size	Dropdown	10 mL (0.1 mL divisions) / 100 mL (1 mL) / 250 mL (5 mL)	100 mL	—	Structural: the smallest division, and therefore the digits you may honestly claim
Liquid volume	Slider	0–100	45	mL	True volume in the cylinder and the height of the meniscus on the scale
Eye level	Slider	–15 to +15	0	cm relative to meniscus	Parallax; how far the apparent reading sits from the true one
Entry unit	Dropdown	mm / cm / m / g / kg / mL / L / cm ³ / s / °C	g	—	The unit your typed reading is interpreted in; a wrong unit fails the check outright
Reveal true value	Toggle	On / Off	Off	—	After you commit a reading, shows the simulator's true value and your error

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Zero first	Instrument=Triple-beam balance; specimen=River pebble; zero-adjust trim=+0.35; bench tilt=0.0	Weigh the pebble, then zero the balance and weigh it again. How big was the error, which way did it go, and would repeating the measurement have caught it?
S2	The meniscus and your eye	Instrument=Graduated cylinder; cylinder size=100 mL; liquid volume=45; eye level stepped +10, 0, -10	Record three readings of the same 45 mL of water. Which one is correct, and what physically changed between them?
S3	Rule against caliper	Instrument=Rule then Vernier caliper; specimen=Steel bolt	Give the bolt's shank diameter to as many digits as each tool honestly allows. Which digit is the estimated one in each case?
S4	Density, twice over	Instrument sequence=Triple-beam balance then Graduated cylinder; specimen=Aluminium cube; cylinder size=100 mL; displacement kit in use	Report the cube's density in g/cm ³ and again in kg/m ³ . Why is one number a thousand times the other when the metal has not changed?

STUDENT ACTIVITIES

1. Zero the balance with an empty pan, then drag the three riders to weigh the pebble. Record the mass with its uncertainty as $\pm$ half the smallest division.
2. Set the eye marker to +10 cm, 0 cm and -10 cm in turn and record the cylinder reading each time, then state which is the true volume.
3. Measure the aluminium cube's edge with the rule, then with the caliper. Record both, each with the correct number of significant figures, and note which digit is estimated.
4. Find the brass nut's volume by displacement: record the cylinder level before and after, then subtract. Record all three numbers.
5. Convert your cube density from g/cm³ to kg/m³ using the conversion ladder, and write down how many rungs you climbed and in which direction.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Committed reading	Live numeric	Selected unit	The value you typed, shown with the unit you chose
Tolerance check	Pass-fail badge	—	Green when your value falls within half the smallest division of the true value
Significant figures check	Pass-fail badge	—	Green only when the digit count matches what the instrument's precision supports
Absolute error	Live numeric	Same as reading	Your value minus the true value, shown only when reveal is on
Uncertainty	Live numeric	Same as reading	$\pm$ half the smallest division of the instrument in use
Precision comparison	Bar chart	mm or g	Uncertainty of each instrument used on the same specimen, side by side
Conversion trace	Data table	mixed	Each conversion attempted, the rungs crossed, and whether the exponent moved the right way
Measurement log	Data table	mixed	Every committed reading with instrument, specimen, unit, uncertainty and pass/fail, exportable as CSV

WHAT THE STUDENT SHOULD REALISE

Students believe a measurement is a number, and that more decimal places mean a better one. Here the 250 mL cylinder cannot support a tenth of a millilitre, eye position changes the reading without changing the water, and the balance lies until it is zeroed. A measurement is a number, a unit and an uncertainty, and the instrument decides how many digits you may claim. The student should be able to say: *"My last digit is a guess, and the instrument tells me which digit that is."*

A5.4 · Organizing and graphing data

Experiment name: The Plotting Bench: Charts That Lie

Render mode: Data Dashboard

Simulation engine: Data-driven model

Interaction level: Investigate

Session length: 18–25 min

NGSS anchor: Supporting — MS-ESS2-6

THEME & SCENE

A dark drafting table lit from within, laid out like a workbench rather than a spreadsheet. Down the left edge, a spool of field data unwinds and rows tick past on a paper ribbon, still warm from the logger. In the centre, a large white chart canvas glows on a light box, empty at first with two bare axis rails waiting. Along the right edge sits a rack of variable chips, one per column, each colour-edged by type: blue for continuous, amber for time, grey for category. Beneath the canvas turns a chart-type carousel of six mark styles. Top right, a small cartoon reader in a chair studies whatever you have drawn and says out loud what they conclude from it.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Data spool and row ticker	Actor	Paper ribbon feeding from a reel at 4 rows/s on load, then parking; hovering a row highlights its mark on the canvas	No
2	Dataset cartridges	Structure	Five slot-in cartridges: Sierra snowpack water content by month (n=48), Monterey kelp canopy area vs sea surface temperature (n=96), class reaction times (n=120), Central Valley annual rainfall 1980–2024 (n=45), dissolved oxygen vs depth in a Delta slough (n=60)	Yes: swap
3	Variable chips	Structure	One 60×28 px chip per column, labelled with name and unit, edge-coloured by data type; drag onto an axis rail to assign	Yes: drag
4	Axis rails	Structure	Horizontal and vertical drop rails flanking the canvas; a chip snapped onto a rail sets that axis, and a mismatched type makes the rail flash amber	Yes: connect
5	Chart canvas	Environment	720×520 px white plotting field on a light box, with a redraw animation that tweens marks whenever any choice changes	No
6	Chart-type carousel	UI-Probe	Rotating strip of six mark styles: scatter, line, bar, histogram, pie, box plot; each shows a thumbnail of the current data drawn that way	Yes: swap
7	Axis scale controls	Instrument	Min, max, tick interval and linear/log switch on each axis, with a tick ruler you can drag to change spacing directly	Yes: drag
8	Truncation ghost	Overlay	Whenever a bar chart's y-axis does not start at zero, a translucent grey version of the same chart drawn from zero is laid behind it	No
9	Data marks	Particle	The plotted points, bars, boxes and slices themselves; 4 px dots, 18 px bars, animated into position on every rebuild, each carrying its source row id	No
10	Trend line and residual sticks	Overlay	Least-squares line with thin vertical sticks from each point to the line; toggled, drawn only for two continuous variables	No
11	Naive reader avatar	Actor	Seated figure with a speech bubble that states, in one sentence, the conclusion the chart as drawn supports, right or wrong	No
12	Chart critique engine	Instrument	Invisible rule set of twelve named faults: truncated axis, wrong mark for the data type, unequal category spacing, pie of non-parts, missing units, missing spread, over-plotting, connected non-series, reversed axis, log without label, dropped rows, misleading aspect ratio	No
13	Fault pins	Overlay	Numbered pins that drop onto the offending region of the chart when a fault fires, each expanding to a one-line explanation	Yes: place
14	Data table view	Instrument	Sortable, filterable grid of the loaded rows, side-by-side with the canvas; selecting rows dims their marks	Yes: drag

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Spread whiskers	Overlay	Range bars or box whiskers drawn per aggregated group, showing what the summary hid	No
16	Export card	Instrument	A framed card holding the chart image, the caption the student types, and the CSV of exactly the rows drawn	Yes: swap

HOW IT WORKS — THE MODEL

Nothing is simulated physically here; the model is the grammar of graphics. Each dataset column is typed as categorical, ordinal, continuous or temporal, and each chart type declares which types it accepts on each axis. Assigning a chip runs a compatibility test, and an incompatible pairing still draws, badly, because seeing the bad chart is the lesson. Scales are pure transforms: a mark's pixel position is $(\text{value} - \text{axis min}) \div (\text{axis max} - \text{axis min})$ times the canvas span, so truncating the axis genuinely multiplies apparent differences without changing a single number. The critique engine scores the chart against twelve faults each frame. The naive reader is the honest part: it reads only pixel geometry, comparing bar heights and slopes as a person would, and reports the conclusion that geometry supports. When the reader's conclusion and the dataset's true relationship disagree, the gap is displayed as a number.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Dataset	Dropdown	Sierra snowpack / Monterey kelp / Class reaction times / Central Valley rainfall / Slough oxygen by depth	Sierra snowpack	—	Structural: which rows, columns and chips exist at all
X variable	Drag-handle	Any chip from the rack onto the horizontal rail	Month	—	Horizontal variable and, through its type, which chart types are sensible
Y variable	Drag-handle	Any chip from the rack onto the vertical rail	Snow water equivalent	—	Vertical variable and the axis units
Chart type	Radio	Scatter / Line / Bar / Histogram / Pie / Box plot	Line	—	Which mark is drawn for each row or group
Y-axis start	Dropdown	Zero / Auto-fit / Custom	Zero	—	Where the vertical axis begins; anything but zero on a bar chart raises the truncation ghost
Y-axis maximum	Numeric field	1–1200	1200	mm	How much vertical room the data gets, and therefore how steep everything looks
Axis scale	Radio	Linear / Logarithmic	Linear	—	How values spanning several orders of magnitude are spaced
Aggregation	Dropdown	Every row / Mean per group / Median per group / Total per group	Every row	—	How many marks are drawn and how much individual spread is hidden
Show spread	Toggle	On / Off	Off	—	Draws range bars or box whiskers on aggregated groups
Reader test	Toggle	On / Off	On	—	Whether the naive reader states the conclusion your chart supports

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	Snow through the year	dataset=Sierra snowpack; X=Month; Y=Snow water equivalent; chart type=Line; Y-axis start=Zero	Try line, bar and pie on the same rows. Which one shows the melt season, and why is a pie chart impossible for these data?
S2	The cliff that is not there	dataset=Central Valley rainfall; chart type=Bar; Y-axis start=Custom; Y-axis maximum=520	The reader says rainfall collapsed. By how many millimetres did it actually change, and what did your axis do to that number?

#	Scenario	Preset	The question the student answers
S3	Kelp and warm water	dataset=Monterey kelp; X=Sea surface temperature; Y=Kelp canopy area; chart type=Scatter, then Line	Two measurements with no time order. Which mark is right, and what does joining the dots falsely claim?
S4	One hundred and twenty people	dataset=Class reaction times; chart type=Histogram, then Bar with aggregation=Every row; show spread=On	What does the histogram reveal about the class that a bar per person hides, and what does the mean alone leave out?

STUDENT ACTIVITIES

1. Drag Month onto the X rail and Snow water equivalent onto Y, then step through all six chart types. Record which faults the pins report for each.
2. Set Y-axis start to Custom and narrow the range until the reader announces a collapse. Record the axis limits you used and the true change in mm.
3. Plot kelp against temperature as a scatter, turn on the trend line, and record the direction of the relationship in one sentence with numbers.
4. Aggregate the reaction times to Mean per group with spread off, then turn spread on. Record the mean and the range, and say which one you would report.
5. Write a caption for your best chart, export it, and check whether the reader's conclusion now matches the true relationship.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Chart canvas	Line graph	Dataset units	The live chart, redrawn on every control change
Chart score	Live numeric	0–100	Starts at 100 and deducts per active fault, weighted by how badly each misleads
Fault list	Data table	—	Every fault currently firing, with the pin number and the region it points at
Reader's conclusion	Live numeric	—	The sentence the naive reader draws from the pixels, updated on each redraw
Misreading gap	Live numeric	%	Difference between the apparent effect size the chart implies and the true effect size in the rows
Spread summary	Bar chart	Dataset units	Mean, median and range for each group when aggregation is on
Rows drawn versus rows loaded	Counter	count	Flags any rows silently dropped by the current axis limits
Export card	Data table	mixed	Chart image, caption and the exact CSV of the rows plotted

WHAT THE STUDENT SHOULD REALISE

Students believe a graph is decoration added after the science, and that a chart cannot be wrong if its numbers are right. Here the same forty-five true rainfall numbers produce a collapse and a flat line depending on one axis setting, and the reader is fooled every time. The chart is the claim. The student should be able to say: *"The numbers did not change, my axis did, so the graph was making the argument, not the data."*

A5.5 · Claim, evidence and reasoning

Experiment name: Argument Bridge: The Delta Fish Kill

Render mode: Hybrid 2D+3D

Simulation engine: Data-driven model + Rigid-body

Interaction level: Argue-from-data

Session length: 18–25 min

NGSS anchor: MS-LS2-4

THEME & SCENE

Split screen. Above, a Sacramento Delta slough at first light: olive water, tule reeds, a thick jade algae mat hugging the north bank, and two dozen bluegill drifting pale-belly-up against the reed line. A concrete outfall pipe from a tomato field dribbles into the channel; a small aluminium survey skiff waits at the ramp for the student to drive it. A timeline ribbon under the scene runs from fourteen days before the kill to two days after. Below, in cross-section, a rocky gap: an evidence pier on the left, a claim pier on the right, and nothing between them but air. Planks and girders lie in a stack. A small red load-test cart waits at the left abutment.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Slough channel and flow field	Environment	400 m reach, 60 m wide, 0–3.5 m deep with a scoured centre; tidal flow reversing twice daily at 0.1–0.4 m/s, drawn as drifting surface streaks	No
2	Algae mat	Actor	Procedural jade-green cover, 0–90 % of the surface, thickening over the fourteen days then collapsing to brown after the kill; blocks light to the water beneath and respire all night	Yes: resize
3	Fish population agents	Actor	Bluegill and mosquitofish, 400 agents, shoaling and depth-seeking; each carries an oxygen-stress state that turns it from silver to pale and finally to a floating body	No
4	Farm outfall and nutrient plume	Field	300 mm concrete pipe on the north bank with a visible discharge cone; nitrate and phosphate scalar fields spreading downstream on the ebb	Yes: drag
5	Survey skiff with sonde	Actor	4 m boat the student steers along the reach; a multiparameter sonde on a winch line reads dissolved oxygen, temperature, nitrate, pH, salinity and turbidity at the set depth	Yes: drag
6	Sampling site pins	Instrument	Up to eight numbered pins dropped anywhere on the reach; each stores a full vertical profile at the moment it was taken	Yes: place
7	Dissolved oxygen field	Field	Three-dimensional scalar, 0–12 mg/L, with a strong diurnal swing under the mat: 11 mg/L at 16:00, crashing to 1.4 mg/L at 05:00, and stratified so the bottom stays lowest	No
8	Timeline scrubber	UI-Probe	Ribbon from day –14 to day +2 in one-hour steps, with the algae cover, temperature and oxygen traces drawn along it as a sparkline	Yes: drag
9	Evidence cards	Structure	Auto-generated on each sample: a 140×90 px plank showing the quantity, value, site, depth and timestamp; its physical thickness in the bridge scales with its relevance to the current claim	Yes: drag
10	Claim rack	Structure	Five hanging claim cards, including one vague, two wrong-cause and one over-broad; selecting one raises it onto the right-hand pier	Yes: swap
11	Reasoning girders	Structure	Twelve candidate linking statements as steel girders of different lengths; each snaps between a plank and the next only if it actually connects those two ideas	Yes: connect
12	Bridge span and piers	Structure	Evidence pier at left, claim pier at right, 12 m gap; planks lay flat, girders truss beneath, and unsupported sections visibly sag	No
13	Load-test cart	Actor	Red four-wheeled cart with adjustable ballast; drives left to right at 1 m/s, falls through any gap or any plank whose relevance weight is below its axle load	No
14	Rebuttal bot	Actor	An opposing scientist at the right abutment who states the strongest alternative explanation the data still allows and sends a heavier cart across	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
15	Gap detector	Instrument	Invisible checker walking the evidence to reasoning to claim chain, reporting each break as a named gap with its position on the span	No
16	Argument scorecard	Overlay	Panel listing every card with its relevance percentage, every girder with its logical validity, and the final claim strength	No

HOW IT WORKS — THE MODEL

Two engines run together. Underneath, a real ecosystem model: algae photosynthesise in daylight and respire day and night, so oxygen swings hard under a thick mat and bottoms out just before dawn; nitrate from the outfall drives mat growth over the fourteen days; water temperature sets oxygen solubility; fish agents die when dissolved oxygen stays below 2 mg/L for more than three hours. Every measurement the skiff takes is genuinely drawn from those fields at that place, depth and hour. On top runs the argument mechanic: each evidence card is scored for relevance to the selected claim, defined as how much the measured quantity actually changed the modelled fish mortality. Relevance becomes plank thickness. Reasoning girders carry a validity flag, and only a valid girder bridges two planks. The cart then physically drives the argument, and it falls exactly where the reasoning does.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
Case	Dropdown	Delta fish kill / Schoolyard heat islands / Oak grove die-off	Delta fish kill	—	Structural: which live scene, dataset, claim rack and girder set load
Claim	Dropdown	Something in the water killed the fish / Low dissolved oxygen before dawn killed the fish / Fertiliser from the farm poisoned the fish / The water got too hot for fish / Algae eat fish	Something in the water killed the fish	—	Which card rises onto the claim pier and therefore how every evidence card is scored
Sampling site	Drag-handle	Anywhere along the 400 m reach, up to 8 pins	Two pins: at the outfall and mid-channel	—	Where the sonde reads, and whether you sample under the mat or in clear water
Sample depth	Slider	0.0–3.5	0.5	m	Which layer the sonde reads; oxygen is stratified and the bottom is worst
Timeline position	Timeline scrubber	Day –14 to day +2, 1 h steps	Day 0, 05:00	—	When the sample is taken; the oxygen crash sits between 03:00 and 07:00
Evidence on the bridge	Multi-select	Any collected cards, maximum 6	The first two cards collected	—	Which planks bear load; irrelevant planks are paper-thin
Reasoning girders	Drag-handle	Twelve candidate statements, snapped into the gaps between planks	None placed	—	Whether each plank actually connects to the next and to the claim
Challenge weight	Dial	1×–5×	2×	—	Ballast on the load-test cart, and therefore how strong the argument must be
Rebuttal bot	Toggle	On / Off	On	—	Whether the strongest alternative explanation is argued against you

Control	Widget	Range / options	Default	Unit	What it changes
Reveal the true cause chain	Toggle	On / Off	Off	—	After a load test, shows the model's actual chain of causes for comparison

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	The vague claim	claim=Something in the water killed the fish; evidence=any 3 cards; challenge weight=2×	The planks are laid but the cart still falls at the right-hand end. What is missing from a claim like this, and how would you make it testable?
S2	Right answer, wrong evidence	claim=Low dissolved oxygen before dawn killed the fish; evidence=nitrate at the outfall + water temperature at noon; challenge weight=2×	Your claim matches the simulator's own cause. Why does the bridge still collapse?
S3	Before dawn, under the mat	timeline=Day 0, 05:00; sample depth=2.5; sampling site=3 pins under the algae mat	What oxygen value did you record, and what can that one number support that a sample taken at noon could not?
S4	The opposing scientist	rebuttal bot=On; challenge weight=5×; claim=Low dissolved oxygen before dawn killed the fish	The rebuttal is that a salty tide pushed in and killed them. Which single extra measurement removes that alternative, and why?

STUDENT ACTIVITIES

1. Drive the skiff to three sites, drop pins, and sample at 0.5 m and 2.5 m at both 05:00 and 16:00. Record all eight oxygen values in a table.
2. Select the vague claim, build the bridge with your three strongest cards, run the load test at 2×, and record where the cart falls.
3. Swap to a specific claim, rebuild, and record the relevance percentage the scorecard gives each of your cards.
4. Place reasoning girders until the gap detector reports zero gaps, then write down the sentence each girder carries in order.
5. Turn the rebuttal bot on at 5×, record the alternative it raises, then collect the one measurement that defeats it and re-run.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Load test	Pass-fail badge	—	Green when the cart crosses at the set challenge weight, red at the plank or gap where it fell
Card relevance	Bar chart	%	One bar per evidence card, showing how much that quantity actually mattered to fish mortality
Gap list	Data table	—	Every break in the evidence to reasoning to claim chain, with its position and its name
Claim strength	Live numeric	0–100	Combines evidence relevance, reasoning validity and how specific the claim is
Rebuttal survival	Pass-fail badge	—	Green only when the argument still crosses after the alternative explanation is loaded on
Sonde profiles	Line graph	mg/L, °C vs m	Vertical profiles of oxygen, temperature and salinity at each pin, per timestamp
Fish mortality	Line graph	count vs h	Cumulative deaths against the oxygen trace, for comparison after the test
Argument transcript	Data table	mixed	Claim, ordered evidence, ordered reasoning and the test result, exportable as CSV

WHAT THE STUDENT SHOULD REALISE

Students believe evidence means any true fact they collected, and that a correct conclusion excuses weak support. The bridge refuses both: a true nitrate reading taken at the wrong hour is a paper-thin plank that carries nothing, and the right claim still collapses when nothing connects it to the data. The student should be able to say: *"Evidence only counts if it is about the thing I am claiming, and reasoning is the part that carries the weight."*

A5.6 · Designing an investigation of a system

Experiment name: Run My Plan: The Ecocolumn Trial

Render mode: Hybrid 2D+3D

Simulation engine: Data-driven model + State machine

Interaction level: Design

Session length: 18–25 min

NGSS anchor: Supporting — MS-LS2-3

THEME & SCENE

Two halves of one room. On the left, a magnetic planning board with nine empty slots down its face, from Question to Recording, and a tray of snap-together procedure blocks in bright plastic below it. On the right, a shelf of ecocolumns: two-litre bottle stacks, each a terrestrial chamber of loam and radish seedlings above a decomposition chamber of oak litter and red worms above an aquatic chamber of pond water, elodea and ramshorn snails. A shop lamp hangs over the left end of the shelf; a window at the right end throws a cooler, brighter wash across the last two columns. Press RUN and the room time-lapses, the plan executes exactly as written, and a review panel slides up.

OBJECTS & COMPONENTS

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
1	Planning board	UI-Probe	1.2 m magnetic whiteboard with nine labelled slots: Question, Independent variable, Dependent variable, Controlled variables, Procedure, Instruments, Replicates, Duration, Recording; empty slots read as gaps, not as errors, until RUN	Yes: place
2	Procedure block tray	Structure	Eighteen snap blocks with fill-in fields: Build N columns; Set variable to value in column i; Hold variable identical; Measure quantity with instrument; Repeat every interval; Randomise shelf positions; Run for D days; Record to table; Photograph daily; Compare means	Yes: connect
3	Ecocolumn assembly	Structure	Three 2 L PET bottles, cut and screw-coupled, 42 cm tall, cotton wick between the lower two, mesh vent in the cap; up to twelve instances on the shelf	Yes: swap
4	Terrestrial chamber contents	Actor	350 g loam, 12 radish seeds sprouting on day 3, six isopods that move to the damp side; height and leaf count tracked per plant	Yes: place
5	Decomposition chamber	Actor	20 g dry oak litter, four red worms, fungal bloom appearing from day 6; litter mass falls on a decay curve and the leachate darkens	Yes: place
6	Aquatic chamber	Actor	600 mL pond water, three elodea sprigs pearling oxygen in light, four ramshorn snails grazing, an algae film thickening on the sunlit face	Yes: place
7	Shop lamp and light field	Field	40 W lamp over the left bay; illuminance field falling from 8000 lx at bay one to 2200 lx at bay twelve	Yes: drag
8	Window draft and temperature gradient	Field	Hidden nuisance field: 3 °C cooler and 1200 lx brighter at the window end, with a daily swing twice as large there	No
9	Instrument rack	Instrument	Steel rule ± 1 mm, digital balance ± 0.01 g, dissolved oxygen probe ± 0.2 mg/L, thermometer ± 0.5 °C, lux meter ± 50 lx, pH strips ± 0.5 , 10 mL syringe ± 0.5 mL; each instrument stamps its own uncertainty onto the data column it fills	Yes: swap
10	Procedure executor	Instrument	Invisible interpreter that walks the block sequence literally; anything the plan does not specify is filled by a random but legal choice, and that choice is logged	No
11	Nuisance event generator	Field	Scheduled realistic disturbances: a four-day cloudy spell, one snail death, 8 % evaporation per week, one seed that never germinates	No
12	Auto data table	Instrument	Grows only the columns the plan asked for, at the intervals the plan set; unmeasured quantities are simply absent, not blank	Yes: drag
13	Design critique panel	Overlay	Ten criteria as coloured rows: testable question, single independent variable, controlled variables, replicates, instrument precision, sampling interval, duration versus process timescale, position randomisation, safe procedure, conclusion supported	No

#	Object	Class	Build note (geometry · appearance · behaviour)	Editable
14	Peer-review avatars	Actor	Three reviewers who each raise the fault their role would raise: a statistician on replicates and spread, a field ecologist on timescale and confounds, a technician on instruments and procedure	No
15	Version comparison rack	Overlay	Side-by-side view of v1 and v2 plans, their datasets and their scorecards, with changed blocks outlined	Yes: swap
16	Report skeleton export	Instrument	Card assembling the question, method blocks in order, the data table, the graph and a conclusion field the student fills	Yes: swap

HOW IT WORKS — THE MODEL

The ecocolumn runs a coupled stock-and-flow model at one tick per simulated hour: light drives elodea and radish photosynthesis, which adds oxygen and biomass; litter decays on a first-order curve, releasing nitrate downward into the aquatic chamber; snails and worms respire, drawing oxygen down overnight so dissolved oxygen swings between 3 and 9 mg/L within a single day. On top of that sits a literal procedure executor. It never repairs a plan. If the plan does not say when to measure, it measures at a random hour and the diurnal swing appears as noise; if the plan does not say to randomise shelf position, the light and temperature gradient is left standing and silently confounds the result. Every unspecified detail is logged as an assumption. The critique engine then scores the plan on ten criteria against what the executor actually had to invent.

CONTROL PANEL

Control	Widget	Range / options	Default	Unit	What it changes
System under study	Dropdown	Ecocolumn / Schoolyard bioswale / Classroom light and heat	Ecocolumn	—	Structural: which physical system, instrument rack and nuisance set load
Procedure sequence	Drag-handle	Up to 20 snap blocks from the tray, in any order	A five-block starter plan	—	Structural: the plan itself, executed literally and in the order written
Independent variable	Dropdown	Light hours / Water added / Litter mass / Snail count / Soil type / None chosen	None chosen	—	What is deliberately varied between columns; None chosen runs an observation study
Controlled variables	Multi-select	Light / Water / Temperature / Shelf position / Seed batch / Water source	Seed batch	—	What the executor forces identical across every column
Columns built	Stepper	1–12	3	count	Treatment levels times replicates; how much a single dead snail matters
Run duration	Slider	3–60	14	days	How long before the answer is read, against processes that take 6 to 30 days
Measurement interval	Slider	1–168	24	h	Sampling rate; a weekly sample cannot see a daily oxygen swing
Instrument per measurement	Dropdown	Rule / Digital balance / Dissolved oxygen probe / Thermometer / Lux meter / pH strips / Syringe	Rule for height, thermometer for temperature	—	The precision stamped on each data column
Randomise shelf positions	Toggle	On / Off	Off	—	Whether the lamp and window gradient is averaged out or left to confound
Time compression	Dial	500×–50000×	10000×	—	Simulated days per real second during the run

SCENARIOS

#	Scenario	Preset	The question the student answers
S1	The plan you would actually write	Procedure sequence=starter five blocks; independent variable=None chosen; columns=1; duration=14; measurement interval=168	Run it. What question can this data answer, and what is the honest answer to "why did the snails die"?
S2	One variable, three columns	independent variable=Light hours; columns=3; controlled=Water, Temperature, Seed batch, Water source; randomise=Off; interval=24	The lamp-end column wins. The ecologist says it was position, not light. Who is right, and which single toggle settles it?
S3	Sampling too slowly	independent variable=Litter mass; interval=168; instrument=Dissolved oxygen probe; duration=21	Oxygen reads almost the same every week. Set the interval to 3 h, re-run, and record what the weekly plan hid.
S4	Revise and re-run	Load the v1 critique; fix the three red criteria; columns=9; randomise=On; interval=6; duration=21	Which of your v1 conclusions survives v2, and which one turns out to have been the shelf, not the treatment?

STUDENT ACTIVITIES

1. Build a plan by snapping blocks onto the board, then press RUN and record every assumption the executor logged because your plan did not say.
2. Set columns to 1 and run, then set columns to 9 and run the identical treatment. Record the mean and the range in both, and say which you would report.
3. Run the same investigation with the measurement interval at 168 h and then at 3 h. Record the dissolved oxygen minimum in each and explain the difference.
4. Turn randomise shelf positions on and re-run S2. Record the difference between the lamp-end and window-end columns before and after.
5. Read all three peer reviews, rewrite the plan to answer them, and record your v1 and v2 scores side by side on the comparison rack.

OUTPUTS & MEASUREMENT

Readout	Type	Unit	Description
Executed dataset	Data table	mixed	Only the quantities the plan asked for, at the intervals it set, each column stamped with its instrument uncertainty
Assumption log	Data table	—	Every detail the plan left unspecified and the value the executor invented instead
Design scorecard	Bar chart	0–10	Ten criteria, each scored and colour-coded, with a one-line reason per criterion
Peer review	Data table	—	Three named critiques, one per reviewer role, each pointing at a specific block
Result with spread	Line graph	Depends on measure	The dependent variable against time, one trace per column, with range bands across replicates
Confound flag	Pass-fail badge	—	Red whenever an uncontrolled gradient could explain the result as well as the treatment can
What this data can and cannot say	Live numeric	—	Two short lists generated from the plan, not from the outcome
Version comparison	Data table	mixed	v1 against v2: blocks changed, criteria fixed, and whether the conclusion changed
Report skeleton	Data table	mixed	Question, method, data, graph and conclusion, exportable as CSV and PDF

WHAT THE STUDENT SHOULD REALISE

Students believe designing an investigation means writing neat steps. The executor takes them at their word: an unstated measurement time becomes a random one, an unrandomised shelf turns the lamp into the result, and one column cannot tell a treatment from bad luck. Every decision left out of the plan is made for you instead. The student should be able to say: *"If my plan does not say it, the experiment decides it, and then my result is about the wrong thing."*

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